Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in Observer-Patch Holography

B. Müller Alexander Osika Kai Xue Ben Cassie Brieuc de La Fournière

September 10, 2026

Paper release: `r2041` **Released:** September 10, 2026

Author affiliations: Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks.

Abstract

Observer-Patch Holography (OPH) needs a finite carrier for screen cuts. This paper turns those cuts into self-reading patch bodies with ports, local state, records, repair channels, synchronization, and checkpoint restoration. Given a declared finite algebra-state, Lüders instrument, and two-wing representation, its public record projectors support the corresponding Born–Lüders identities and the Tsirelson bound. The effect table does not select that instrument, and the carrier does not derive the quantum representation. A reversible proposal layer reconstructs a finite Gibbs potential when cycle affinities vanish and gives an exact $1, 3, 3, 5$ degeneracy theorem for the source-derived twenty-four-channel embedding.

On the declared echosahedral carrier lineage, an integer atom-counting grammar and normalized central-readback cost give twelve unit lines and an exact gap. Oriented incidence and refinement maps derive inverse pairing, the proper action of the alternating group A_5 on five letters, and a rank-three frame. Target-blind impulse and port readback derive the inverse-port response. Complete reversible response and endogenous overlap transport force the abstract compact Lie-algebra type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. The twelve-port orbit annihilates every multipole of rank one through five and carries a unique rank-six invariant with an exact sixty-two-point Morse census; any icosahedrally invariant finite-range cosine kinetics is isotropic through angular rank five, with rank six the least symmetry-allowed nonzero anisotropic rank. Under the declared fifteen-state matter contract, anomaly balance and tensor descent give the Standard Model charge lattice and maximal faithful matter image $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$ on the declared tensors. Source reconstruction of the matrix current and matter action, physical global-form selection, laboratory current identification, physical seam and family attachment, scalar multiplicity, and continuum quantum-field realization enter only as named premises; this paper constructs none of them. Declared conservative-record populations have a controlled flat causal limit, with interval counts recovering proper-time ratios. Finite signed-seam operators admit exact kinetic and kernel checks. Hardware evidence enters only with public device, calibration, circuit, data, and independent checks.

One typed construction and its source assumptions

The finite carrier, accepted repair, support geometry, gravity, abstract compact current, and conditional matter belong to one typed construction. Accepted repair produces the quotient-visible normal form. A1's complete twelve-port response and A2's endogenous holonomy force the abstract local Standard Model gauge Lie algebra. A declared matrix current and matter packet provide conditional realizations, while the source producer derives only the inverse-port response. Each composition uses the objects and premises displayed here.

Three meanings of screen

The word "screen" is used for three related objects that must not be identified without a receipt.

1. The *local carrier boundary* is the twelve-port oriented interface of one Echosahedral carrier on the declared branch. Its incidence has $(V, E, F) = (12, 30, 20)$.
2. The *federation screen* is the routed system of interfaces, records, repairs, and checkpoints of many carriers at finite cutoff.
3. The *support screen* is the observer-facing geometric chart. On the spherical branch it is the refined conformal S^2 used for caps, collars, modular flow, and Lorentz reconstruction.

Local icosahedral incidence does not determine the topology of the federation nerve. A federation of identical local carriers can be routed as a path, a cycle, a higher-genus complex, or a spherical complex. The map from routed carriers to a support-visible spherical nerve is therefore a physical bridge, not a change of notation.

Structure-sensitive, presentation-invariant physics

OPH is not neutral under arbitrary changes of substrate. It is invariant under changes of presentation that preserve the complete observer-visible carrier signature. On the Echosahedral branch that signature contains

$$\mathcal{C}_{i,r} = (\mathcal{A}_{i,r}, \rho_{i,r}, P_{i,r}, I^{\mathrm{or}}_{i,r}, \mathcal{R}_{i,r}, \mathcal{U}_{i,r}, \mathsf{Chk}_{i,r}, \mathsf{Resp}_{i,r}, c_{sr}),$$

where $P_{i,r}$ is the port set, $I^{\mathrm{or}}_{i,r}$ is oriented incidence, $\mathcal{R}_{i,r}$ is the record algebra, $\mathcal{U}_{i,r}$ is the repair or feedback interface, $\mathsf{Resp}_{i,r}$ is the visible response law, and c_{sr} is the refinement lineage. Hidden coordinates, port names, worker partitions, materials, and wiring presentations are silent when an isomorphism preserves this whole tuple and its error model. A change in port number, incidence, orientation, accessible algebra, response, repair law, clock, or refinement lineage need not be silent. A cube and an icosahedron are therefore different carrier contracts even when both are built from the same material.

A carrier body is not automatically an observer. It realizes an observer only when it supplies bounded access, self-readback, durable records, record-conditioned feedback, boundary prediction against controls, and checkpoint continuation. One carrier may pass that test. A connected sub-federation may pass it instead. No theorem fixes primitive observer size by counting carrier bodies.

The common finite computation

At cutoff r, source-bound carrier data are routed into an observer-patch federation. Accepted repair then acts on the physical quotient:

$$\mathsf{SourceCarrierTower}_r \xrightarrow{\text{realize/route}} \mathsf{ObserverFederation}_r$$
$$\xrightarrow{\pi_r} \mathsf{PhysicalQuotient}_r \xrightarrow{\mathrm{Rep}_r} \mathsf{PublicNormalForm}_r.$$

The last arrow is the consensus result only under semantic-dependency-complete transactions, coherent union-collar payloads, repair completeness, local diamonds, protected records, and the stated endpoint conditions. A collection of oscillators with equal frequency does not supply those clauses.

Physical phase locking can instantiate one synchronization layer. For a routed edge $e = ((i,a),(j,b))$, a source-produced phase record may certify frequency entrainment and a stable relative phase,

$$\dot{\theta}_{i,a} - \dot{\theta}_{j,b} \longrightarrow 0, \qquad d_{S^1}(\theta_{i,a} - \theta_{j,b}, \delta_e) \le \varepsilon_e.$$

That certificate becomes a consensus parent only when the phase record fixes a commensurability map for the exposed packets and is tied to the accepted repair ledger, semantic records, an independently calibrated clock, and the confluence premises. Phase locking can synchronize an interface. It does not by itself make the interface an observer, settle semantic disagreement, or produce physical time.

Two projections of one source

The public normal form has two separately typed projections:

$$\begin{aligned}
\mathsf{AuthenticatedSemanticHistory}_r &\longrightarrow \big(\mathsf{FiniteInformationalCauset}_r, \mathsf{CanonicalSourceHeight}_r\big),\\
\mathsf{ExactPortResponse}_r &\longrightarrow \big(\mathsf{RankThreeCarrier}_r,\ \mathsf{UnitDirections}_r \simeq S^2 \simeq \mathsf{FutureNullRays}\big),\\
\mathsf{PublicNormalForm}_r &\xrightarrow{\text{carrier-to-support}} \big(\mathsf{Support}_{S^2,r}, \mathsf{FiniteCapBWCertificate}_r\big),\\
\left.\begin{array}{r}\mathsf{FiniteCapBWCertificate}_r\\ \mathsf{CompatibleModularStateTower}_r\end{array}\right\}_{\text{same tower}} &\longrightarrow \mathsf{BW/KMS}_r \longrightarrow \mathsf{Lorentz/H^3Kinematics}_r.
\end{aligned}$$

$$\begin{aligned}
\mathsf{FiniteStrictPoset}_r &\longrightarrow \mathsf{ExactAuthenticatedCausetLog}_r,\\
\left.\begin{array}{r}\mathsf{FiniteInformationalCauset}_r\\ \mathsf{CanonicalSourceHeight}_r\\ \mathsf{RankThreeCarrier}_r\end{array}\right\} &\longrightarrow \mathsf{AmbientLorentzCarrier}_{1+3,r},\\
\left.\begin{array}{r}\mathsf{FiniteInformationalCauset}_r\\ \mathsf{AmbientLorentzCarrier}_{1+3,r}\end{array}\right\} &\longrightarrow \mathsf{AuxiliarySeparatedForwardPlacement}_r,\\
\left.\begin{array}{r}\mathsf{AmbientLorentzCarrier}_{1+3,r}\\ \mathsf{SuppliedSpatialReadback} + \mathsf{EdgeSpeed}_r\end{array}\right\} &\longrightarrow \mathsf{ForwardConePlacement}_r,\\
\mathsf{ForwardConePlacement}_r &\xrightarrow{\substack{\text{equal-height separation,}\\ \text{incomparable spacelike}}} \mathsf{FaithfulFiniteCausalPlacement}_r,\\
\mathsf{FaithfulFiniteCausalPlacement}_r &\xrightarrow{\substack{\text{physical signals+operational clocks,}\\ \text{source refinement+count-volume,}\\ \text{dimension+manifoldlikeness,}\\ \text{topology+uniqueness}}} \mathsf{EffectiveSpacetime}_{3+1},\\
\left.\begin{array}{r}\mathsf{NineSourceDirectionBalances}_r\\ \mathsf{SymmetricFields} + \mathsf{Ward/Bianchi}_r\\ \mathsf{AlgebraicCoupling} + \mathsf{Steps} + \mathsf{Connectedness}_r\end{array}\right\} &\longrightarrow \mathsf{FiniteEinsteinForm}_r,\\
\left.\begin{array}{c}\mathsf{EffectiveSpacetime}_{3+1}\\ \mathsf{FiniteEinsteinForm}_r\\ \mathsf{curvature} + \mathsf{physical\ stress}\\ \mathsf{vacuum} + 8\pi\mathsf{G\ scale}\\ \mathsf{small\ ball} + \mathsf{smooth\ convergence\ control}\end{array}\right\} &\xrightarrow{\text{physical promotion}} \mathsf{SmoothPhysicalEinstein}_{3+1}.
\end{aligned}$$

$$
\left.\begin{array}{r}\textsf{CompleteResponse}_{12}\\ \textsf{EndogenousHolonomy}_{A_5}\end{array}\right\}\longrightarrow \mathfrak{u}(1)\oplus\mathfrak{su}(2)\oplus\mathfrak{su}(3)
$$
$$
\xrightarrow{\text{conditional matrix and matter packet}} \textsf{MaximalFaithfulMatterImage}
$$
$$
\dashrightarrow \textsf{PhysicalGauge/QFT}.
$$

The dashed arrow denotes source and laboratory maps not constructed here.

The finite carrier-to-support leg is constructed from one oriented icosahedral incidence nerve: twelve carrier charts, thirty seam algebras, and twenty nonvacuous triple restrictions. The same bound artifact supplies confluent seam repairs, an operational observer receipt, and a refinement-natural oriented S^2 support limit. The separate geometric 2π-KMS comparison and FiniteCapBWCertificate are not outputs of that finite bridge. The state tower, common-comparison maps, compatible state/vector data, modular controls, and cofinal modulus form an independent compatible modular-state tower. The BW theorem consumes both inputs on the same refinement tower. Independently, authenticated read-from provenance supplies the finite event carrier and generated order. Every finite strict partial order also compiles exactly into an abstract authenticated log; the supplied relation and unthreaded snapshots make this grammar expressivity, not OPH dynamics or physical selection. Its canonical source height is zero at roots and one plus the maximum direct-parent height otherwise, equals the attained longest authenticated-parent-chain length, and strictly increases on generated ancestry. The exact source Gram quotient supplies a positive rank-three carrier. Its direct sum with an independent real axis,

$$
W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}, \qquad Q(t,x) = t^2 - g_{\mathrm{src}}(x,x),
$$

is a four-dimensional ambient target carrier with Lorentz inertia $(1,3)$, and source-unit directions are exactly its future-null rays. Canonical source height enters only through a scaled event placement. This finite construction removes the free event order, supplied placement rank, rank-four chart, and fitted signature. Once the support leg produces a conformal S^2, the independent classical identity $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ gives the same celestial Lorentz cone and the three-dimensional observer-frame fiber $H^3 = \mathrm{SO}^+(3,1)/\,\mathrm{SO}(3)$.

Every finite event log admits an auxiliary injective one-way realization: enumerate its events along one source axis and take a height scale larger than the enumeration diameter. Distinct same-height events are then spacelike. The enumeration is arbitrary and generally creates unsupported cone comparisons, so it supplies neither source selection nor faithful physical coordinates. A supplied event-local spatial readback and edge speed bound send generated precedence into the future cone. Equal-height spatial separation and a strict spacelike inequality for every increasing-height pair unsupported by ancestry derive converse cone support and exact two-way order–cone equivalence. Event separation follows from antisymmetry, and the two-way equivalence gives exact interval preservation. The exact-embedding route requires one source-selected refinement family preserving order, placement, and source directions, with dense and isotropic physical links on S^2, calibrated $\#I/\rho \to \mathrm{Vol}(I)$, independent dimension and manifoldlikeness tests, stable thickened-antichain topology, and convergence to a unique distinguishing Lorentzian limit. Event count is not public capacity N.

A controlled effective route uses vanishing causal discrepancy and compatible count-volume convergence. Conservative Fibonacci-record populations with a specified complete-neighbour read law have such a flat $1+3$ limit when their fill error is small relative to the shrinking read radius. Equal-mass assignments with vanishing displacement error give normalized event counts converging to volume. On fixed interior timelike diamonds, fourth roots of interval-count ratios recover proper-time ratios, with a reference interval fixing the unit and complete ancestry access supplying the readout. Within the class of nonzero closed convex pointed displacement cones, operational invariance under source rotations and every boost along one axis forces the Lorentz cone up to time orientation. The covariance is an assumption about actual influences and readouts. Physical selection of the population, read law and common matter realization is separate from these conditional results.

On the same finite event type, the source-order Einstein theorem replaces all-null balance by nine supplied balances on fixed algebraic source-direction representatives and, with symmetric fields, an algebraic coupling, four step maps, Ward/Bianchi identities, and connectedness, derives the all-null and Einstein-form tensor relations. The directions are algebraic rather than observed signals; the informational order does not select a field, step, or balance. The matrix fields and steps remain supplied. Reading them as smooth curvature and physical stress requires a same-family tensor-curvature reconstruction converging to the smooth Einstein tensor, or the separately stated continuum small-ball/null-balance identification, together with same-source stress, Ward/Bianchi, coupling, scale, and remainder data on the physical continuum family. Scalar-curvature convergence is only a diagnostic. The separate `composedEinsteinBranch` returns the conditional Einstein-form composition under those typed inputs; it does not supply them. Any admitted continuum spacetime is an emergent effective description, not a fundamental object inserted into the finite carrier.

The second projection begins with an exact finite result on the certified Echosahedral lineage. The twelve-port module decomposes as

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

The declared integer counting and normalized central-readback cost realization gives the twelve unit lines and exact gap. An append-only signed-event machine generates those integer loads. Conservative whole-unit seam repairs preserve total load and strictly decrease $V(N) = \sum_i N_i^2$; minimum move count is natural under every carrier rotation and the declared refinements. Divisibility of total load by twelve is necessary for consensus, while an explicit eighteen-move path settles the declared full-pile packet. A half-unit display rescales event values and the repair threshold together, so it is a units convention on the same move graph. Oriented incidence independently gives the antipodal pairing, proper A_5 action, and rank-three Gram frame, and determines the antipode J. Complete reversible response and endogenous overlap transport force the abstract local Standard Model gauge Lie algebra. Under the explicit inverse-port contract, the signed central involutive responses are exactly $\pm J$, with common sign conventional. Conditional also on the matrix current and rank-15 matter contract with its unique charge-conjugate projector pair, anomaly and tensor-descent certificates give the hypercharge lattice, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The target-blind producer derives the inverse-port response, without selecting the matrix current. Separate band premises select the rank-three response. Tensoring it with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal $\mathbb{Z}_6$ action come from that table. A distinct local-domain receipt checks the declared tensor-identity operator and conditional gap inheritance without source-selecting the matter action or transporting the twelve-port Spin packet. Laboratory identification of the finite matter carriers, current, and line sectors, exclusion of extra light sectors, physical matter-pole and continuum family identification, scalar attachment and dynamics, and quantum-field construction require separate maps.

The compact sector-category/Tannaka route conditionally reconstructs a compact group by a logically independent route. Physical family identification and extra-sector exclusion require separate identifications; they do not enter the contract-conditional finite gauge implication. Physical family identification requires an additional map on the charged-lepton and quantitative selector/gap branches. Physical unification requires a source-bound commuting square identifying its reconstructed compact group with the group acting through the Echosahedral current response:

$$\begin{array}{ccc} \mathsf{A5PortResponse}_r & \longrightarrow & G_r^{\mathrm{screen}} \\ \downarrow & & \downarrow\simeq \\ \mathsf{TransportableSectorCategory}_r & \longrightarrow & G_r^{\mathrm{DR}}. \end{array}$$

On those premises the abstract Lie-type agreement is exact. The construction supplies neither the physical vertical maps nor an identification of the two group actions. In the same way, the rank-three response band and declared generation table form a conditional complex rank-45 candidate, while three physical generations require matter-pole, continuum, seam-selection, persistence, and complement-complete refinement receipts. The rank-three result depends on the complete-band and cost-order premises; it is not a consequence of the icosahedral graph alone.

Finite controls and scope

The finite A_5 evaluator control has 60 reachable correctable public records on $\mathcal{H}_k = \ell^2(A_5) \otimes \mathbb{C}^k$:

$$M_0 = 60, \qquad D_{\mathrm{raw}} = 60k, \qquad \Delta_{\mathrm{raw}} = 60(k-1).$$

Raw equality occurs only at $k = 1$. Publicly inert multiplicity makes D_{raw} implementation-dependent, so the result is an evaluator control rather than physical capacity closure.

The unified claim has a precise scope. Consensus and geometry/gravity are composable branches of one source-bound self-reading carrier tower. The finite gauge branch uses the same carrier architecture, and A1–A2 force its abstract local Standard Model gauge Lie algebra. The matrix current and matter action are conditional realizations and are not joined to that tower by a common source construction. Local icosahedral incidence by itself constrains only the carrier module. The physical maps that turn these constraints into one inhabited universe are additional assumptions. Matching dimensions or symmetry labels does not supply them.

What This Paper Contributes

The companion OPH papers use observer patches, records, repairs, and screen cuts. This paper gives those words a finite carrier. It separates the abstract observer patch from its geometric support chart and from its physical or digital implementation. The separation removes dependence on hidden coordinates, labels, materials, and wiring choices only when the complete observer-visible carrier signature is preserved. OPH is therefore presentation-invariant rather than carrier-neutral: quotient-visible port incidence, orientation, response, repair, clock, and refinement data can change the physical branch.

The concrete contribution is the regulated screen architecture. Echosahedral carriers supply a twelve-port reference interface, edge sectors supply the heat-kernel/Casimir weights used by the quantitative branch, and central records give the commuting public indicator algebra. A separately declared quantum algebra-state, Lüders instrument, and two-wing representation give the Born–Lüders and Tsirelson identities; the effect table does not select the Lüders instrument. Checkpoint restoration states what it means for an observer to continue after repair. The finite modular-gearing package keeps strict-descent normalization separate from reversible proposal rates, tests whether the oriented 24-slot register is an invariant channel space, and fixes only symmetry-protected degeneracies rather than numerical gaps or clock units. The finite echosahedral selector gives the exact unit split within the declared integer counting and normalized readback-cost realization. Atomic signed record events generate the integer loads, while conservative whole-unit repairs terminate by strict quadratic descent. Their minimum move count is natural under rotations and the declared refinements. Oriented incidence separately gives the proper-A_5 result without using downstream particle or gauge targets. Incidence and target-blind port readback derive the inverse-port response and refinement maps. The complete-response and endogenous-transport clauses force the abstract local Standard Model gauge Lie algebra, while the declared matrix current is a conditional realization. Under the declared fermionic Spin category, the anomaly and descent certificates imply the Standard Model charge lattice and common $\mathbb{Z}_6$ tensor kernel from the stated premises alone. The corresponding quotient is the maximal faithful matter image on the declared tensors. Source reconstruction of the matter action, physical global-form selection, laboratory identification, the physical seam mechanism, scalar multiplicity, and equality with the distinct Tannaka current are premises of any stronger reading; this paper proves none of them. The generation-count, extra-sector, charged-lepton, and quantitative selector/gap branches carry their own declared completions. The twelve-port orbit additionally carries an exact multipole fixed point: ranks one through five annihilate, the unique rank-six invariant has a certified sixty-two-point Morse census, and every icosahedrally invariant finite-range cosine kinetics confines its lowest directional residue to that one rank-six line. Hardware claims are kept behind a public evidence rule: manifests, hashes, calibration records, raw or reduced traces, and exact verifier receipts carry the weight.

Contents

1 Scope

OPH microphysics uses finite observer patches with echosahedral interfaces. A spherical screen is a geometry chart for observer-visible cuts. The cellulated-sphere gauge-register construction is a regulator and digital calibration chart. The premise is:

> Observers access finite observer-visible cuts. A spherical screen is a geometry chart for such cuts. The underlying fixed-cutoff implementation surface is a federation of finite overlap-facing observer patches, with echosahedral local interfaces and recurrent toroidal subchannels.

At fixed cutoff, federated patch carriers provide the microphysical architecture. Mathematical statements use the finite algebras and declared interfaces given below. Hardware statements require separate public evidence with stable provenance and exact checks. The octahedral $\mathbb{Z}_2/S_3$ finite-group model is a calibration chart rather than a physical identification.

2 Finite Local Action and Operator Domain

The finite carrier can be tested as a common home for causal, seam, and field data without identifying that home with a continuum spacetime.

Proposition 2.1 (Finite signed-seam operator realization)**.** *On the observer-visible seam complex of* 8,662 *carriers and* 11,816 *seams, the declared reversing transport frustrates all* 38 *triangles. Two coordinate routes through the same* $\mathbb{F}_2$ *rank implementation agree with the exact rank identity and give lift-ambiguity rank* 3,117*. Scalar, chiral, and gauge sections carry a sign-twisted derivative of support radius one. In the finite receipt, the unit-counting measure is a declared realization choice. Exact integer evaluations on the declared deterministic sections test the adjoint and kinetic identities, gauge covariance, observer-cover gluing, subcomplex restriction naturality, and Dirichlet boundary restriction. The signed-graph rank theorem gives twisted-kernel dimension zero.*

Proof. The seam holonomies and both boundary-rank eliminations are evaluated over $\mathbb{F}_2$. The kernel statement reduces to the signed-graph consistency theorem; the other operator checks use deterministic integer-valued sections. The declared transport and unit-counting measure define this finite realization. Independent replay checks its source artifacts and exact operator identities [19]. □

An authenticated semantic log defines a finite locally finite poset [1, 2]. Its proved meaning here is authenticated read-from dependence. The semantic log fixes the finite event carrier and generated order. The canonical source height computed from authenticated parenthood—zero at roots and one plus the maximum direct-parent height otherwise—is the attained longest authenticated-parent-chain length and strictly increases on generated ancestry. Independently, the exact rank-three positive source Gram quotient and a real axis define

$$W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}, \qquad Q(t, x) = t^2 - g_{\mathrm{src}}(x, x),$$

as a four-dimensional ambient target carrier with Lorentz inertia $(1, 3)$. Every finite strict partial order compiles exactly into an abstract authenticated semantic log; the generic authenticated-parent relation is the whole supplied order rather than its Hasse reduction alone. The supplied relation and unthreaded snapshots make this a grammar-expressivity result, not a dynamics-selection theorem. Source height enters only as the temporal coordinate of a positively scaled event placement. An enumeration-dependent placement along one source axis gives every finite log an injective one-way forward-causal realization; it is neither order-reflecting nor physical. This is not an intrinsic dimension estimate for the finite poset. Source-unit directions are exactly future-null rays and hence form S^2. A supplied spatial readback satisfying the edge speed bound sends every generated ancestry into the future cone. Equal-height spatial separation and strict spacelikeness of unsupported increasing-height pairs imply converse support and exact order–cone equivalence. Antisymmetry derives injectivity, while the two-way equivalence preserves every source interval exactly on the placed image. The machine-checked diamond witness is non-chain, has spacelike branches, and maps every parent edge to a null displacement. Thus a freely declared precedence relation, rank-four chart, and fitted signature are absent from the finite precursor.

The exact-embedding route from this precursor requires a source-causal continuum certificate. One source-selected refinement family must identify physical events and links, preserve exact two-way order/cone agreement, make the physical link directions dense and isotropic on S^2, calibrate $\#I/\rho \to \mathrm{Vol}(I)$, pass independent dimension, interval-abundance, height/profile, manifoldlikeness, and topology tests, and converge to a unique distinguishing Lorentzian limit [3, 4, 5, 6]. Under the appropriate continuum hypotheses order fixes conformal geometry and volume fixes the conformal factor [8, 9]. Standard Lorentz-invariant causal-set approximation uses Poisson sprinkling [7]; the deterministic OPH log is not a Poisson sprinkling, public capacity N is not event number, and event number is not spacetime volume without calibration.

The conservative-record carrier also supports a controlled flat causal construction. Fibonacci residues give q^3 distinct spatial sites; repeating them with complete shrinking-radius neighbour reads and a declared model clock produces $1 + 3$ causal-order and normalized count-volume limits. Interior timelike intervals have ordering fraction tending to $1/10$ [22]. This theorem specifies a population and read law whose realization by native patch updates and whose agreement with a field action require separate arguments. The same histories admit a retrospective clock. For intervals approaching fixed interior timelike diamonds, the fourth root of their inclusive count ratio converges to their proper-time ratio. A reference interval fixes the unit, and complete ancestry access supplies the count. This readout needs no oscillator or numerical timestamps. Operational covariance gives a separate cone criterion: a nonzero closed convex pointed cone invariant under the source icosahedral rotations and all boosts along one axis must be one of the two Lorentz cones. A time orientation selects the future cone. The covariance must act on possible influences and readouts. Algebraic direction maps alone do not establish that operational premise[22].

The exact same-event-type source-direction Einstein theorem, in a separately supplied $3+1$ tensor interface, takes nine fixed algebraic source-direction balances, supplied symmetric fields, discrete step maps, an algebraic coupling, Ward/Bianchi identities, and connectedness, and derives all-null balance and the finite Einstein-form tensor relation. The directions are not observed provenance links or sky samples; no provenance edge selects a tensor field, step, or balance, and the theorem does not identify its coordinate differences with those of the constructed carrier. Reading the fields as smooth curvature and physical stress requires a same-family tensor-curvature reconstruction converging to the smooth Einstein tensor, or the separately stated continuum small-ball/null-balance identification, together with same-source stress, Ward/Bianchi, coupling, scale, and remainder data on the same physical continuum family. Scalar-curvature convergence is only a diagnostic. The finite-regulator ambient carrier dimension and signature are constructed, while poset dimension, the fields, and their physical interpretation remain explicit.

Corollary 2.2 (Exact positive finite-domain signed-seam gap)**.** *With the declared unit-counting measure, the sign-twisted kinetic Hamiltonian is integer symmetric and positive semidefinite. Its kernel is trivial on the connected frustrated visible seam complex, so its smallest eigenvalue is strictly* *positive. The determinant argument gives an exact positive floor. A separate sparse computation gives* $0.1175367033\ldots$ *with relative residual below* 1.5×10^{-14}*; this decimal is a numerical refinement rather than the certificate.*

Corollary 2.3 (Declared rank-forty-five tensor extension)**.** *The declared fifteen-state Standard Model charge table has vanishing mixed gravitational, weak, color, and cubic abelian anomaly sums, even weak-doublet parity, and the diagonal* $\mathbb{Z}_6$ *action fixes every state. Its chiral charge multiset is disjoint from its conjugate. Under the separate complete-band and cost-order premises, tensoring the selected rank-three response band with this table gives a conditional complex rank-*45 *extension. The local-domain receipt tests a separately declared operator* $D_\sigma\otimes I_{45}$*. It has support radius one, zero kernel, and the same exact positive finite-domain gap as the signed seam operator, with multiplicity* 45*. The source does not select this matter action. The twelve-port Spin packet and the 8,662-node operator domain have no certified source, domain, or transport bridge. The receipt contains no Yukawa coefficient, particle mass, or pole datum [19].*

Remark 2.4 (Physical boundary)*.* The finite result supplies none of the continuum, field-selection, or laboratory-scale structure a physical reading would require. The finite sign layer follows the declared reversing convention.

3 Canonical Observer-Patch Semantics

This paper defines the fixed-cutoff observer vocabulary used by the OPH framework. An observer patch is first an operational algebraic object:

$$\mathsf{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e\ni i}, \mathcal{U}_i, \mathrm{Chk}_i)\,.$$

Here $\mathcal{A}_i$ is the finite or regulated accessible algebra, ρ_i is the accessible state, $\mathcal{R}_i \subseteq Z(\mathcal{A}_i)$ is the exact central record algebra or a declared approximate record algebra, $\mathcal{I}_e$ is an overlap-visible interface algebra, $\pi_{i,e} : \mathcal{A}_i \to \mathcal{I}_e$ is the visible restriction map, $\mathcal{U}_i$ is the allowed family of update and repair instruments, and Chk_i is checkpoint data sufficient for the target reconstruction problem.

Definition 3.1 (Abstract observer patch)**.** *An abstract observer patch is the tuple* O_i *above, considered up to isomorphism of accessible algebras, record statistics, visible restrictions, repair instruments, and checkpoint continuation.*

Definition 3.2 (Support patch)**.** *A support patch is a geometric chart for an abstract observer patch on a branch with reconstructed geometry. Examples are a cap* $P_i \subset S^2$*, a collar, or a causal diamond. The support patch supplies chart data for modular flow, entropy variation, and overlap comparison.*

Definition 3.3 (Carrier patch)**.** *A carrier patch is a physical or digital implementation satisfying*

$$\mathrm{Realize}_\varepsilon(H_i) \longrightarrow \mathsf{O}_i,$$

meaning that the implementation induces the declared accessible algebra, visible interface statistics, record readout, repair instruments, and checkpoint continuation within error ε*.*

The echosahedral body used below is a reference carrier architecture. It is a finite implementation surface for twelve-port interfaces, records, repair channels, and recurrent subchannels. The observer identity lives in the quotient data exposed by O_i, so a change of hidden coordinates, port labels, or substrate presentation is physically silent when it preserves the visible interface, oriented incidence where declared, response, record, repair, checkpoint, and refinement processes.

Definition 3.4 (Algebraic audit instrument)**.** *An algebraic audit instrument is an exact computable map on finite quotient-visible representation data. It certifies invariance, nonconjugacy, orbit structure, lattice preservation, or automorphism fusion inside a declared regulator chart. It is not a physical actuator unless a carrier implementation realizes the same visible map with a public evidence bundle.*

The tuple O_i does not impose equal primitive observer size. Fixed-cutoff OPH requires bounded accessible algebras and declared interfaces; it also permits different connected support regions and different finite or regulated algebras. In the carrier picture, an observer-supporting object may be a variable finite federation of screen cells. A theorem selecting isomorphic fixed-capacity elementary carriers would need extra hypotheses: homogeneous UV cellulation, isomorphic local cell algebras, equivariant overlaps, refinement preservation of the cell type, and a proof that one cell or a fixed block of cells is the primitive carrier.

Claim 3.5 (Presentation invariance and carrier sensitivity)**.** *Two carrier patches are physically equivalent for the declared OPH observables when a quotient-visible isomorphism preserves their accessible algebras and states, interface algebras, record statistics, repair maps, checkpoint continuation, response laws, and refinement lineage. On the Echosahedral branch it must also preserve the twelve-port oriented incidence and its antipodal pairing. A material, coordinate, or wiring change satisfying this contract is silent. A change of the preserved carrier signature is outside the equivalence claim and may change the physical branch.*

This claim is theorem-level only on a branch that supplies the required quotient isomorphism and error bound. It fails if a predicted OPH observable depends on hidden carrier coordinates after the full visible signature has been held fixed. It says nothing about a cube and an icosahedron whose visible incidence structures differ.

Definition 3.6 (Operational observer test)**.** *A candidate physical system realizes an observer patch when it supplies a bounded accessible interface, durable re-readable records, self-read or internal state estimation, record-conditioned future behavior, boundary prediction better than shuffled-record controls, and checkpoint continuation within a declared error.*

The formal finite receipt implements seven corresponding algebraic clauses at one regulator. Two operational observers have unequal owner-region records whose typed restrictions agree on a nonzero common section accessible to both; a one-corner mutation breaks the agreement. This establishes a bounded overlap-readable observer receipt, without supplying a physical instrument, source realization, refinement naturality, or higher-overlap coherence.

4 Mathematical and Empirical Scope

The finite-algebra results concern patches, overlaps, records, repair interfaces, event algebras, and checkpoint laws. Finite representation and lattice calculations support those results when their matrices, exact checks, and quotient-visible invariants are public. A generic overlap network is only a finite constraint code. Quantum error-correcting distance, min-cut resilience, spectral convergence, Byzantine-fault-tolerant liveness, and hardware speedup require their own hypotheses and certificates.

Spherical cellulations and digital finite-group models are calculational charts. Federated echosahedral carriers are candidate implementations with multi-port interfaces, recurrent toroidal subchannels, exposed overlap data, records, and repair loops. A hardware result enters only through public raw or reduced evidence with stable hashes, manifests, calibration files, and exact verification. These finite results do not identify laboratory hardware, select a unique ultraviolet completion, solve the Yang–Mills mass gap by hardware, collapse a complexity class, or derive first-principles hadron masses.

5 The Public Evidence Rule

Hardware-facing language is admissible only under the following rule.

> Any hardware evidence cited as evidence in this paper must be represented by a public evidence bundle in the OPH repository, or by a public pinned subrepository commit, with enough raw data and metadata for an external reader to check the claim.

Private notebooks, local runtime logs, unpublished bench transcripts, and development-repo notes may motivate architecture choices, but they do not carry evidential weight inside the paper. A hardware evidence bundle must include, at minimum:

1. a manifest with stable bundle identifier, date, operator, body identifier, controller identifier, firmware hash, mesh/body hash, and wiring map hash;
2. raw readout files, such as coupling matrices, MDD or discharge-timing traces, dark baselines, low-power sweeps, ring-diversity scans, and calibration logs;
3. body and board provenance, including photographs or signed photo hashes where relevant;
4. the task definition, scorebook, repair or rerank law, and exact-verifier program;
5. exact-verifier receipts for any high-level task claim;
6. negative controls, shuffle or replay controls where applicable, and a statement of non-claims.

This rule prevents dependence on hidden laboratory state. The theorem sections use finite algebras, declared patch interfaces, and stated branch assumptions.

6 From Spherical Screens to Federated Patch Carriers

The spherical screen is useful because an observer-accessible cut often has an effective closed two-surface description. A cellulated sphere can encode finite capacity, caps, collars, edge centers, and observer-visible cuts. The microscopic carrier is the finite patch federation. The chart is the observer-facing presentation of its visible data.

The sphere also carries the symmetry bridge used by the rest of OPH. Caps on S^2 are the geometric support regions whose modular flows become Lorentz boosts in the controlled scaling branch. The imported classical isomorphisms $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$ identify the orientation-preserving conformal group with proper orthochronous Lorentz-frame kinematics. These isomorphisms are not source-derived and are not new Lean theorems. Attachment of the chart to a populated $3+1$-dimensional event spacetime remains conditional on the physical-causality, manifoldlikeness, dimension, and volume receipts above. Finite cellulations of the chart supply the regulator side: patch ports, edge sectors, collars, and overlap checks.

At fixed cutoff, the fundamental object is a finite patch federation. Each patch has:

1. an internal finite state algebra;

2. a bounded family of exposed overlap ports;
3. a local record algebra;
4. a readout map from internal state to port-visible packets;
5. a repair interface that changes local state in response to mismatch;
6. a checkpoint interface that exposes enough observer-accessible data to define continuation.

The abstract observer patch and its carrier realization are separate typed objects. Write

$$\mathsf{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e\ni i}, \mathcal{U}_i, \mathsf{Chk}_i)\,, \qquad \mathrm{Realize}_\varepsilon(H_i) \longrightarrow \mathsf{O}_i.$$

The elementary carrier H_i may be an Echosahedron on the homogeneous reference branch. Its port names, body coordinates, and hardware details are absent from the abstract observer whenever they disappear under the physical quotient. A fixed-cutoff simulator is therefore typed as

$$\mathsf{Sim}_r = (G_r, \{\mathsf{O}_{i,r}\}, \Sigma_r, \Gamma_r, Q_r, \mathsf{Prop}_r, \to_r^{\mathrm{acc}}, \mathrm{Rep}_r, \pi_r^{\mathrm{eq}}, L_r^{\mathrm{eq}}, \mathsf{Chk}_r, c_{sr}).$$

Here Prop_r is a reversible proposal menu, $\to_r^{\mathrm{acc}}$ is the strict-descent accepted-repair relation, Rep_r is its normalizer, and $(\pi_r^{\mathrm{eq}}, L_r^{\mathrm{eq}})$ is a separate equilibrium law and reversible or Euclidean generator. The refinement maps are c_{sr}.

Definition 6.1 (Typed simulator ledger)**.** *A simulator run has one immutable root manifest and seven typed ledgers:*

A. *the quotient/normalizer ledger owns presentation states, hidden-representative quotient, accepted transactions, semantic histories, normal forms, and checkpoint continuation;*

B. *the source-ensemble ledger owns the base measure, source action, MaxEnt constraints, proposal and acceptance rates, stationary law, detailed balance, and cycle affinities;*

C. *the quantum/modular ledger owns the noncommutative algebra, density matrices, spectral support, modular Hamiltonians, frequency-resolved jumps, conditional expectations, and mixed-GNS comparison data;*

D. *the geometry/BW ledger owns support incidence, cap normals, held-out cross-ratios, geometric dilation, wrong-temperature controls, event-chart complexes, quadratic cones, and tetrads;*

E. *the operational-history ledger owns semantic event identities, the observer registry groupoid, worldlines, event correspondences, instrument readings, and affine synchronization;*

F. *the physical-current ledger owns reversible port perturbations, unitary response, commutators, A_5-equivariance, and determinant, spin, and deck descent;*

G. *the transport ledger owns free and collision kernels, first and second conditional moments, mass-shell data, phase-space measure, and stress transfer.*

The ledgers share source hashes and refinement identifiers through the root manifest. Their typed objects are distinct. In particular, the accepted strict-descent relation is not reused as the equilibrium generator, a central register density emits trivial modular action, a finite register operator is not a physical current, and a deterministic transport claim requires a vanishing second-moment receipt.

The public large-run archive makes part of this ledger concrete [19]. OPH-FPE instantiates 81,920 finite patch rows on the level-six icosahedral cell tower, routes 122,880 gauge-labelled seams through three of each row's twelve local slots, commits all patch records, and materializes 2,048 patch-observer neighborhoods. One immutable, source-bound, distinct integer authority per row selects the endpoint preserved by an inconsistent seam repair. The archived source has 102,415 mismatches. Each enabled repair removes one, and sixteen shuffled replays reach the same authority-bound terminal hash. A standalone verifier reconstructs the exact normal form and hashes from the primitive arrays without importing simulator code.

The authority-decorated source is an explicit finite premise. Its axiomatic selection and refinement naturality have not been derived. The large-run records provide custody and readout without causing later port writes. A separate eight-carrier receipt implements 96 literal record-conditioned probe/read/write transactions and checks their ablation, record-counterfactual, A_5-covariance, idempotence, and disjoint commutation properties under a producer-free verifier. That component has not been integrated into the large run. Neither archive constructs the physical quotient or laboratory readout required by the definition above.

Definition 6.2 (Sphere fold)**.** *Fix a regulator r. Let Σ_r be the finite presentation space of a patch federation, let Γ_r be the presentation-redundancy groupoid, and let*

$$Q_r = \Sigma_r/\Gamma_r$$

be the physical quotient. Let $\pi_r : \Sigma_r \to Q_r$ be the quotient map, let

$$n_r : Q_r \to Q_{r,\mathrm{nf}}$$

be the accepted repair normal-form map on the declared physical quotient, and let

$$\chi_{S,r} : Q_{r,\mathrm{nf}} \to \mathsf{Chart}_S(r)$$

be the support-visible spherical screen chart. Its outputs are caps, collars, cuts, edge-sector records, boundary data, and geometric readouts. The sphere fold of a finite presentation $s \in \Sigma_r$ is

$$\mathrm{Fold}_{S,r}(s) := \chi_{S,r}(n_r(\pi_r(s)))\,.$$

Two finite presentations have the same sphere fold exactly when their repaired quotient normal forms induce the same support-visible caps, collars, edge-sector records, and observer-facing screen readouts.

For Lorentz-branch chart receipts, the support chart also exposes a derived cap-normal readout

$$\chi^{\mathrm{cap}}_{S,r} : Q_{r,\mathrm{nf}} \to \mathsf{CapChart}_r$$

with output

$$\mathsf{CapChart}_r = (C_r, \partial C_r, \mathbf{c}_r, \alpha_r, n_{C,r}, \varepsilon_{\mathrm{round},r}, \varepsilon_{\mathrm{inc},r})\,.$$

On the exact analytic round-cap branch,

$$n_{C,r} = (\cot \alpha_r, \csc \alpha_r\, \mathbf{c}_r).$$

On a finite fitted branch the output is labeled approximate round cap chart. A global bound requires a separate round-cap rigidity/convergence certificate; if the fit is not round it emits cap not round. The normal $n_{C,r}$ is support-chart data derived from the repaired quotient. It is not a microscopic carrier coordinate, hidden port label, or new local degree of freedom. A cap normal determines a geodesic plane and half-space in H^3; selecting a particular observer point $u \in H^3$ requires a separate observer-frame, tetrad, or clock receipt. On the spacetime and Einstein paper's producer branch, the cap chart itself is produced from the support-visible incidence complex of the repaired normal form under the spherical-incidence, mesh, cross-ratio, and 2π-normalization receipts. The independent source-causal theorem constructs the finite $1+3$ carrier; its promotion to an effective event base requires the spacetime and Einstein paper's physical continuum certificate.

Definition 6.3 (Carrier-to-support screen bridge)**.** *A carrier-to-support screen bridge at cutoff r consists of a source-bound map from routed carrier ports and seams to the support projections of the repaired observer-patch net, together with: (i) unital interface-algebra homomorphisms on every seam; (ii) coherent recovery maps on higher overlaps; (iii) a quotient-visible nerve whose support incidence is computed from the normal form; and (iv) refinement maps commuting with routing, quotienting, incidence, records, and support projection. A spherical bridge additionally certifies that the global nerve satisfies the spherical-incidence, cap-mesh, oriented cross-ratio, and normalization receipts consumed by the spacetime and Einstein paper. Local carrier coordinates are excluded from the support output unless this map derives them as quotient-visible data.*

Proposition 6.4 (Local Icosahedral incidence does not select the global support)**.** *The certified twelve-port incidence of every local carrier does not determine the topology or conformal structure of the federation support.*

Proof. Fix one certified local carrier and make one copy at every vertex of an arbitrary finite graph. Route selected port pairs along that graph while leaving each local port algebra, oriented incidence, antipode, and A_5 action unchanged. A path, a cycle, a tree, and a cellulation of a closed surface therefore have identical local carrier certificates and different federation nerves. The local certificate cannot distinguish them. Full interface maps, higher-overlap coherence, and the global incidence/refinement receipt are necessary parents of a support-screen claim. □

Finite bridge theorem. The source-derived incidence nerve uses the twelve ports as carrier charts, the thirty edges as seam algebras, and the twenty oriented faces as nonvacuous triple restrictions. It verifies every interface homomorphism and face cocycle. On the same bound artifact, phase readback produces accepted confluent seam repairs, and the operational observer passes record, feedback, prediction, and checkpoint controls. The carrier nerve maps refinement-naturally to an oriented S^2 support limit. This closes the finite carrier, federation, observer, and support bridge of Definition 6.3. It does not produce an H^3 frame, physical source-causal continuum, BW/KMS clock, physical scale, laboratory attachment, or continuum fields.

The fold adds no independent dynamics. Repair is the finite patch operation that compares overlaps, updates records, and reaches the accepted normal form. Folding names the spherical chart presentation of that repaired quotient state. In plain language, sphere folding is what overlap repair looks like on the observer-facing screen.

Strict descent prevents the accepted relation from serving as a reversible equilibrium law. If $x \to_r^{\mathrm{acc}} y$ lowers the well-founded repair measure, the reverse move cannot be another accepted descending step. The simulator consequently keeps three operators distinct:

$$\mathrm{Rep}_r, \qquad e^{-tL_r^{\mathrm{eq}}}, \qquad \sigma_t^\rho.$$

They are, respectively, the irreversible normal-form map, an equilibrium or Euclidean relaxation, and the reversible modular automorphism group. A common local proposal menu may contribute data to all three constructions. This shared input does not identify the operators.

Theorem 6.5 (Finite rate reconstruction on the reversible layer). *Let $G = (X, E)$ be a connected finite state graph with positive rates $q^{\mathrm{eq}}_{u\to v}$ in both directions of every edge. Define*

$$a_{u\to v} := \log \frac{q^{\mathrm{eq}}_{v\to u}}{q^{\mathrm{eq}}_{u\to v}}.$$

The following conditions are equivalent:

(i) *every oriented cycle has zero affinity;*

(ii) *there is a potential $K : X \to \mathbb{R}$, unique up to an additive constant, with $a_{u\to v} = K_v - K_u$;*

(iii) *$\pi_x \propto e^{-K_x}$ satisfies detailed balance $\pi_u q^{\mathrm{eq}}_{u\to v} = \pi_v q^{\mathrm{eq}}_{v\to u}$.*

The rates belong to the reversible equilibrium layer. They are not rates of the accepted strict-descent relation.

Proof. Condition (ii) telescopes around every cycle. Under (i), fix a root and define K_x by summing a along any path from the root to x; zero cycle affinity gives path independence. Exponentiating $K_v - K_u = a_{u\to v}$ gives detailed balance, and taking logarithms of detailed balance gives (ii). □

K in this theorem is a classical Gibbs potential. On the commutative state algebra its modular automorphism is trivial, so rate reconstruction does not create a nontrivial modular clock. The finite potential-theory core of the theorem, with condition (i) read over closed walks, is machine-checked in `Lean/Thermodynamics/KolmogorovCriterion.lean`; the source-derived rates and the channel realization are outside that check.

6.1 Finite modular and register theorems

The finite equilibrium layer supports several exact statements before any continuum or physical current interpretation is made. They also fix which information a simulator has to produce.

Theorem 6.6 (Exact modular coarse graining). *Let $c : X_s \to X_r$ be a surjective coarse-graining map between finite sets and let $\mu_s(x) > 0$. The pushed-forward law and its modular potential are*

$$\mu_r(y) = \sum_{c(x)=y} \mu_s(x), \qquad K_r(y) = -\log\left(\sum_{c(x)=y} e^{-K_s(x)}\right), \qquad K_s = -\log \mu_s.$$

Thus modular potentials coarse-grain by a log-sum-exp rule, up to the common additive normalization convention. More generally, if $\mu_s(x) = Z_s^{-1} m_s(x) e^{-S_s(x)}$ and $m_r(y) > 0$ is the declared coarse reference weight, then

$$S^{\mathrm{eff}}_r(y) = -\log\left(\sum_{c(x)=y} m_s(x) e^{-S_s(x)}\right) + \log m_r(y)$$

gives the exact pushed-forward law. For a finite quantum channel Φ_{sr}, the corresponding statement is $\rho_r = \Phi_{sr}(\rho_s)$ and $K_r = -\log \rho_r$ whenever the output is faithful. An arithmetic average of K_s is not the coarse modular potential in general.

Proof. Both classical formulas follow by substituting $\mu_s = e^{-K_s}$ or $\mu_s = Z_s^{-1} m_s e^{-S_s}$ into the definition of the pushforward. The quantum formula is the definition of the modular Hamiltonian of the pushed-forward faithful state. □

Proposition 6.7 (A faithful equilibrium law does not select its generator). *Let $G = (X, E)$ be a connected finite graph and let $\mu_x > 0$. For every collection of positive symmetric edge conductances $c_{xy} = c_{yx}$,*

$$q_{x\to y} = \frac{c_{xy}}{\mu_x}$$

defines a continuous-time Markov generator that is reversible with stationary law μ. Distinct conductance families give distinct generators with the same state. A geometry, MaxEnt state, or repair normal form therefore selects no equilibrium dynamics unless an additional rate or conductance rule is supplied.

Proof. The identity $\mu_x q_{x\to y} = c_{xy} = \mu_y q_{y\to x}$ is detailed balance. The diagonal entries are fixed by row-sum zero. Varying any conductance varies the generator while preserving the same stationary law. □

Theorem 6.8 (Weighted graph Hodge split and entropy production). *Choose an orientation of a connected finite graph, positive vertex and edge weights, and let D be the vertex-to-edge coboundary. Every real edge one-cochain has the unique orthogonal decomposition*

$$a = DK + h, \qquad D^* h = 0,$$

with K unique up to an additive constant. For an irreducible stationary Markov law π whose transition graph is E, assume the rate support is bidirectional: $q_{x\to y} > 0$ and $q_{y\to x} > 0$ for every $\{x, y\} \in E$. Define on each chosen edge

$$J_{xy} = \pi_x q_{x\to y} - \pi_y q_{y\to x}, \qquad F_{xy} = \log \frac{\pi_x q_{x\to y}}{\pi_y q_{y\to x}}.$$

Then the steady entropy-production rate is

$$\sigma = \sum_{\{x,y\}\in E} J_{xy} F_{xy} \geq 0,$$

and it vanishes exactly when detailed balance holds on every edge. The bare logarithmic rate ratio omits the stationary weights and is therefore not, by itself, the thermodynamic force.

Proof. Finite-dimensional orthogonal projection gives the Hodge split, since $(\operatorname{im} D)^\perp = \ker D^*$. Each entropy-production summand has the form $(A - B)\log(A/B) \geq 0$ for positive A, B. Equality holds exactly when $A = B$ edge by edge. □

Theorem 6.9 (Finite KMS-symmetric jump generator). *Let $\rho = Z^{-1}e^{-K}$ be faithful on a finite matrix algebra. Suppose a Heisenberg-picture Lindblad generator has jump operators $V_{\omega,a}$ satisfying*

$$[K, V_{\omega,a}] = -\omega V_{\omega,a}, \qquad V_{-\omega,a} = V^*_{\omega,a}, \qquad \gamma_a(-\omega) = e^{-\omega}\gamma_a(\omega),$$

with positive Kossakowski matrices in every frequency sector. Suppose also that its Lamb-shift Hamiltonian commutes with K, and that the generator is self-adjoint for the declared symmetric GNS inner product

$$\langle A, B\rangle_{\rho,1/2} = \operatorname{Tr}(\rho^{1/2} A^* \rho^{1/2} B).$$

Then the semigroup is completely positive and unital, preserves ρ, commutes with the modular flow $A \mapsto e^{itK} A e^{-itK}$, and respects the displayed frequency-sector decomposition. The sign in the KMS rate relation is tied to the displayed commutator convention.

Proof. Positive Kossakowski matrices give complete positivity. Unitality and GNS self-adjointness imply $\mathrm{Tr}(\rho L(A)) = \langle \mathbf{1}, L(A)\rangle_{\rho,1/2} = \langle L(\mathbf{1}), A\rangle_{\rho,1/2} = 0$, hence stationarity. The eigenoperator relation, the paired KMS rates, and $[K, H_{\mathrm{LS}}] = 0$ make every term modular covariant and preserve its frequency sector. $\square$

Remark 6.10 (Finite modular receipt boundary). The source-ensemble ledger reports the pushed-forward laws or channels, independent conductances and rates, stationary weights, cycle-affinity residual, and the weighted Hodge remainder. The quantum/modular ledger reports the faithful density, modular spectrum, frequency projectors, Kossakowski positivity, KMS-pair residuals, GNS-symmetry residual, and modular-covariance residual. These finite data instantiate Theorems 6.6–6.9; the theorems do not manufacture those data.

Theorem 6.11 (Exact $A_5 \times \mathbb{Z}_2$ register commutant)**.** *For the twelve-port permutation module,*

$$\mathbb{R}[\mathsf{P}_{12}] \cong \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3'} \oplus \mathbf{5}$$

is multiplicity-free. The icosahedral adjacency operator has four distinct eigenvalues $5, \sqrt{5}, -\sqrt{5}, -1$ on these summands. Consequently its real A_5-commutant is $\mathbb{R}[A_{\mathrm{ico}}]$, represented by polynomials of degree at most three. On the oriented register $\mathsf{R}_{24} = \mathbb{R}[\mathsf{P}_{12}] \otimes \mathbb{R}[\mathbb{Z}_2]$, every self-adjoint operator commuting with A_5 and orientation reversal is specified by one real scalar on each irreducible summand in each parity sector, eight scalars in total.

Proof. Schur's lemma gives one scalar on each real irreducible summand. The four distinct adjacency eigenvalues make the spectral projectors degree-three interpolation polynomials in A_{ico}. Diagonalizing the commuting reversal involution gives two parity copies of the same multiplicity-free decomposition. $\square$

Proposition 6.12 (No equivariant directed 24-cycle)**.** *Regard the oriented register as the transitive $(A_5 \times \mathbb{Z}_2)/(C_5 \times \{1\})$-set. It has no $A_5 \times \mathbb{Z}_2$-equivariant successor permutation that is a single directed 24-cycle.*

Proof. Its equivariant automorphism group is

$$N_{A_5 \times \mathbb{Z}_2}(C_5 \times \{1\})/(C_5 \times \{1\}) \cong \mathbb{Z}_2 \times \mathbb{Z}_2,$$

because $N_{A_5}(C_5) \cong D_{10}$. Every equivariant permutation therefore has order at most two, whereas a 24-cycle has order 24. $\square$

Definition 6.13 (Source-derived oriented-channel realization)**.** *Let E^{or} be the oriented transitions of the reversible quotient-state layer and set*

$$\mathcal{H}_E = \ell^2(E^{\mathrm{or}}), \qquad F_E|e\rangle = a_e|e\rangle, \qquad a_e = \log \frac{q^{\mathrm{eq}}_{\bar{e}}}{q^{\mathrm{eq}}_e}.$$

A source-derived 24-channel realization is a full-rank map

$$C : \mathcal{H}_{24} := \mathbb{C}[\mathsf{P}_{12}] \otimes \mathbb{C}^2_{\mathrm{or}} \longrightarrow \mathcal{H}_E$$

whose coefficients, aggregation rule, and provenance are emitted by the physical proposal ledger. After replacing C by $C(C^*C)^{-1/2}$, assume $C^*C = \mathbf{1}$. Edge reversal is denoted $R_E|e\rangle = |\bar{e}\rangle$, and register reversal by R_{24}. The state set X, transition set E^{or}, and label map into the 24 channel types are distinct objects.

Theorem 6.14 (Oriented-register modular compression)**.** *For a source-derived realization C, the register is invariant under the edge modular grading exactly when*

$$\epsilon_{\mathrm{ch}} := \|(\mathbf{1} - CC^*)F_E C\|_{\mathrm{HS}} = 0.$$

Equivalently, there is a unique operator Ω_{24} with

$$F_E C = C\Omega_{24}, \qquad \Omega_{24} = C^* F_E C.$$

Suppose in addition that the edge rates are A_5-equivariant, C intertwines the edge and register actions of A_5 and reversal, and hence $R_E F_E R_E = -F_E$. Then

$$\Omega_{24} \cong \bigoplus_{\lambda \in \{1,3,3',5\}} \mathbf{1}_{V_\lambda} \otimes M_\lambda, \qquad \mathrm{spec}(M_\lambda) = \{-\omega_\lambda, +\omega_\lambda\},$$

for four nonnegative source-derived values $\omega_1, \omega_3, \omega_{3'}, \omega_5$. Thus the paired frequencies have degeneracies $1, 3, 3, 5$. The symmetry fixes this block pattern and these degeneracies, but not the four numerical values.

Proof. The residual vanishes precisely when $F_E \operatorname{im} C \subseteq \operatorname{im} C$. Since C is isometric, restriction to this invariant subspace gives the unique compression $\Omega_{24} = C^* F_E C$. A_5-equivariance and Schur's lemma give one Hermitian 2×2 orientation matrix M_λ on each of the multiplicity-free summands $1, 3, 3', 5$. Reversal covariance gives $R_{24}\Omega_{24}R_{24} = -\Omega_{24}$, so each orientation block is traceless and has eigenvalues $\pm\omega_\lambda$. Tensoring with V_λ gives the displayed degeneracies. ☐

Corollary 6.15 (Raw labels are modular eigenchannels only at equal affinity)**.** *If each raw register slot is realized as a normalized sum of transitions with that label and distinct labels have disjoint edge support, exact channel closure requires*

$$a_e = a_{e'} \quad \text{whenever } e \text{ and } e' \text{ carry the same raw label.}$$

Otherwise the raw slot basis is not a modular eigenbasis; any invariant 24-channel space must instead use source-derived linear combinations.

Proposition 6.16 (Operator-frame identification boundary)**.** *Let $V_a \in \mathcal{A} \subseteq B(\mathcal{H})$ be realized transition operators and let ω_a be source-derived frequencies. If self-adjoint K and K' both solve*

$$[K, V_a] = \omega_a V_a, \qquad [K', V_a] = \omega_a V_a \quad \text{for every } a,$$

then $K - K'$ lies in the commutant of the algebra generated by the V_a. Hence the channel frame identifies a noncommutative modular Hamiltonian only modulo that represented commutant. If, in addition, $K, K' \in \mathcal{A}$ and the V_a generate $\mathcal{A}$, the ambiguity lies in $\mathcal{A} \cap \mathcal{A}' = Z(\mathcal{A})$.

Proof. Subtracting the two commutator equations gives $[K - K', V_a] = 0$ for every a; the generated-algebra statement follows. ☐

Remark 6.17 (Rate scale, clock scale, and geometric modular identification). Multiplying every equilibrium rate by one constant leaves every affinity, the reconstructed stationary law, K, and Ω_{24} unchanged, while rescaling the relaxation generator. Rate ratios therefore determine a dimensionless potential; rate magnitudes determine kinetics. Neither fixes a frequency in hertz. Operational time requires the independent clock instrument and affine calibration used later in this paper. A physical Bisognano–Wichmann (BW) identification further requires a refinement-natural comparison showing that the gear-derived state is the same support-visible cap state used by the geometric modular theorem, the independently complete mixed Gelfand–Naimark–Segal common-comparison package on that same tower, its regularized cutoff schedule, and the independent 2π-normalization receipt.

The register receipt reports character-projector multiplicities, equivariance residual, within-irrep spectral spread, orientation-parity leakage, the adjacency-polynomial residual, and equivariant-permutation order. The modular-gearing receipt additionally reports the oriented rate ledger, quotient lumpability defect, fundamental-cycle affinities, stationary-law and detailed-balance residuals, C^*C, $\epsilon_{\rm ch}$, the four paired gaps, and the commutant dimension of any operator-valued lift. Exact data give Theorems 6.11 and 6.14 and Proposition 6.12; approximate data are diagnostics with declared tolerances.

These theorems classify finite register states and operators. They do not construct a physical gauge-current algebra. That step requires the separate full-rank, compact skew-adjoint, commutator-closed current lift, its inner A_5 action, and refinement-natural physical descent used by the compact and particle-sector papers.

On the literal transitive twelve-port set $\mathsf{P}_{12} = A_5/C_5$, an irreducible A_5-equivariant rate law has a unique invariant equilibrium distribution, hence $\pi_p = 1/12$ and constant K_p. On $\mathsf{P}_{12} \times \{+,-\}$, with A_5 acting on the port coordinate, an invariant potential has at most two values K_+ and K_-. The geometry can therefore carry one orientation gap $\Delta K = K_- - K_+$. It cannot supply twenty-four independent energy levels. The state set X, oriented transition set $E^{\rm or}$, and channel label map $\lambda : E^{\rm or} \to \mathsf{R}_{24}$ are separate objects; λ need not be a bijection. In particular, twenty-four oriented edge instances contain only twelve unoriented edges and cannot connect a graph on twenty-four states.

The finite carrier vocabulary is compatible with the layered functional carrier of the consensus paper: the carrier is a quotient-visible finite implementation surface, and hidden port labels or substrate coordinates are silent when visible interface data, record statistics, repair maps, and checkpoint continuation are preserved.

This record/repair architecture realizes two consistency requirements. A world that is the fixed point of its own description must contain observers, records of their readings, and a mechanism that keeps those records consistent under continued reading; the patch federation above, with its record algebras, readout maps, and repair interfaces, is that mechanism made explicit at fixed cutoff. A screen at exact self-similar balance carries no events, so records require detuning; the register constructions below give the fixed-cutoff surface on which that detuning has a declared carrier. Both mappings hold on the declared branches of this paper, including the named MaxEnt-hypothesis caveat where it applies.

6.2 Edge-center scalar-slot register

The downstream dark-sector and coherent-matter papers need one fixed-cutoff microphysics export: on the declared scalar quotient of an edge-center collar register, first-order scalar observables have only the canonical opportunity count as their nonconstant direction.

Definition 6.18 (Finite scalar-slot register)**.** *Fix a regulator r, physical quotient $Q_r = \Sigma_r/\Gamma_r$, and collar or local support region C. Let $E_{\nu,r}(C)$ be the finite set of edge-center scalar slots on the physical quotient. For each $e \in E_{\nu,r}(C)$, let*

$$p_{e,\nu} : Q_r \to \{0, 1\}$$

be the quotient-visible central indicator that slot e carries an active scalar repair opportunity, and let $a_e > 0$ be the declared scalar-slot weight. The canonical quotient-edge scalar opportunity count is

$$\mathcal{N}_{\nu,r,C}(q) = \sum_{e \in E_{\nu,r}(C)} a_e \, p_{e,\nu}(q).$$

The normalized scalar readout is

$$S_{\nu,r,C}(q) = \frac{\mathcal{N}_{\nu,r,C}(q)}{\sum_{e\in E_{\nu,r}(C)} a_e}.$$

The indicators $p_{e,\nu}$ live on Q_r, not on hidden presentations in Σ_r. Gauge representatives, port labels, implementation coordinates, and accepted repair schedules are removed by the quotient before $\mathcal{N}_{\nu,r,C}$ is read.

Theorem 6.19 (Rank-one scalar-slot completeness on the quotient-edge branch)**.** *Let G_C be the quotient-preserving relabeling group of scalar slots in collar C. Assume:*

(i) G_C preserves the physical quotient and the scalar-slot weights a_e;

(ii) the scalar branch keeps only quotient-visible central observables invariant under G_C;

(iii) the scalar slots form one scalar orbit after transverse finite-thickness averaging, equivalently the declared scalar observable is the weighted orbit sum.

Then every first-order quotient-visible scalar observable on the edge-center scalar register is an affine function of $\mathcal{N}_{\nu,r,C}$:

$$F_{\rm sc}(q) = \alpha + \beta\,\mathcal{N}_{\nu,r,C}(q).$$

Equivalently, after constants are removed,

$$\mathcal{S}^{(1)}_{\rm sc}(C) = \mathrm{span}\{\mathcal{N}_{\nu,r,C}\}.$$

Proof. A first-order scalar observable on the slot register has the form

$$F(q) = \alpha + \sum_{e\in E_{\nu,r}(C)} b_e\, p_{e,\nu}(q).$$

Quotient visibility of a scalar requires invariance under G_C. If G_C acts transitively on the scalar slot orbit, invariance forces b_e/a_e to be constant on that orbit. Therefore

$$F(q) = \alpha + \beta \sum_e a_e p_{e,\nu}(q) = \alpha + \beta\,\mathcal{N}_{\nu,r,C}(q).$$

With finite transverse thickness the same argument applies orbitwise before the declared weight integral over the transverse coordinate. The finite-thickness scalar is the weighted orbit sum, so the first-order scalar subspace is one-dimensional after constants are removed. □

This theorem is only about the scalar quotient of the edge-center register. It does not select a primitive observer size, a unique hardware carrier, a probability law on Q_r, or a full finite covariant stress parent.

6.3 Quotient-edge scalar register and protected center reserve

The scalar-slot theorem above is the rank-one first-order export. The dark-sector and coherent-matter papers also need the local event-algebra realization: scalar activation, the protected $\mathbb{Z}_6$ reserve, and finite-thickness collar survival have to live on one quotient-visible edge-center register before any physical coefficient follows.

Trace convention. Throughout this subsection, $\tau_{q,r,C}$ denotes the normalized quotient expectation on the finite physical collar quotient algebra. Thus

$$\tau_{q,r,C}(1) = 1.$$

For a projection p with $\tau_{q,r,C}(p) > 0$, define the reciprocal reserve depth by

$$d^{\#}_{q,r,C}(p) := \tau_{q,r,C}(p)^{-1}.$$

The reserve survival coefficient always uses the normalized mean τ_q, not the reciprocal depth. Thus

$$\tau_q(Z_{6,r,C}) = \frac{P}{24} \quad \text{is the presence probability,}$$

while

$$d^{\#}_q(Z_{6,r,C}) = \frac{24}{P} \quad \text{is the reciprocal slot depth.}$$

The notation $\mathrm{Tr}_q(Z_{6,r,C}) = 24/P$ is valid only when Tr_q is explicitly defined as a legacy reciprocal slot-count statistic. It is not an operator-algebra trace; the reciprocal quantity used here is $d^{\#}_q$. Here and below the unindexed $Z_{6,r,C}$ denotes one declared protected class projector, not the sum over all six class projectors. The indexed and total projectors are defined in Definition 6.27.

Definition 6.20 (Physical collar quotient and edge-center map)**.** *Fix a finite regulator r and a connected collar cut C. Let*

$$Q_{r,C}$$

be the physical collar quotient obtained from the fixed-cutoff presentation after quotienting gauge representatives, hidden carrier coordinates, port labels, repair-schedule identifiers, mesh labels, shard or worker identifiers, and inert ancillary labels.

Let

$$\mathcal{A}^{\mathrm{phys}}_{r,C}$$

be the finite physical collar algebra, with center

$$Z(\mathcal{A}^{\mathrm{phys}}_{r,C}).$$

The connected cut exposes a quotient-visible edge-center sector map

$$\eta_{r,C} : Q_{r,C} \longrightarrow \Xi_{r,C},$$

where $\Xi_{r,C}$ is the finite set of edge-center cut sectors visible on the repaired collar normal form.

For each $\xi \in \Xi_{r,C}$, let

$$e_{\xi,r,C}$$

be the central sector projection corresponding to the fiber $\eta^{-1}_{r,C}(\xi)$. In the classical quotient algebra this is the indicator function of the fiber. In the finite quantum quotient algebra this is the central summand projection for the same quotient-visible sector.

Definition 6.21 (Edge-center quotient algebra)**.** *The edge-center quotient algebra of the connected collar cut C is*

$$\boxed{\mathcal{E}_{r,C} := \mathrm{Alg}\{e_{\xi,r,C} : \xi \in \Xi_{r,C}\} \subseteq Z(\mathcal{A}^{\mathrm{phys}}_{r,C}).}$$

Equivalently,

$$\mathcal{E}_{r,C} \cong \ell^{\infty}(\Xi_{r,C}), \qquad 1_{\mathcal{E}_{r,C}} = \sum_{\xi \in \Xi_{r,C}} e_{\xi,r,C}.$$

Theorem 6.22 (Edge-center quotient algebra)**.** *On the fixed-cutoff connected-collar branch, $\mathcal{E}_{r,C}$ is a finite commutative central subalgebra of $\mathcal{A}^{\mathrm{phys}}_{r,C}$.*

Proof. The fibers $\eta^{-1}_{r,C}(\xi)$ partition the physical collar quotient. Therefore the corresponding sector projectors satisfy

$$e_{\xi,r,C}e_{\xi',r,C} = \delta_{\xi\xi'}e_{\xi,r,C}, \qquad \sum_{\xi\in\Xi_{r,C}} e_{\xi,r,C} = 1.$$

Because the edge-center label is quotient-visible superselection data of the connected cut, its sector projectors commute with every physical collar observable and lie in $Z(\mathcal{A}^{\mathrm{phys}}_{r,C})$. The algebra generated by finitely many mutually orthogonal central projections is finite, commutative, and isomorphic to $\ell^\infty(\Xi_{r,C})$. □

Definition 6.23 (Quotient scalar readout algebra)**.** *Let*

$$\mathsf{Scal}_{r,C} \subseteq \mathcal{A}^{\mathrm{phys}}_{r,C}$$

be the subalgebra generated by all quotient-local collar readouts whose values are invariant under orientation choice, vector/tensor frame choice, bulk-coordinate choice, hidden carrier coordinate, port relabeling, accepted repair schedule, mesh label, shard label, and inert ancillary label.

Elements of $\mathsf{Scal}_{r,C}$ are the physical scalar collar readouts of the connected cut.

Assumption 6.24 (Edge-center scalar completeness)**.** *On the declared fixed-cutoff scalar branch, every quotient-local scalar collar readout on a connected cut is resolved by the edge-center sector map:*

$$\boxed{\mathsf{Scal}_{r,C} = \mathcal{E}_{r,C}.}$$

Equivalently, there is no independent scalar event algebra carried by bulk coordinates, screen-area labels, noncentral matrix directions, hidden carrier coordinates, port labels, or repair-schedule identifiers after the physical quotient has been taken.

Definition 6.25 (Scalar recoverability defect and activation projector)**.** *Let*

$$R_{r,C} \in \mathsf{Scal}_{r,C,+}$$

be the quotient-local scalar recoverability defect on the collar cut. On the positive scalar source branch,

$$R_{r,C} = I(A : D \mid B)$$

or the corresponding quotient-local finite-collar representative of that recoverability defect.

Define the scalar activation projector by

$$\boxed{\Pi^{\mathrm{scal}}_{r,C} := \mathbf{1}_{(0,\infty)}(R_{r,C}).}$$

Inside this subsection we may also write

$$S_{r,C} := \Pi^{\mathrm{scal}}_{r,C}$$

when matching the dark-sector notation.

Theorem 6.26 (Scalar activation is an edge-center event)**.** *Assume edge-center scalar completeness. Then*

$$\boxed{\Pi^{\mathrm{scal}}_{r,C} \in \mathcal{E}_{r,C}.}$$

Proof. By definition, $R_{r,C}$ is a quotient-local scalar collar readout, hence

$$R_{r,C} \in \mathsf{Scal}_{r,C}.$$

By Assumption 6.24,

$$\mathsf{Scal}_{r,C} = \mathcal{E}_{r,C}.$$

Therefore $R_{r,C} \in \mathcal{E}_{r,C}$. Since $\mathcal{E}_{r,C}$ is a finite commutative algebra, functional calculus is internal to $\mathcal{E}_{r,C}$. Thus

$$\Pi^{\mathrm{scal}}_{r,C} = \mathbf{1}_{(0,\infty)}(R_{r,C}) \in \mathcal{E}_{r,C}.$$

□

Definition 6.27 (Protected Z_6 reserve projector)**.** *Let the declared faithful matter image on this finite protected-center branch be*

$$G_{\mathrm{packet}} = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}.$$

This branch declaration does not select the physical global form. On the protected-center finite-collar branch, let

$$\zeta^{(j)}_{6,r,C} : \Xi_{r,C} \to \{0,1\}, \qquad j \in \mathbb{Z}_6,$$

be six pairwise disjoint quotient-visible indicators for the protected $\mathbb{Z}_6$ reserve classes. Their sum is the indicator of the total protected-reserve support. Define

$$Z^{(j)}_{6,r,C} := \sum_{\xi : \zeta^{(j)}_{6,r,C}(\xi)=1} e_{\xi,r,C}, \qquad Z^{\mathrm{tot}}_{6,r,C} := \sum_{j \in \mathbb{Z}_6} Z^{(j)}_{6,r,C}.$$

Fixing one declared class j_0, write

$$\boxed{Z_{6,r,C} := Z^{(j_0)}_{6,r,C}.}$$

Then

$$Z^{(j)}_{6,r,C}, Z^{\mathrm{tot}}_{6,r,C}, Z_{6,r,C} \in \mathcal{E}_{r,C}.$$

This notation does not select j_0 as the physical blocked class. Any use of one class rather than the total projector carries that attachment separately.

Theorem 6.28 (Same-register and commutation theorem)**.** *Assume edge-center scalar completeness and protected-center realization. Then*

$$\boxed{\Pi^{\mathrm{scal}}_{r,C}, Z_{6,r,C} \in \mathcal{E}_{r,C}}$$

and therefore

$$\boxed{[\Pi^{\mathrm{scal}}_{r,C}, Z_{6,r,C}] = 0}$$

inside the physical quotient collar algebra.

Proof. Theorem 6.26 gives

$$\Pi^{\mathrm{scal}}_{r,C} \in \mathcal{E}_{r,C}.$$

Definition 6.27 gives

$$Z_{6,r,C} \in \mathcal{E}_{r,C}.$$

The algebra $\mathcal{E}_{r,C} \cong \ell^\infty(\Xi_{r,C})$ is finite and commutative. Hence any two of its elements commute in $\mathcal{A}^{\mathrm{phys}}_{r,C}$. □

Assumption 6.29 (Scalar activation channel completeness)**.** *The scalar activation channel on the connected collar cut is the quotient event algebra generated by the nonzero recoverability-defect event:*

$$\boxed{\mathcal{C}^{\mathrm{scal}}_{r,C} = \mathrm{Alg}\{\Pi^{\mathrm{scal}}_{r,C}\} \subseteq \mathcal{E}_{r,C}.}$$

Equivalently, there is no second independent scalar activation generator after the physical quotient has been taken.

Theorem 6.30 (Scalar-channel exhaustion)**.** *Assume edge-center scalar completeness, scalar activation channel completeness, and protected-center realization. Then the scalar activation channel is exhausted by $\Pi^{\mathrm{scal}}_{r,C}$, and the declared class j_0 splits it as*

$$\boxed{\Pi^{\mathrm{scal}}_{r,C} = \Pi^{\mathrm{scal}}_{r,C}(1 - Z_{6,r,C}) + \Pi^{\mathrm{scal}}_{r,C} Z_{6,r,C}.}$$

The first summand survives the declared j_0 event. The second summand is blocked by that event. Calling j_0 the physical blocked class is a separate attachment, and replacing it by the total projector gives a different split.

Proof. By Assumption 6.29, every scalar activation event is generated by $\Pi^{\mathrm{scal}}_{r,C}$. Thus no scalar activation support exists outside this projector.

By Theorem 6.28, $\Pi^{\mathrm{scal}}_{r,C}$ and $Z_{6,r,C}$ commute. Therefore

$$\Pi^{\mathrm{scal}}_{r,C}(1 - Z_{6,r,C}) \quad \text{and} \quad \Pi^{\mathrm{scal}}_{r,C} Z_{6,r,C}$$

are commuting orthogonal projections. Their sum is

$$\Pi^{\mathrm{scal}}_{r,C} - \Pi^{\mathrm{scal}}_{r,C} Z_{6,r,C} + \Pi^{\mathrm{scal}}_{r,C} Z_{6,r,C} = \Pi^{\mathrm{scal}}_{r,C}.$$

Hence the reserve split exhausts the scalar activation channel. □

Theorem 6.31 (No separate scalar carrier)**.** *Assume edge-center scalar completeness. Let B be any proposed alternative scalar activation projector, possibly represented on a bulk, screen-area, noncentral, implementation-local, or hidden-carrier algebra.*

If B changes scalar activation but does not factor through $\mathcal{E}_{r,C}$, then the physical quotient observable algebra has been changed. If B does not change quotient observables, then its scalar content is the same as its edge-center central support in $\mathcal{E}_{r,C}$, and the extra carrier is inert.

Proof. A physical scalar activation event must be a quotient-local scalar observable. By Assumption 6.24, every such observable factors through the edge-center sector map

$$\eta_{r,C} : Q_{r,C} \to \Xi_{r,C}.$$

Equivalently, every such event is represented by a projection in

$$\mathcal{E}_{r,C}.$$

Suppose B is a bulk, screen-area, hidden-carrier, implementation-local, or noncentral event that does not factor through $\eta_{r,C}$. Then there are physical presentations that agree on the edge-center quotient sector but differ in the value of B. If that difference changes scalar activation, then scalar activation is not determined by the declared physical scalar quotient. The observable algebra has therefore been enlarged or changed.

If that difference does not change any quotient observable, then the degrees of freedom in B outside the edge-center quotient are physically silent. The scalar part of B is obtained by passing to the central edge-center event with the same quotient-visible activation support. That central support lies in $\mathcal{E}_{r,C}$.

For a noncentral operator B, the same argument is internal to $\mathcal{A}_{r,C}^{\mathrm{phys}}$. If the noncentral matrix part affects scalar readout, then the readout is not a scalar central event on the declared quotient. If it does not affect scalar readout, only its central support contributes to scalar activation, and that support lies in $\mathcal{E}_{r,C}$.

Thus a separate scalar carrier either changes quotient observables or collapses to the edge-center scalar event represented in $\mathcal{E}_{r,C}$. □

6.4 Coherent-matter scalar-source forcing

The scalar-slot exhaustion theorem fixes the available local scalar carrier. It does not make raw mass, heat, stored energy, or an engineering substrate factor into a scalar source. The source is a quotient-visible coherent-material receipt with explicit readback and boundary-prediction closure.

Definition 6.32 (OPH-coherent material subfederation)**.** *Fix a screen collar C, a finite material region $U \subset C$, and a refinement scale r. An OPH-coherent material subfederation is a finite family*

$$\mathfrak{M} = (\mathfrak{F}_U, \mathsf{R}_U, \mathsf{P}_U, \mathsf{C}_U)$$

where $\mathfrak{F}_U$ is a self-reading subfederation over U, R_U is a stable internal record readout, P_U is a boundary prediction operator against neighboring screen states, and C_U is a coherent mismatch-reduction certificate. The datum is nondegenerate when all four factors survive quotienting and refinement,

$$\mathbf{1}_{\mathrm{self-read}}(\mathfrak{F}_U)\, \mathsf{R}_U\, \mathsf{P}_U\, \mathsf{C}_U > 0.$$

Presentations with the same quotient record, boundary prediction, and coherence certificate define the same subfederation class.

Definition 6.33 (Canonical coherent-matter scalar source)**.** *For an OPH-coherent material subfederation $\mathfrak{M}$, define the canonical scalar source readout*

$$S_{\mathrm{coh}}^{\mathrm{can}}(U, t; h) := \mathbf{1}_{\mathrm{self-read}}(\mathfrak{F}_U)\, \mathsf{R}_U(U, t; h)\, \mathsf{P}_U(U, t; h)\, \mathsf{C}_U(U, t; h).$$

The readout is zero exactly when one of the self-read, record, boundary prediction, or coherent mismatch-reduction receipts vanishes. Rest mass, heat, and arbitrary coherent energy are outside this definition.

Claim 6.34 (Quotient descent of coherent-material source)**.** *The functional $S_{\mathrm{coh}}^{\mathrm{can}}$ descends to the screen quotient. It is invariant under relabelings of internal coordinates, duplicate micro-presentations, and record refinements that preserve $(\mathsf{R}_U, \mathsf{P}_U, \mathsf{C}_U)$ in the quotient.*

Proof. Each factor in Definition 6.33 is a screen-record observable: self-read existence is a boundary-stable record property, R_U is the internal record value, P_U is read through neighboring boundary records, and C_U is the observed reduction of prediction mismatch. The quotient identifies presentations with identical boundary-visible records and predictions. Hence the product is constant on quotient classes and descends. □

Theorem 6.35 (Nonzero coherent scalar source)**.** *Every nondegenerate OPH-coherent material subfederation has*

$$S_{\mathrm{coh}}^{\mathrm{can}}(U,t;h) > 0.$$

The zero branch is precisely the degenerate branch in which at least one coherent-material receipt vanishes.

Proof. Nondegeneracy is the strict positivity of the product in Definition 6.32. The canonical source is that product. □

Theorem 6.36 (Coherent-matter same-channel forcing)**.** *Assume Scalar Edge-Center Exhaustion scalar edge center exhaustion, Theorem 6.30. For every nondegenerate OPH-coherent material subfederation on a collar C, the source*

$$S_{\mathrm{coh}}^{\mathrm{can}}(U,t;h)$$

is valued in the unique scalar edge-center register $\mathcal{E}_{r,C}$. Consequently the coherent-material scalar source uses the same scalar channel that is priced by the Z_6 edge-center collar theorem.

Proof. By Lemma 6.34, $S_{\mathrm{coh}}^{\mathrm{can}}$ is a quotient-local scalar observable: it has no boundary direction, no spin frame, and no unscreened vector index. Scalar Edge-Center Exhaustion states that every quotient-local scalar perturbation that can affect a collar record factors through $\mathcal{E}_{r,C}$. The source therefore lies in $\mathcal{E}_{r,C}$. Theorem 6.28 identifies this register with the same edge-center scalar slot used by the Z_6 collar computation, so the channel identity follows. □

Corollary 6.37 (Finite scalar channel bridge)**.** *On the same branch, the coherent-material record slots and the Z_6 protected-reserve collar slices may be packaged as one finite indexed channel E. The record panel consists of scalar-slot activity, opportunity weights, and normal-form activation maps; the collar panel consists of finite-thickness slice weights and reserve means. With this packaging, the phrase "the same scalar channel" means that both panels are functions of the same finite family E. Equality of two separately defined numerical counters is insufficient.*

Proof. Theorem 6.36 places the coherent-material source in $\mathcal{E}_{r,C}$, while Theorem 6.28 places the protected Z_6 reserve in the same edge-center scalar register. Choosing the scalar slots of $\mathcal{E}_{r,C}$ as the index family E gives one finite channel carrying both panels. The packaging adds no new physical premise; it removes a bookkeeping ambiguity. □

Theorem 6.38 (Single scalar response form with undetermined effective coefficient)**.** *On the weak-field branch, assume that the frequency response is differentiable at zero source and that its quotient-local scalar dependence factors through the edge-center register of Theorem 6.30. Its first-order term then has the form*

$$\delta\nu_C = \chi_{\nu,C}^{\mathrm{eff}}\,\langle \eta_C, S_{\mathrm{coh}}^{\mathrm{can}}\rangle_C + O\Big((S_{\mathrm{coh}}^{\mathrm{can}})^2\Big)\,,$$

for a local edge-center test functional η_C and an undetermined effective coefficient $\chi_{\nu,C}^{\mathrm{eff}}$. The one-register result fixes the scalar form only. It neither derives this coefficient nor excludes several physical mechanisms contributing additively in the same scalar channel.

Proof. The factorization hypothesis and Theorem 6.30 place the perturbation in $\mathcal{E}_{r,C}$. Differentiability gives a linear functional on that scalar register, which may be written in the displayed form after choosing its normalization. Distinct same-channel contributions simply add into the effective coefficient, so no uniqueness of a microscopic contribution or susceptibility follows. □

The finite evidence packet records the coherent material source, its scalar edge-center binding, nonzero activation, same-channel forcing, and the conditional single-channel linear-response form proved above.

Definition 6.39 (Oriented 24-slot repair register)**.** *On the twelve-port carrier-sieve branch, let*

$$\mathsf{P}_{12,r,C}$$

be the twelve exposed central carrier ports of the connected cut. Reversible write/check orientation gives the oriented repair-slot set

$$\mathsf{R}_{24,r,C} := \mathsf{P}_{12,r,C} \times \{+,-\},$$

so

$$|\mathsf{R}_{24,r,C}| = 24.$$

The corresponding oriented slot algebra is

$$\mathcal{R}^{\mathrm{or}}_{24,r,C} := \ell^{\infty}(\mathsf{R}_{24,r,C}).$$

The oriented register is a bookkeeping refinement of the same quotient-edge cut surface. It does not create an independent scalar carrier.

Proposition 6.40 (The oriented register carries no modular clock)**.** *Every faithful state on $\mathcal{R}^{\mathrm{or}}_{24,r,C} = \ell^{\infty}(\mathsf{R}_{24,r,C})$ is tracial, so its modular automorphism group is trivial. The twenty-four slots can carry channel names, orientations, central event projectors, and orbit counts. They do not determine a modular Hamiltonian, a set of clock ticks, or a frequency. A nontrivial modular lift requires a noncommutative transition algebra, for example $B(\ell^2 X)$ with operators $V_{u\to v} = |v\rangle\langle u|$, together with a faithful physical state on that algebra.*

Proof. The algebra is finite and commutative. Its density operator is central in every faithful representation, hence $\rho^{it} a \rho^{-it} = a$ for all a and t. □

The shared-edge protected-reserve branch carries two named branch inputs: the shared cut entropy density $\bar{\ell}_{\mathrm{shared}} = P/4$ and the $\mathbb{Z}_6$ class equidistribution stated in Theorem 6.41 are declared assumptions of the branch, not derived quantities. Every value downstream of the $P/24$ class mean inherits both inputs.

Theorem 6.41 (Protected reserve mean and reciprocal depth)**.** *On the shared-edge protected-reserve branch, assume $P > 0$, the declared total reserve expectation*

$$\tau_{q,r,C}(Z^{\mathrm{tot}}_{6,r,C}) = \bar{\ell}_{\mathrm{shared}} = \frac{P}{4}$$

and the declared $\mathbb{Z}_6$ class-equidistribution hypothesis. Then for every $j \in \mathbb{Z}_6$,

$$\boxed{\tau_{q,r,C}(Z^{(j)}_{6,r,C}) = \frac{P}{24}.}$$

In particular the same formula holds for the declared unindexed $Z_{6,r,C} = Z^{(j_0)}_{6,r,C}$. Equivalently, for the reciprocal reserve depth,

$$\boxed{d^{\#}_{q,r,C}(Z_{6,r,C}) = \frac{24}{P}.}$$

At the comparison coordinate

$$P_C = 1.630968209403959,$$

one has

$$\frac{P_C}{24} = 0.06795700872516496,$$

$$\frac{24}{P_C} = 14.715185655746689,$$

and, under the presence reading of Lemma 6.46,

$$1 - \frac{P_C}{24} = 0.9320429912748350.$$

Proof. The total protected reserve expectation on this branch is the declared branch input

$$\tau_q(Z^{\mathrm{tot}}_{6,r,C}) = \bar{\ell}_{\mathrm{shared}} = \frac{P}{4}.$$

By the declared class-equidistribution hypothesis, each of the six pairwise disjoint class projectors has the same normalized expectation. Additivity and the total expectation therefore give, for every j,

$$\tau_q(Z^{(j)}_{6,r,C}) = \frac{\tau_q(Z^{\mathrm{tot}}_{6,r,C})}{|\mathbb{Z}_6|} = \frac{P/4}{6} = \frac{P}{24}.$$

For the declared class j_0, the normalized quotient reserve mean is

$$\tau_q(Z_{6,r,C}) = \frac{P}{24}.$$

By definition of the reciprocal reserve depth,

$$d^{\#}_q(Z_{6,r,C}) = \frac{1}{\tau_q(Z_{6,r,C})} = \frac{24}{P}.$$

□

The total projector and the declared one-class projector are distinct:

$$\tau_q(Z^{\mathrm{tot}}_{6,r,C}) = \frac{P}{4}, \qquad \tau_q(Z_{6,r,C}) = \frac{P}{24}.$$

The $1 - P/24$ survival coefficient uses the latter. A physical application must therefore identify one class as the blocked event; class equidistribution alone does not make that choice.

Remark 6.42 (Oriented-register denominator coincidence). The oriented 24-slot register of Definition 6.39 also displays the denominator 24 on the local screen side. By that definition the register creates no independent scalar carrier, so the coincidence of denominators carries no corroborating weight for the class-equidistribution hypothesis; the $P/24$ value rests on the two declared branch inputs alone.

Definition 6.43 (Finite-thickness scalar opportunity profile). *Let $Y_{r,C}$ be the finite transverse collar-coordinate set at regulator r. Let*

$$p_y, \qquad y \in Y_{r,C},$$

be the transverse slice projectors, with

$$\sum_{y\in Y_{r,C}} p_y = 1.$$

Let

$$\mathcal{A}^{\mathrm{thick}}_{r,C}$$

be the finite-thickness collar algebra containing the edge-center algebra and the transverse slice algebra:

$$\mathcal{A}^{\mathrm{thick}}_{r,C} \supseteq \mathcal{E}_{r,C} \vee \mathrm{Alg}\{p_y : y \in Y_{r,C}\}.$$

Let

$$\Pi^{\mathrm{scal,thick}}_{r,C}$$

be the scalar activation projector lifted to the thickened collar, and let

$$Z^{\mathrm{thick}}_{6,r,C}$$

be the protected reserve projector lifted to the thickened collar.

Assume that $\Pi^{\mathrm{scal,thick}}_{r,C}$*,* $Z^{\mathrm{thick}}_{6,r,C}$*, and every* p_y *belong to one common finite commutative event algebra* $\mathcal{C}_{r,C} \subseteq \mathcal{A}^{\mathrm{thick}}_{r,C}$*. Their products are then projections and the positive trace below gives ordinary event probabilities. Mere containment in the thick algebra does not imply this commutation clause.*

Define the scalar activation mass at slice y *by*

$$a_r(y) := \tau^{\mathrm{thick}}_{q,r,C}\left(\Pi^{\mathrm{scal,thick}}_{r,C} p_y\right).$$

Assume

$$A_r := \sum_{y\in Y_{r,C}} a_r(y) > 0.$$

The normalized finite-thickness scalar activation profile is

$$\boxed{w_r(y) := \frac{a_r(y)}{A_r}.}$$

Then

$$\sum_{y\in Y_{r,C}} w_r(y) = 1.$$

For every scalar-active slice, $a_r(y) > 0$*, define the scalar-opportunity conditional reserve profile by*

$$\boxed{\epsilon_{\mathbb{Z}_6,r}(y) := \frac{\tau^{\mathrm{thick}}_{q,r,C}\left(\Pi^{\mathrm{scal,thick}}_{r,C} Z^{\mathrm{thick}}_{6,r,C} p_y\right)}{\tau^{\mathrm{thick}}_{q,r,C}\left(\Pi^{\mathrm{scal,thick}}_{r,C} p_y\right)}.}$$

Slices with $a_r(y) = 0$ *do not contribute to* w_r *and may be assigned any harmless value.*

The scalar-weighted protected reserve mean is

$$\boxed{\bar{\epsilon}^{\mathrm{scal}}_{\mathbb{Z}_6,r} := \sum_{y\in Y_{r,C}} w_r(y)\epsilon_{\mathbb{Z}_6,r}(y) = \frac{\tau^{\mathrm{thick}}_{q,r,C}\left(\Pi^{\mathrm{scal,thick}}_{r,C} Z^{\mathrm{thick}}_{6,r,C}\right)}{\tau^{\mathrm{thick}}_{q,r,C}\left(\Pi^{\mathrm{scal,thick}}_{r,C}\right)}.}$$

For continuum notation, first embed the finite coordinate sets in one measurable space and form $\mu_r = \sum_y w_r(y)\delta_y$. A continuum branch must separately certify that μ_r converges to $w(y)\,dy$ and that the weighted reserve means converge,

$$\sum_y w_r(y)\epsilon_{\mathbb{Z}_6,r}(y) \longrightarrow \int dy\, w(y)\epsilon_{\mathbb{Z}_6}(y),$$

with $0 \le \epsilon_{\mathbb{Z}_6} \le 1$ and

$$\int dy\, w(y) = 1.$$

Pointwise arrows on regulator-dependent finite sets do not by themselves imply this integral convergence. Only under the stated convergence receipt do we write

$$\bar{\epsilon}^{\mathrm{scal}}_{\mathbb{Z}_6} = \int dy\, w(y)\epsilon_{\mathbb{Z}_6}(y).$$

Theorem 6.44 (Finite-thickness mean reserve disintegration)**.** *At fixed regulator,*

$$\boxed{\sum_{y\in Y_{r,C}} w_r(y)\epsilon_{\mathbb{Z}_6,r}(y) = \frac{\tau^{\mathrm{thick}}_{q,r,C}\left(\Pi^{\mathrm{scal,thick}}_{r,C} Z^{\mathrm{thick}}_{6,r,C}\right)}{\tau^{\mathrm{thick}}_{q,r,C}\left(\Pi^{\mathrm{scal,thick}}_{r,C}\right)}.}$$

Consequently, if the scalar-weighted protected reserve mean receipt holds,

$$\boxed{\frac{\tau^{\mathrm{thick}}_{q,r,C}\left(\Pi^{\mathrm{scal,thick}}_{r,C} Z^{\mathrm{thick}}_{6,r,C}\right)}{\tau^{\mathrm{thick}}_{q,r,C}\left(\Pi^{\mathrm{scal,thick}}_{r,C}\right)} = \frac{P}{24},}$$

then

$$\boxed{\sum_{y\in Y_{r,C}} w_r(y)\epsilon_{\mathbb{Z}_6,r}(y) = \frac{P}{24}.}$$

If a cofinal regulator tail satisfies the scalar-weighted receipt with value $P/24$ and the continuum convergence receipt of Definition 6.43, then

$$\boxed{\int dy\, w(y)\epsilon_{\mathbb{Z}_6}(y) = \frac{P}{24}.}$$

Proof. Temporarily write $\tau = \tau^{\mathrm{thick}}_{q,r,C}$, $\Pi^{\mathrm{scal}} = \Pi^{\mathrm{scal,thick}}_{r,C}$, and $Z_6 = Z^{\mathrm{thick}}_{6,r,C}$. Using the definitions,

$$\sum_y w_r(y)\epsilon_{\mathbb{Z}_6,r}(y) = \sum_y \frac{\tau(\Pi^{\mathrm{scal}} p_y)}{\tau(\Pi^{\mathrm{scal}})} \cdot \frac{\tau(\Pi^{\mathrm{scal}} Z_6 p_y)}{\tau(\Pi^{\mathrm{scal}} p_y)}.$$

Canceling the slice factors gives

$$\sum_y w_r(y)\epsilon_{\mathbb{Z}_6,r}(y) = \frac{1}{\tau(\Pi^{\mathrm{scal}})} \sum_y \tau(\Pi^{\mathrm{scal}} Z_6 p_y).$$

Since $\sum_y p_y = 1$,

$$\sum_y \tau(\Pi^{\mathrm{scal}} Z_6 p_y) = \tau(\Pi^{\mathrm{scal}} Z_6).$$

Therefore

$$\sum_y w_r(y)\epsilon_{\mathbb{Z}_6,r}(y) = \frac{\tau(\Pi^{\mathrm{scal}} Z_6)}{\tau(\Pi^{\mathrm{scal}})}.$$

If that scalar-weighted quotient mean is $P/24$, the finite displayed identity follows. On a cofinal tail, the separately assumed convergence of the weighted means gives the continuum expression. No pointwise-limit interchange is inferred. □

Remark 6.45 (Unconditioned trace is not enough). The identity

$$\tau_q(Z_{6,r,C}) = P/24$$

does not by itself imply

$$\frac{\tau_q(\Pi^{\mathrm{scal}}_{r,C} Z_{6,r,C})}{\tau_q(\Pi^{\mathrm{scal}}_{r,C})} = P/24.$$

The latter is a scalar-weighted reserve mean receipt. It follows from scalar-reserve unbiasedness. Co-registration or commutation alone is insufficient.

Lemma 6.46 (Presence reading of the reserve profile)**.** *On the common commutative event algebra of Definition 6.43, the thick reserve, scalar, and slice projectors have projection-valued products. Thus the finite profile $\epsilon_{\mathbb{Z}_6,r}(y)$, a ratio of positive traces of nested events, lies in $[0,1]$ and is the conditional probability of the reserve-presence event in a co-registered scalar slot at transverse coordinate y. The continuum profile has this reading only on a branch carrying the stated measure-convergence receipt. A scalar opportunity survives exactly when the presence event does not occur. The local scalar survival factor is therefore*

$$\lambda_{\mathrm{slot}}(y) = 1 - \epsilon_{\mathbb{Z}_6}(y)$$

for every occupancy law whose presence event has probability $\epsilon_{\mathbb{Z}_6}(y)$. No occupancy variable with that presence probability attains the Poisson zero-count factor $e^{-\epsilon}$: $1-\epsilon < e^{-\epsilon}$ strictly for $\epsilon > 0$, and every finite m-fold sub-slot refinement $(1-\epsilon/m)^m$ sits strictly below $e^{-\epsilon}$, which is the supremum of that family, attained by no finite regulator. A mean-count reading, in which $\epsilon(y)$ is consumed as the expectation of an $\mathbb{N}$-valued occupancy rather than as the presence probability, severs the link to the projection trace and consumes a mean-count or projective-limit carrier beyond $Z_{6,r,C}$. That carrier is not supplied by the finite-projection branch; its assumption-minimal consequence is recorded in Remark 6.49.

Theorem 6.47 (Finite-thickness collar survival)**.** *Under the presence reading of Lemma 6.46, the finite-regulator collar survival coefficient is*

$$\boxed{\lambda_{\mathrm{collar,r}} = 1 - \sum_{y\in Y_{r,C}} w_r(y)\,\epsilon_{\mathbb{Z}_6,r}(y).}$$

On a branch carrying the continuum convergence receipt, its limit is

$$\boxed{\lambda_{\mathrm{collar}} = \int dy\, w(y)(1-\epsilon_{\mathbb{Z}_6}(y)) = 1 - \int dy\, w(y)\,\epsilon_{\mathbb{Z}_6}(y).}$$

Proof. The finite local survival factor is $1-\epsilon_{\mathbb{Z}_6,r}(y)$ by Lemma 6.46. Averaging over the normalized finite profile gives the finite sum. The separately assumed convergence of the weighted means then gives the continuum integral and its linear rewrite. □

Corollary 6.48 (Exact finite-thickness coefficient from the scalar-weighted receipt). *If*

$$\int dy\, w(y)\epsilon_{\mathbb{Z}_6}(y) = \frac{P}{24},$$

then

$$\boxed{\lambda_{\mathrm{collar}} = 1 - \frac{P}{24}.}$$

Numerically,

$$\boxed{\lambda_{\mathrm{collar}} = 0.9320429912748350\ldots.}$$

The scalar-weighted mean receipt alone fixes the coefficient, by linearity; no uniformity clause on the profile is consumed. More generally, if the scalar-weighted mean is $\bar{\epsilon}^{\mathrm{scal}}_{\mathbb{Z}_6}$*,*

$$\boxed{\lambda_{\mathrm{collar}} = 1 - \bar{\epsilon}^{\mathrm{scal}}_{\mathbb{Z}_6}.}$$

Proof. Substitute the receipt value into the survival formula of Theorem 6.47. □

Remark 6.49 (Mean-count reading and its Markov band). Under the mean-count reading, with $\epsilon_{\mathbb{Z}_6}(y)$ the expectation of an $\mathbb{N}$-valued reserve occupancy and survival the zero-count event, Markov's inequality bounds the survival factor for any occupancy law: $\Pr[N \geq 1] \leq \epsilon$ gives $1 - \epsilon \leq \Pr[N = 0] \leq 1$, hence with the scalar-weighted $P/24$ receipt

$$1 - \frac{P}{24} \leq \lambda_{\mathrm{collar}} \leq 1.$$

The coefficient is order one and bounded away from zero on every reading; the exact value is reading-dependent, and the presence reading is the one the receipt semantics of this branch supply. Which reading nature uses is a physical premise of the susceptibility branch, recorded there.

Definition 6.50 (Uniform product-thickening branch). *The uniform product-thickening branch consists of the following clauses.*

1. *Product collar algebra:*
$$\mathcal{A}^{\mathrm{thick}}_{r,C} \cong \mathcal{E}_{r,C} \otimes \mathcal{T}_r(Y).$$

2. *Product quotient trace:*
$$\tau^{\mathrm{thick}}_{q,r,C} = \tau^{\mathcal{E}}_{q,r,C} \otimes \tau^{Y}_{r}.$$

3. *Reserve pullback from the edge-center register:*
$$Z^{\mathrm{thick}}_{6,r,C} = Z_{6,r,C} \otimes 1_Y.$$

4. *Scalar activation disintegration:*
$$\Pi^{\mathrm{scal,thick}}_{r,C} = \sum_{y \in Y_{r,C}} \Pi^{\mathrm{scal}}_{r,C}(y) \otimes p_y,$$

   with each $\Pi^{\mathrm{scal}}_{r,C}(y) \in \mathcal{E}_{r,C}$*.*

5. *Slice-wise scalar-reserve unbiasedness: for every scalar-active slice,*

$$\boxed{\frac{\tau^{\mathcal{E}}_{q,r,C}\Big(\Pi^{\mathrm{scal}}_{r,C}(y)Z_{6,r,C}\Big)}{\tau^{\mathcal{E}}_{q,r,C}\Big(\Pi^{\mathrm{scal}}_{r,C}(y)\Big)} = \tau^{\mathcal{E}}_{q,r,C}(Z_{6,r,C}) = \frac{P}{24}.}$$

The fifth clause is essential. Product algebra and product trace alone do not force the scalar-active edge sampler to be reserve-unbiased.

Theorem 6.51 (Exact uniform product-thickening coefficient)**.** *On the uniform product-thickening branch, under the presence reading of Lemma 6.46,*

$$\boxed{\lambda_{\mathrm{collar}} = 1 - \frac{P}{24}.}$$

Numerically,

$$\boxed{\lambda_{\mathrm{collar}} = 0.9320429912748350\ldots.}$$

Proof. By slice-wise scalar-reserve unbiasedness,

$$\epsilon_{\mathbb{Z}_6}(y) = P/24$$

for every scalar-active transverse slice. Substituting this into the finite-thickness survival formula gives

$$\lambda_{\mathrm{collar}} = \int dy\, w(y)\left(1 - \frac{P}{24}\right).$$

Since w is normalized,

$$\int dy\, w(y) = 1.$$

Therefore

$$\lambda_{\mathrm{collar}} = 1 - \frac{P}{24}.$$

On this branch the coefficient coincides with the finite-thickness value of Corollary 6.48, since the scalar-weighted receipt fixes it by linearity on its own; the uniform branch adds slice-wise constancy of the profile, which the exact value does not consume. □

Remark 6.52 (Exact value licensing). The exact value $1 - P/24$ is licensed by the scalar-weighted $P/24$ receipt under the presence reading. Without that receipt the theorem-grade coefficient is

$$\lambda_{\mathrm{collar}} = 1 - \int dy\, w(y)\, \epsilon_{\mathbb{Z}_6}(y),$$

and under the mean-count reading of Remark 6.49 the available theorem-grade band is

$$1 - \frac{P}{24} \le \lambda_{\mathrm{collar}} \le 1.$$

6.5 Local scalar certificates and review checks

The exact coefficient requires one quotient scalar-readout algebra, a complete and exhaustive scalar channel, the oriented 24-slot register, the normalized $\mathbb{Z}_6$ trace, compatible scalar and reserve profiles through the finite thickness, and unbiased survival under disintegration. Under these conditions Theorem 6.51 gives $\lambda_{\text{collar}} = 1 - P/24$. Without them the theorem-grade expression is

$$\lambda_{\text{collar}} = 1 - \int dy\, w(y)\, \epsilon_{\mathbb{Z}_6}(y),$$

and under the mean-count reading it may add only the Markov band

$$1 - \frac{P}{24} \leq \lambda_{\text{collar}} \leq 1.$$

The coefficient ledger imposes the following checks. No survival coefficient may use $24/P$; occurrences of $\mathrm{Tr}_q(Z_6)$ with $24/P$ must be renamed to $d_q^{\#}$ or explicitly declared a reciprocal statistic; every exact $1 - P/24$ coefficient claim must cite Theorem 6.51; every Markov lower bound must cite the weighted-mean result; and every statement that the Z_6 reserve fixes scalar opportunities must cite the scalar-channel exhaustion theorem and its no-separate-carrier corollary.

Definition 6.53 (Finite source-bridge readouts)**.** *For cosmology runs that attempt a first-principles primordial bridge, the carrier must also expose quotient-visible maps*

$$\begin{gathered}
\mathsf{StressRead}_r,\quad \mathsf{ScalarSourceMap}_r,\quad \mathsf{CoherentMatterScalarRead}_r,\quad \mathsf{ClockRead}_r,\quad \mathsf{RepairTangent}_r,\\
\mathsf{VolumeJacobianRead}_r,\quad \mathsf{ScreenMassMatrixRead}_r,\quad \mathsf{LowModeProjectorRead}_r,\\
\mathsf{ScalarPrecisionRead}_r,\quad \mathsf{QuotientMeasureRead}_r,\quad \mathsf{ScalarReleaseEnergyRead}_r,\\
\mathsf{PrimitiveCollarLedgerRead}_r,\quad \mathsf{ReserveGeneratorRead}_r,\quad \mathsf{OrientationHalfRead}_r,\\
\mathsf{DiamondRead}_r,\quad \mathsf{ProperScaleRead}_r,\quad \mathsf{ModularRemainderRead}_r,\quad \mathsf{StressMomentRead}_r,\\
\mathsf{FaceFluxRead}_r,\quad \mathsf{ReactionRead}_r,\quad \mathsf{TetradRead}_r,\quad \mathsf{ScalarRefinementRead}_{sr},\\
\mathsf{RadialWindowRead}_r,\quad \mathsf{SpatialGeometryRead}_r,\quad \mathsf{AreaVolumeRead}_r,\quad \mathsf{LapseShiftRead}_r,\\
\mathsf{LineageRead}_r,\quad \mathsf{ScreenEmbeddingRead}_r,\quad \mathsf{ModeFormRead}_r,\quad \mathsf{CosmoGeomRead}_r,\\
\mathsf{ProperTimeRead}_r,\quad \mathsf{ComovingMetricRead}_r,\quad \mathsf{SpatialModeRead}_r,\quad \mathsf{AngularProjectorRead}_r,\\
\mathsf{FreezeoutMapRead}_r,\quad \mathsf{FreezeoutSurfaceRead}_r,\quad \mathsf{ReleaseSurfaceRead}_r,\\
\mathsf{HomogeneousProjectorRead}_r,\quad \mathsf{AnomalyLoadRead}_r,\quad \mathsf{LoadNoDataLedgerRead}_r.
\end{gathered}$$

They are defined on the same settled-form carrier used for the geometric record packet and must factor through the physical quotient: hidden carrier coordinates, port relabelings, and accepted repair schedules cannot change their output. The finite packet must expose enough data to reconstruct, for every active sector,

$$T_I^{ab},\quad Q_I^a,\quad u_I^a,\quad \rho_I,\quad p_I,\quad \pi_I^{ab}.$$

Under Scalar Edge-Center Exhaustion,

$$\mathsf{CoherentMatterScalarRead}_r(U,t;h) = S_{\text{coh}}^{\text{can}}(U,t;h)$$

is an edge-center scalar readout of OPH-coherent material support. It is a material scalar receipt, not a rest-mass, heat, or raw energy counter. For anomaly abundance selection, the added release maps supply the release hypersurface, tetrad normal and physical cell volumes, FLRW zero-mode projector, finite anomaly load observable $\mathsf{L}_{A,r}$, and no-data-use manifest used by anomaly abundance source receipt. *The source-bridge maps are instrumentation for the screen-to-radial theorem. They do not turn a screen covariance into a primordial spectrum unless the source-stress, single-clock, repair-gap, freeze-out, geometric-screen-scalar, scalar-precision, primitive-collar-law, source-release-energy, infinitesimal reserve-generator, orientation-half, refinement-tilt, radial-null, finite-window, and forward-residual receipts also pass. Radial uniqueness requires either a scale-labelled source-refinement orbit with a scale-natural physical embedding, source/physical covariance residual equality, strong finite-to-continuum convergence, and safe-band leakage control, or a cofinal radial cross-covariance family with spherical-Hankel convergence and held-out reconstruction. On the physical dilation branch the source-facing generator gives $\theta = P_\star/48$ and the continuum intertwiner restricts the radial power to a one-dimensional family. No concrete finite carrier satisfying the full receipt set is claimed in this paper; the receipt set is the complete premise list for any such claim.*

Theorem 6.54 (Carrier invariance of the anomaly load readout)**.** *If two carrier presentations induce the same quotient state in $Q^{\mathrm{rel}}_{A,r}$, the same parent stress readout, the same release hypersurface, and the same homogeneous projector, then they emit the same $\mathsf{L}_{A,r}$.*

Proof. Every term in

$$\mathsf{L}_{A,r} = \Pi_r^{(0)} \sum_c V_{c,r}^{\mathrm{phys}} T_{A,r}^{ab} n_a n_b$$

is quotient-visible by hypothesis. Hidden carrier coordinates, port labels, repair schedule identifiers, and worker metadata are quotiented out by the physical quotient. Therefore the load readout is invariant. □

Definition 6.55 (Einstein-branch geometry readout)**.** *An Einstein-branch geometry readout is a cofinal family of maps*

$$\mathsf{Geom}_r : Q_{r,\mathrm{nf}} \to \mathsf{GeoData}_r, \qquad Q_{r,\mathrm{nf}} = n_r(Q_r),$$

where the emitted data include

$$\begin{gathered} \mathsf{Chart}_{S,r}, \quad \mathcal{A}_r^{\mathrm{geo}}(C), \quad \omega_r^{\mathrm{geo},C}, \quad \lambda_{C,r}, \quad B_{C,r}, \quad L_{C,r}, \\ \mathsf{AreaRead}_r, \quad \mathsf{StressRead}_r, \quad \mathsf{DiamondRead}_r, \quad \mathsf{TetradRead}_r, \quad \mathsf{ScaleRead}_r, \quad c_{sr}^{\mathrm{geo}}. \end{gathered}$$

It must be insensitive to hidden carrier coordinates, gauge representatives, port relabelings, worker/shard labels, and accepted repair schedules. It must also be refinement-compatible:

$$c_{sr}^{\mathrm{geo}} \circ \mathsf{Geom}_s = \mathsf{Geom}_r \circ c_{sr}^{\mathrm{nf}} + O(\epsilon_{sr}^{\mathrm{geo}}), \qquad \epsilon_{sr}^{\mathrm{geo}} \to 0.$$

Theorem 6.56 (Geometry-readout quotient factorization)**.** *If the geometry-facing readouts of Definition 6.53 are packaged as an Einstein-branch geometry readout Geom_r that factors through $Q_{r,\mathrm{nf}}$, then every geometry-facing observable extracted from Geom_r is independent of hidden carrier coordinates, gauge representatives, port relabelings, worker/shard labels, accepted repair path, and repair schedule.*

Proof. By construction, every extracted geometry-facing observable is a function of $\mathsf{Geom}_r(q_{\mathrm{nf}})$ for a repaired quotient normal form $q_{\mathrm{nf}} \in Q_{r,\mathrm{nf}}$. Any hidden carrier coordinate, gauge representative, port label, worker/shard label, accepted repair path, or repair schedule that presents the same quotient normal form therefore has the same geometry readout. The conclusion is exactly quotient factorization. □

The factorization theorem proves presentation independence of a declared readout; the complementary production direction, deriving the Geom_r payload itself from the repaired normal form instead of declaring it, is the spacetime and Einstein paper's quotient-intrinsic geometry-producer theorem, which constructs the screen/cap data from the support-visible incidence complex on its computable-receipt branch and proves the receipt selection underdetermined by bare confluence.

In particular, a finite carrier patch, graph vertex, implementation point, or worker-local shard is not one canonical OPH screen cell unless a separate measure-realization theorem proves that identification. Evidence sufficient for that identification must expose a geometry manifest, metric-form hashes, orientation and topology, boundary conditions, lineage maps, lapse and shift, source embedding, operator assembly, scale certificate, and refinement maps.

A spherical screen is then a coarse chart over a federation of such patches. It is what a particular observer-facing support cut looks like after quotienting hidden implementation details and choosing a geometric presentation. The microscopic carrier may be graph-like, federated, multi-patch, and locally polyhedral.

The P, N closures supply the geometric side of that chart. Write

$$P_\star = \frac{a_{\mathrm{cell}}}{\ell_\star^2}, \qquad N_\Lambda = \frac{A_{\mathrm{screen}}}{4\ell_\star^2},$$

where $N_\Lambda = 3.31 \times 10^{122}$ is the Λ-located working capacity. An equal-area cellulation then has

$$K_{\mathrm{cell}} = \frac{A_{\mathrm{screen}}}{a_{\mathrm{cell}}} = \frac{4N_\Lambda}{P_\star} \simeq 8.12 \times 10^{122}.$$

At the conditional bridge capacity the same count reads 8.66×10^{122}. The Λ-located value is a measured-side comparison display and is not an input to the closure. This counts canonical geometric screen cells. It does not assign an independent Hilbert factor to each cell. In particular $P_\star/4 \simeq 0.408$ nats is about 0.588 bits, so it is not by itself the logarithm of an integer-dimensional autonomous cell algebra. The area law lives on shared cuts, edge centers, and constrained overlap data.

Definition 6.57 (Federated patch carrier)**.** *A fixed-cutoff federated patch carrier is a tuple*

$$\mathfrak{F} = (V, E, \{\mathcal{A}_i\}_{i\in V}, \{\mathcal{I}_e\}_{e\in E}, \{\pi_{i,e}\}, \{\mathcal{R}_i\}, \{\mathcal{U}_i\}),$$

where V is a finite set of patches, E is a finite overlap graph, $\mathcal{A}_i$ is the local finite algebra of patch i, $\mathcal{I}_e$ is the finite interface algebra on overlap $e = \{i, j\}$, $\pi_{i,e} : \mathcal{A}_i \to \mathcal{I}_e$ is the declared visible restriction, $\mathcal{R}_i \subseteq Z(\mathcal{A}_i)$ is the patch record algebra, and $\mathcal{U}_i$ is the allowed local update and repair interface.

The carrier is *observer-facing* when every physical claim is made through the visible restrictions $\pi_{i,e}$, record algebras $\mathcal{R}_i$, and the quotient-local observables specified by the OPH consensus package. Hidden coordinates may exist, but they are not directly physical.

6.6 Canonical multiresolution regulator chart

Definition 6.58 (Reference multiresolution carrier)**.** *Choose the nested geodesic icosahedral subdivisions of S^2. A regulator is*

$$r = (m, L, b),$$

where m is the subdivision depth, L is the finite support or volume scale of the exported carrier, and b is a boundary/phase label. Every added cell, collar, edge-center, or observer-visible channel j carries a finite algebra

$$D_j = \bigoplus_{\alpha \in S_j} M_{d_{j,\alpha}}(\mathbb{C})$$

and a faithful detail state τ_j. At regulator r,

$$\widetilde{M}_r = \bigotimes_{j \in \mathcal{J}_r} D_j, \qquad \widetilde{\omega}_r = \bigotimes_{j \in \mathcal{J}_r} \tau_j.$$

A finite-depth local presentation circuit W_r maps these multiresolution coordinates to the declared patch, port, collar, and gauge coordinates:

$$M_r = \mathrm{Ad}(W_r)(\widetilde{M}_r), \qquad \widehat{\omega}_r = \widetilde{\omega}_r \circ \mathrm{Ad}(W_r^*).$$

For $r \preceq s$, refinement adds detail factors. The refinement embedding and coarse-graining are

$$\iota_{rs}(A) = W_s\big[(W_r^* A W_r) \otimes \mathbf{1}\big]W_s^*,$$

and

$$Q_{sr}(X) = W_r\big[(\mathrm{id} \otimes \tau_{s \setminus r})(W_s^* X W_s)\big]W_r^*.$$

The embedded map $E_{sr} = \iota_{rs} Q_{sr}$ is the canonical state-preserving conditional expectation.

Theorem 6.59 (Fixed-cutoff regulator certificate)**.** *The maps in Definition 6.58 are unital, completely positive, composition-compatible, and preserve the faithful reference states. The refinement embeddings are unital injective $*$-homomorphisms, the E_{sr} are faithful conditional expectations, and the reference modular groups restrict exactly along the tower. The inductive limit has canonical finite-regulator renormalization maps given by these conditional expectations; their martingale tails are the finite-volume and lattice-spacing Cauchy errors used by the compact and main papers.*

Proof. In bare coordinates all statements reduce to tensoring an observable with the identity and averaging the detail factors against the faithful product state. Composition follows from product-state associativity. Conjugating by the presentation circuits gives the physical patch coordinates without changing the algebraic identities. □

Remark 6.60 (Reference state, ground state, and vacuum). The state $\widehat{\omega}_r$ is the finite-regulator reference state. It may be called a finite-volume ground state only after a positive self-adjoint transfer or Hamiltonian has been constructed and shown to have that state as its ground state. The term continuum vacuum is reserved for the GNS/OS limit after positive energy and the declared vacuum-sector uniqueness or superselection condition have been proved.

Remark 6.61 (Regulator-to-continuum identification rule). A finite evidence bundle implements this regulator branch only when it exports the factor list, presentation circuit, refinement maps, conditional-expectation identities, transported state errors, renormalized observable tails, and positive-transfer or reflected-Gram certificates. A finite cellulation, finite-group score, or stable numerical extrapolation by itself is a regulator-chart result.

Definition 6.62 (Clocked spatial readout package)**.** *A finite cosmological run that claims FLRW spatial curvature must export the clock and spatial geometry readouts separately from screen/collar defect bookkeeping:*

$$\mathsf{SpatialMetricRead}_r, \quad \mathsf{SpatialFrameRead}_r, \quad \mathsf{SpatialConnectionRead}_r,$$
$$\mathsf{SpatialTransportRead}_r, \quad \mathsf{CurvatureHolonomyRead}_r,$$
$$\mathsf{ProperRadiusRead}_r, \quad \mathsf{NestedCapacityRead}_r.$$

The transport readout uses the spatial Levi–Civita connection on the clock slice. Finite permutation holonomy of an S^2 screen triangulation or S_3 collar defect is a repair/sieve diagnostic unless these readouts identify it with a spatial Levi–Civita holonomy under quotient, transport, and refinement naturality.

7 The Echosahedral Patch Object

An echosahedral patch is the reference local body for this microphysics. The term is architectural: a bounded observer patch with a highly symmetric multi-port interface.

Definition 7.1 (Echosahedral patch)**.** *An echosahedral patch is a finite patch object*

$$\mathcal{E} = (\mathcal{A}, \{P_a\}_{a=0}^{11}, \{\rho_a\}_{a=0}^{11}, \mathcal{R}, \mathcal{M}, \mathcal{U}, \mathcal{G}_{\mathcal{E}}),$$

where:

1. $\mathcal{A}$ *is the internal finite algebra;*
2. $P_0, \ldots, P_{11}$ *are twelve labeled overlap ports;*
3. $\rho_a : \mathcal{A} \to \mathcal{I}_a$ *are port readout maps;*
4. $\mathcal{R} \subseteq Z(\mathcal{A})$ *is the observer-accessible record algebra;*
5. $\mathcal{M}$ *is a finite mismatch-score family on exposed port packets;*
6. $\mathcal{U}$ *is a finite family of local update and repair instruments;*
7. $\mathcal{G}_{\mathcal{E}}$ *is a declared finite symmetry or approximate-symmetry group of the port arrangement.*

Definition 7.2 (Certified echosahedral counting realization)**.** *A certified local lineage augments the quotient-visible echosahedral patch by:*

(i) *twelve primitive orthogonal port-center atoms $e_{r,p}$, each of normalized trace* $1/12$*;*

(ii) *the oriented* 12*-vertex,* 30*-edge,* 20*-face triangular incidence packet, with degree-five vertices and five-cycle links;*

(iii) *a separately declared integer atom-counting grammar with total-charge fiber $\mathcal{Q}_r = \{q \in \mathbb{Z}^{12} : \sum_p q_p = 12\}$, and the normalized central-readback Hilbert–Schmidt cost $H_r(q) = \sum_p q_p^2$;*

(iv) *an append-only twelve-register event packet in which writes and retractions carry signs* $+1$ *and* -1*, together with conservative whole-unit seam repairs and minimum-move settling cost. An accepted repair orients a seam from i to j only when $d = q_i - q_j \geq 2$, then replaces (q_i, q_j) by $(q_i - 1, q_j + 1)$; uphill and equalizing-reversal moves are not repair steps;*

(v) *every refinement carries an orientation-preserving incidence isomorphism. It preserves the port centers and readback packet, and these isomorphisms obey the refinement cocycle; and*

(vi) *a source manifest that rejects Standard Model, product-adjoint, gauge, coupling, particle, measured-target, and fitted-coordinate fields, is closed to undeclared fields, and encodes trace fractions with exact JSON integers.*

The signed-event packet realizes integer loads and its separate dynamic settling cost. The total-charge fiber, quadratic cost normalization, and positive readback scale are named realization data. An arbitrary carrier satisfying the three axioms need not select them.

Theorem 7.3 (Echosahedral counting-and-incidence selector theorem)**.** *On every lineage of Definition 7.2, the declared realization packet canonically determines:*

(i) *the twelve coefficient lines* $\mathbb{R}e_{r,p}$ *and the unique all-one defect allocation, with* $H_{\min} = 12$*, next floor* 14*, and exact gap* 2*;*

(ii) *integer loads generated by the signed atomic events, termination of every accepted settling path by strict descent of* $V(q) = \sum_p q_p^2$*, and a minimum settling-move cost natural under every carrier rotation and declared refinement;*

(iii) *the unique fixed-point-free inverse pairing by graph distance three and the resulting six axes;*

(iv) *the full incidence group* $A_5 \times C_2$ *and its orientation-preserving subgroup* A_5*;*

(v) *the exact incidence Gram matrix*

$$G_{pq} \in \{1, 1/\sqrt{5}, -1/\sqrt{5}, -1\} \quad \text{at distances } 0, 1, 2, 3,$$

with

$$G^2 = 4G, \qquad \operatorname{tr} G = 12, \qquad \operatorname{rank} G = 3,$$

whose oriented factorization is the regular six-axis icosahedral frame up to $SO(3)$*; and*

(vi) *refinement naturality and arbitrary simultaneous port-relabeling equivariance of all these objects.*

Thus this realization gives the exact twelve-line unit allocation and gap, integer settling dynamics, inverse pairing, proper A_5 *action, and rank-three Gram frame stated above.*

Proof. Primitive central atoms give the intrinsic line family. On the total-charge fiber,

$$H(q) = 12 + \sum_p (q_p - 1)^2,$$

so the all-one allocation is unique and the least nonzero integral zero-sum deviation has squared norm two. A seam repair with oriented mismatch $d \geq 2$ changes the load square by $-2(d-1)$, which proves termination. Every declared rotation and refinement is an isomorphism of the repair graph and therefore preserves shortest-path cost. The explicit full-pile witness settles in eighteen moves. The exact incidence distance profile is $(1, 5, 5, 1)$, giving a unique distance-three involution. Enumeration gives 120 incidence automorphisms and 60 orientation-preserving ones; conjugation on the five Klein-four subgroups identifies the latter faithfully with all even permutations of five objects, hence with A_5. Exact multiplication in $\mathbb{Q}(\sqrt{5})$ gives $G^2 = 4G$, and the trace fixes rank three. Every declared refinement or complete relabeling preserves the source formulas. The exact implementation and countermodels are carried by the echosahedral selector certificate and its accompanying analysis. □

Remark 7.4 (Load-bearing hypotheses)*.* Without primitive atoms the line split is continuous; without equal trace or total charge twelve the all-unit minimizer is not unique; a linear cost has $\binom{23}{11}$ minimizers; twelve labels without incidence have 10,395 fixed-point-free pairings; without orientation the selected group is $A_5 \times C_2$; and non-incidence or noncocyclic refinement maps fail naturality.

The theorem derives the selector from the declared echosahedral-carrier packet. It does not prove that bare Euler incidence or every possible OPH carrier selects that packet, and it does not construct a physical gauge-current algebra. The executable controls recompute the finite countermodel values and reject both an opaque undeclared numeric input and a floating-point trace fraction that could otherwise be truncated.

An append-only record machine supplies an operational realization of the integer loads. Each write contributes $+1$, each retraction contributes -1, and the port readback sums the signed atomic events. A repair transfers one whole unit along an oriented seam when its mismatch has magnitude at least two. The load square $V(N) = \sum_i N_i^2$ decreases by $2(d-1)$ on a repair with mismatch $d \geq 2$, which proves termination. The total load modulo twelve is a necessary consensus obstruction, with an explicit eighteen-move settling path for the declared full-pile state. Minimum settling cost is preserved by the sixty rotations and the declared refinement maps. The family $\{\frac{1}{2}e_{r,p}\}_{p=0}^{11}$ rescales the event values, seam readback, and repair threshold together. It is the same event mechanism in different readback units, not a half-event countermodel.

For a separately declared uniform twelve-port reference $\tau_p = 1/12$, a feasible perturbation $\rho = \tau + \varepsilon v$, with $\sum_p v_p = 0$, satisfies

$$D(\rho\|\tau) = 6\varepsilon^2 \sum_p v_p^2 + O(\varepsilon^3).$$

The second-order Taylor coefficient is $6I$, while the Hessian, equivalently the Fisher matrix, is $12I$. The A3 weight fixes their common scale. This infinitesimal curvature supplies neither the exact discrete cost H_r nor the minimum settling-move cost. Exact separating witnesses distinguish all three objects.

Theorem 7.5 (Alternative variational icosahedral screen sieve)**.** *Assume the fixed-cutoff OPH screen branch is represented by a quantum-link triangulation of S^2, with finite Hilbert spaces on links, Gauss constraints at vertices, boundary-gauge-invariant physical algebras, and a refinement-stable quotient ensemble locally six-valent away from curvature defects. Let*

$$q_v = 6 - \deg(v).$$

Assume integer charges, a feasible twelve-unit configuration, and an additive cost with $h(0) = 0$, $h(1) > 0$, $h(k) \geq h(1)|k|$, strictly for $|k| \geq 2$. Let edge-center collars expose each unit defect as one central port, with a fixed-point-free inverse-port involution giving six axes u_i. Assume the minimum-cost locus contains a six-axis configuration with $F_1 = 2I_3$ and $F_2 = (4/5)I_5$. If the source selector first minimizes the additive cost and, among those minimizers, maximizes

$$\det F_1 \det F_2, \qquad F_1 = \sum_i u_i u_i^{\mathsf{T}}, \quad F_2 = \sum_i |Q_i\rangle\langle Q_i|, \quad Q_i = u_i u_i^{\mathsf{T}} - I_3/3,$$

then the screen has twelve unit ports on the regular icosahedral orbit. An equal-weight additive invariant load X is read locally as $X/12$.

Proof. For a triangulated sphere with V vertices, E edges, and F faces, Euler and triangular incidence give

$$V - E + F = 2, \qquad 3F = 2E, \qquad E = 3V - 6.$$

Therefore

$$\sum_v q_v = \sum_v (6 - \deg(v)) = 6V - 2E = 12.$$

Thus $\sum_v q_v = 12$. The cost satisfies

$$\sum_v h(q_v) \geq h(1) \sum_v |q_v| \geq 12h(1).$$

Equality requires nonnegative unit charges, and feasibility supplies the twelve-unit minimum. The selector's first stage therefore returns exactly twelve unit defects. The collar rule turns them into twelve central ports. For six axes, $\operatorname{tr} F_1 = 6$ and $\operatorname{tr} F_2 = 4$. Determinant AM–GM gives $\det F_1 \leq 2^3$ and $\det F_2 \leq (4/5)^5$, with equality exactly at $F_1 = 2I_3$ and $F_2 = (4/5)I_5$. The assumed feasible isotropic frame attains both bounds, so the selector's second stage does too. The six equal-norm Q_i then form a regular simplex in $\mathrm{Sym}_0(3)$, so $(u_i \cdot u_j)^2 = 1/5$ for $i \neq j$. The associated Seidel matrix satisfies $S^2 = 5I_6$; after switching one vertex is isolated and the remaining negative-edge graph is a five-cycle. This is the unique switching class and gives the six axes of a regular icosahedron, whose proper rotation group A_5 is transitive on the twelve ports. Equal-weight additivity gives the local read $X/12$. □

Remark 7.6 (External identification of the load). The identification $X = \log(N/\pi)$ is imported from the companion capacity relation; this paper does not construct that object. The theorem itself constrains invariant screen loads that are additive across the twelve ports.

Remark 7.7 (Source-selector boundary). For Theorem 7.5, the D-optimal vector/quadrupole functional is a branch premise, and a strictly completely monotonic pair cost is another sufficient selector by universal optimality. Neither is needed on the certified lineage of Theorem 7.3, where oriented incidence supplies the antipode, A_5 action, and frame, while the declared integer counting grammar and normalized central-readback cost supply the all-unit selector. In the alternative variational theorem, $\sum_v(6 - \deg(v)) = 12$ follows from spherical incidence, while integer charge typing and the strict additive cost are assumptions. Only the twelve-port load normalization follows from unit splitting and placement. The cell value $\bar{\ell}_{\text{shared}} = P/4$ and the $\mathbb{Z}_6$ class equidistribution of Theorem 6.41 have separate premises; the port and slot counts do not imply either statement.

Theorem 7.8 (Exact multipole fixed point of the twelve-port orbit)**.** *Let $u_1, \ldots, u_{12}$ be the twelve unit port directions of the regular icosahedral orbit and let P_l denote the degree-l Legendre polynomial. Then for every unit vector n,*

$$\sum_{i=1}^{12} P_l(u_i \cdot n) = 0, \qquad l = 1, \ldots, 5,$$

and the normalized rank-six invariant

$$\mathcal{I}_6(n) = \frac{25}{132} \sum_{i=1}^{12} P_6(u_i \cdot n)$$

takes the exact values 1 *on the twelve port directions,* $-5/16$ *on the thirty edge midpoints, and* $-5/9$ *on the twenty face centers. Its critical set on the sphere is exactly the sixty-two axis points: twelve maxima at the ports, thirty saddles at the edge midpoints, and twenty minima at the face centers, every tangent Hessian eigenvalue nonzero in exact arithmetic, with Euler count* $12-30+20 = 2$*. On the multiplicity-free band split* $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3'} \oplus \mathbf{5}$ *of Theorem 6.11, the lowest surviving multipole ranks are* $0, 1, 3, 2$*, the antipodal partner of each port is derived from adjacency as its unique distance-three port, and the blind antipodal response* $R = -J$*, with* J *that derived antipodal permutation, acts with sign vector* $(-1, +1, +1, -1)$ *in band order* $(\mathbf{1}, \mathbf{3}, \mathbf{3'}, \mathbf{5})$.

Proof. The icosahedral invariant dimension at angular rank L is

$$m_L = \frac{1}{60}\Big[(2L+1) + 15\,U_{2L}(0) + 20\,U_{2L}(\tfrac{1}{2}) + 12\,U_{2L}(\tfrac{\varphi}{2}) + 12\,U_{2L}(\tfrac{\varphi-1}{2})\Big]$$

with U the Chebyshev polynomials of the second kind; the exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$. Each orbit sum $\sum_i P_l(u_i \cdot n)$ is an invariant of rank l, hence vanishes for $l = 1, \ldots, 5$, and the rank-six invariant line is one-dimensional. The special values, the sixty-two-point Morse census, the derived antipode, and the band signs are certified in exact arithmetic by the machine-readable certificate [20]. □

Proposition 7.9 (Quantitative stability bounds for the equal-weight cosine member)**.** *Let $Q_x(n) = \sum_{i=1}^{12}[1 - \cos(xu_i \cdot n)]$, where $n, u_i \in S^2$, and let $0 < x \leq 1$. After removal of the isotropic part, the terms through eighth order are*

$$A(x)\mathcal{I}_6(n), \qquad A(x) = \frac{2x^6(30 - x^2)}{118125} > 0.$$

Writing the complete x^{10}-and-higher angular tail after division by $A(x)$ as E_x, exact bounds are

$$\|\nabla E_x\| \leq \frac{6875}{101152} < \frac{7}{100}, \qquad \|\operatorname{Hess}_{S^2} E_x\| \leq \frac{383125}{562658} < \frac{7}{10}.$$

The 62 critical directions of $\mathcal{I}_6$ form 31 projective axes, with minimum squared-sine separation $1/2 - \sqrt{5}/6$.

Proof. Exact reduction on the unit sphere gives

$$\sum_i (u_i \cdot n)^6 = \frac{12}{7} + \frac{64}{175}\mathcal{I}_6(n), \qquad \sum_i (u_i \cdot n)^8 = \frac{4}{3} + \frac{256}{375}\mathcal{I}_6(n).$$

The sixth- and eighth-order terms of the cosine series therefore have anisotropic coefficients $4x^6/7875$ and $-2x^8/118125$, which combine to the stated positive amplitude. For every term of order $2m \geq 10$, the spherical gradient is bounded by $12x^{2m}/(2m-1)!$. The Euclidean Hessian is bounded by $12x^{2m}/(2m-2)!$, and the intrinsic Hessian adds at most the gradient bound through the radial correction. On $x \leq 1$, the successive ratios are at most $1/110$ and $1/90$, while $A(x) \geq 58x^6/118125$. Summing the two geometric majorants and reducing the rational numbers gives the displayed bounds. Exact enumeration of the three carrier orbits gives the axis count and separation. The machine receipt replays these reductions independently [20]. □

Theorem 7.10 (Exact persistence of the equal-weight cosine fingerprint)**.** *Under the assumptions of the preceding proposition, the full kernel Q_x has exactly sixty-two stationary directions for every $0 < x \leq 1$. They are the twelve port maxima, twenty face-center minima, and thirty edge-center saddles of $\mathcal{I}_6$. Every stationary point is Morse with the same index as the corresponding rank-six critical point.*

Proof. Equal port weights make Q_x invariant under the proper carrier rotations. The order-five, order-three, and order-two stabilizers of the vertex, face, and edge axes have no fixed tangent vector. Every one of the sixty-two axis directions is therefore stationary for the full cosine kernel.

After division by the positive amplitude $A(x)$, three exact projective charts cover the sphere modulo antipodes. A deterministic dyadic subdivision into 2,624 leaves proves that the tangent-gradient norm of $\mathcal{I}_6$ exceeds the complete tail bound away from the disjoint neighborhoods $\sin^2\theta \leq 1/4096$ of the 31 axes. No further stationary direction occurs there.

For one representative of each axis orbit, normalized local coordinates give the exact factorization $G(y) = C(y)y$, with $G(0) = 0$. Exact interval bounds on $C(y) - C(0)$, using $\|E\|_2 \le \sqrt{\|E\|_1 \|E\|_\infty}$, leave a strictly positive smallest-singular-value margin after the tail Hessian is included. The local chart obeys $\|J_y\| \le 1$ and $\mathrm{dist}(0, y) = \arctan\|y\| \le \|y\|$, so the intrinsic C_2 tail bound is the required coordinate-gradient bound. Each neighborhood contains its axis center as its unique stationary direction. Separate exact Hessian margins preserve the two definite signatures and the indefinite edge signature. Carrier symmetry transports the three local certificates to all 31 axes. The interval arithmetic and an independent implementation are recorded in the machine receipt [20]. □

Theorem 7.11 (Spin-six residue universality on the invariant carrier class)**.** *Let the carrier kinetics be any operator $\lambda_a(k) = a^{-2} \sum_d w(d)\,[1 - \cos(a\,k \cdot d)]$ with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank five at every order in a, and every directional term below angular rank ten is one multiple of $\mathcal{I}_6$. The rank-six Legendre coefficient of t^{2m} is $13\,(2m)!/((2m-6)!!\,(2m+7)!!)$, strictly positive for every $m \ge 3$. For each fundamental orbit the sixth-moment $\mathcal{I}_6$ multiple is nonzero: $64/175$ on the twelve port directions, $-64/189$ on the twenty face directions, and $-2/7$ on the thirty edge directions. Whenever the weighted sixth-moment coefficient is nonzero, the first directional artifact appears at order a^4k^6 and one binary refinement step suppresses it by exactly $1/16$ at that order. The exact same-radius positive-weight member with weight 1 on the twelve port directions and $27/25$ on the twenty face directions cancels the $\mathcal{I}_6$ residue at every order. It is isotropic through k^8 and first populates angular rank ten at order a^8k^{10}. A separate control, with port radius 1 and weight 1, face radius 2 and weight $27/1600$, cancels the sixth-moment $\mathcal{I}_6$ term while retaining the raw eighth-moment $\mathcal{I}_6$ multiple $-256/125$. Rank six is the least symmetry-allowed angular rank in the class, although mixtures can cancel it.*

Proof. The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$ with harmonic components at even ranks $L \le 2m$; the invariant table of Theorem 7.8 forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$. The positivity of the level-six coefficients, the per-orbit sixth-moment multiples, the refinement factor, the same-radius angular-rank-ten control, and the two-radius eighth-moment control are certified in exact arithmetic [20]. □

On the primitive twelve-port propagation branch, whose premises and custody live in the synthesis paper [17], these identities give the dispersion coefficients $C_4 = -a^2/20$, $B_0 = a^4/840$, and $B_6 = 2a^4/7875$, hence $B_6/C_4^2 = 32/315$, $B_0/C_4^2 = 10/21$, and $B_6/B_0 = 16/75$, with the unique rank-six icosahedral shape oriented in $SO(3)/A_5$. The scalar or polarization-independent sector bridge, coherent frame transport, and exclusivity are declared physical premises of that branch. Here spin six denotes spherical-harmonic angular rank, not particle spin.

The finite repair operator acts on thirty internal seams rather than spatial translations. An exhaustive classification of the simulator's declared serialized finite artifacts finds the local-domain kinetic operators and the twelve-port response on separate domains, with no registered accepted packet joining a complete twelve-port translation operator to a physical readout of that same operator. Equal-weight stencils on the twelve vertex, twenty face, and thirty edge direction orbits have distinct rank-six coefficient rays. Proper-carrier transitivity fixes weights within a chosen orbit and does not choose among these spatial alternatives.

7.1 The source-seam record carrier and its homogeneous action

The complete seam packet admits a second exact reading that keeps the finite record mechanism separate from physical spacetime. Give each of the thirty undirected seams a bookkeeping orientation and let an integer current carry the corresponding signed incidence boundary on the twelve ports. Pair each port with its graph-distance-three inverse and take the antipodal-odd part of that boundary. The result is a six-coordinate integer record.

Theorem 7.12 (Exact seam-current image and response completion)**.** *The image of the complete integer seam-current map is exactly*

$$D_6 = \left\{ z \in \mathbb{Z}^6 : \sum_{i=1}^{6} z_i \equiv 0 \pmod 2 \right\}.$$

The six-by-thirty integer map has Smith invariants $(1,1,1,1,1,2)$*. Its residual cokernel is one parity bit, and an explicit six-seam section reaches every element of* D_6*. Pull back the response-selected Gram metric of Theorem 7.3 to* D_6*. The resulting metric image is injective and dense in the same three-dimensional Euclidean response carrier, so its abstract metric completion is canonically isometric to that carrier. Cumulative* D_6*-records act simply transitively by isometries on the integer record carrier, and this action extends to translations of the completion.*

Proof. Exact integer incidence has rank eleven on the twelve-port boundary space and cycle rank nineteen. Composing it with the antipodal-odd readout gives an even coordinate sum. Conversely, the explicit six-seam section reconstructs an arbitrary even-sum vector, which proves the image statement. The parity map supplies the two cokernel cosets and the Smith form. The response Gram map is injective on the integer controls. The golden-ratio coordinate lines are dense, and doubled control vectors belong to D_6, so the restricted image remains dense in the rank-three response carrier. Record addition and the pullback metric give the stated homogeneous isometric action. All finite incidence, section, parity, density, and action identities are checked in Lean [20]. The thirteen finite covariance/isometry table identities in the proper-action subpacket use `native_decide`; those subreceipts therefore trust Lean's native compiler and runtime as well as its kernel and are not kernel-only proofs. ☐

Theorem 7.13 (Finite seam continuity and Ward-promotion boundary)**.** *Let* $q, q', s \in \mathbb{Z}^{12}$ *be stored load, updated load, and source, and let* $J \in \mathbb{Z}^{30}$ *be a current on the bookkeeping-oriented seams. Adopt the net-inflow incidence convention. If*

$$q' - q = s + \partial J,$$

then for every port region R*,*

$$\sum_{p \in R} (q'_p - q_p) = \sum_{p \in R} s_p - \Phi_{\partial R}(J),$$

where $\Phi_{\partial R}$ *is signed outward flux. Internal seams cancel. For the complete port set, a zero-total source preserves total load.*

After scalar extension to $\mathbb{Q}$*, a port load* z *has a solution of* $\partial J = z$ *exactly when* $\sum_p z_p = 0$*. A declared port-zero spanning-tree section constructs one solution; the full solution fibre is its translate by* $\ker \partial$*, which has dimension nineteen. This choice is explicit and carries no naturality claim.*

Separately, let Φ *and* $\Phi^\dagger$ *be real linear maps on a finite state space that are dual for the expectation pairing. A charge expectation is preserved for every signed state exactly when* $\Phi^\dagger Q = Q$*.*

Channel covariance is insufficient: uniform averaging on two states commutes with their swap and erases the nonzero swap-odd charge. Antisymmetric transfer, action symmetry, channel covariance, and charge conservation are therefore distinct statements.

Proof. Regional summation of the oriented incidence map pairs every internal seam with opposite endpoint signs. Rank-nullity for the connected twelve-vertex, thirty-edge graph gives incidence rank eleven and cycle rank nineteen, and the declared tree section proves surjectivity onto the neutral hyperplane. Finite expectation-pairing nondegeneracy proves the dual fixed-point criterion by testing signed point masses. The two-state averaging computation is an exact negative control. The regional, Gauss-fibre, duality, and control statements are checked in Lean [20]. □

The theorem assumes the pointwise finite update and assigns no physical meaning to load, current, update order, or charge. A continuum Ward identity requires a source-derived family of these finite witnesses, directed cofinal refinement with a positive scale tending to zero, realization as components of one stress tensor, weak convergence on an admissible separating test-field class, a finite residual declared on the continuity witness and independently shown by the downstream continuum construction to be derived from it, convergence of that residual, chart covariance, and one inhabited common tower. The generic Lean function may otherwise ignore its witness. The Lean `WardLimitManifest` requires a not-all-zero finite datum and a nontrivial residual; it derives residual vanishing at the candidate limit by uniqueness of limits. The candidate limit may be zero. That construction must identify the declared residual with physical distributional divergence and supply the source, transport, chart, and tower evidence; the structure alone makes no physical identification. In particular, residual convergence must follow from an independent compatibility estimate and cannot be justified by assuming the target Ward equality.

The metric in Theorem 7.12 is the response Gram metric. It is not the inherited Euclidean metric of the integer tuple $z \in \mathbb{Z}^6$. The theorem constructs an abstract three-dimensional completion of records. It supplies no identification with physical position, no ruler, and no global observer gluing.

One composed machine-checked surface carries the seam-current chain in the Lean development (`LightSignalAdequacySurface`): a premise bundle naming the equal source-counting measure, the auxiliary oscillator lift, and the physical-frequency identification as declared inputs, through which the forced weight, the dimensionless generator, the rank-two transverse completion with its exact zero mode, and a second-order wave-shaped evolution with the identified frequency are re-exported as one package. A companion module (`LightSignalMaxwellComposition`) extends this to the composed modal-Maxwell receipt on the same bundle: an explicit boundary map carries the supplied mode dynamics onto the opposite-sign curl pairing at the bundle's chart scale, exactly, on every transverse state at every nonzero chart momentum; the pairing squares to minus the identified frequency on both amplitudes, preserves both modal divergence constraints, commutes with the global phase action, and the same-sign mutation fails the identified wave law. The intertwining is a factorization, not an equivalence, and no potential dynamics, covariance, or continuum control is claimed there; the real position-space seam field, its gauge quotient, and its conserved-source quadratic action are committed separately, as exact finite mathematics on the seam carrier, in Propositions 7.20 and 7.21; those propositions supply no physical attachment. The same module threads the discrete Gauss receipts through the shared seam alphabet: the forced counting measure is orientation-balanced across the committed double cover of the thirty seams and its expected signed port boundary vanishes on the same incidence object that carries the solvability, spanning-tree, cycle-fibre, and rank-nineteen receipts. A separate premise-free classification (`SeamU1HolonomyClassification`) commits the finite $U(1)$ object the boundary lists named as missing: seam connections modulo port gauge are classified exactly by their nineteen chord holonomies: the first finite cohomology group of the seam graph with circle coefficients is $U(1)^{19}$, and the moduli count equals the committed Gauss cycle rank. The connection is a finite kinematic object with no curvature dynamics and no physical field identification; its real de Rham companion carries the committed quadratic action and conserved-source coupling of Proposition 7.21. The attained content is a conditional free-vector law; the composed surface supplies no complete Maxwell-shaped field equation and makes no photon claim.

Theorem 7.14 (Conditional homogeneous Dirichlet generator). *Suppose the admissible directed-seam move laws and their objective are natural under the proper carrier action, and suppose the normalized admissible minimizer is strictly positive and unique. The weight on each of the sixty directed seams is then* $1/60$*. Let*

$$(Pf)(x) = \frac{1}{60}\sum_e f(x+v_e), \qquad L = I - P,$$

where v_e is the exact response-coordinate translation of a directed seam. Then P preserves constants and positivity, L is translation invariant, and

$$2f(x)(Lf)(x) - L(f^2)(x) = \frac{1}{60}\sum_e \big(f(x+v_e) - f(x)\big)^2 \geq 0.$$

Plane waves diagonalize L. If $a_{\text{edge}}^2 = 2 - 2/\sqrt{5}$ denotes the exact squared response-Gram norm of a seam translation, then

$$\frac{6}{a_{\text{edge}}^2}\lambda_L(k)$$

is exactly the normalized thirty-edge cosine character in the response coordinates. A supplied nonzero coordinate dilation a gives

$$\Lambda_a(k,n) = \frac{1}{5a^2}\sum_{j=1}^{30}\left[1 - \cos(ak\, w_j \cdot n)\right], \tag{1}$$

where the w_j are the unit edge directions. The dilation is a coordinate choice until a physical position map fixes its meaning.

Proof. Transitivity and uniqueness make the selected weight constant on the directed seam orbit, and normalization fixes it to $1/60$. Reindexing the directed orbit gives one positive and one negative translation for each of the thirty seams. The square identity follows by expanding $L = I - P$. The plane-wave identity is the exact character calculation for those translation pairs. The response Gram theorem gives the displayed seam norm. Exact source incidence, selection, translation, and Fourier identities are machine-checked in the finite proof chain [20]. □

The naturality and unique-minimizer clauses in Theorem 7.14 are explicit premises. The finite source fixes the orbit and its response metric; it does not prove that this homogeneous internal action is a physical field action.

Theorem 7.15 (Global contraction of a complete positive cosine auxiliary frequency). *Let* $\{(\rho_i, v_i)\}_{i\in I}$ *be a finite weighted family in the Euclidean carrier chart with $\rho_i > 0$, and suppose that*

$$\sum_{i\in I}\rho_i (v_i\cdot x)^2 = t|x|^2 \qquad \textit{for every } x,$$

where $t > 0$. For $a \neq 0$, set

$$\Lambda_a(k) = \frac{2}{ta^2} \sum_{i \in I} \rho_i [1 - \cos(a v_i \cdot k)]$$

and define the sine-feature coordinates

$$\Phi_{a,i}(k) = \sqrt{\frac{4\rho_i}{a^2 t}} \sin\left(\frac{a v_i \cdot k}{2}\right).$$

Then

$$\Lambda_a(k) = \|\Phi_a(k)\|^2, \qquad \|\Phi_a(k) - \Phi_a(p)\| \leq |k - p|.$$

Consequently the nonnegative auxiliary frequency $\Omega_a(k) = \|\Phi_a(k)\|$ satisfies

$$|\Omega_a(k) - \Omega_a(p)| \leq |k - p| \qquad \textit{for all } k, p.$$

The complete primitive-port support has $t = 4$ and prefactor $1/(2a^2)$. The complete seam support has $t = 10$ and prefactor $1/(5a^2)$, reproducing Equation (1).

Proof. The half-angle identity gives $\Lambda_a(k) = \|\Phi_a(k)\|^2$. The global sine-difference bound controls each coordinate difference by $\sqrt{\rho_i/t}\,|v_i \cdot (k - p)|$. Summing the squared bounds and applying the tight-frame identity gives the contraction. The reverse triangle inequality for norms gives the frequency bound. The two support constants follow from their exact second-moment identities. The feature identity, contraction, support constants, and symbol bindings are machine-checked, and an independent exact verifier reconstructs the two finite supports [20]. □

The unit constant is a certified upper bound. No optimality theorem is used. The variables in Theorem 7.15 belong to the selected Euclidean carrier chart. A physical reading requires position and frequency maps, a clock, a field sector, wave-packet or signal- front dynamics, a frame and boost law, a finite physical scale, source-lag and detector models, and a frozen comparison contract.

Proposition 7.16 (Exact unit-domain remainder)**.** *Put $q = ak$, let $-5/9 \leq \mathcal{I}_6(n) \leq 1$, and write*

$$\widehat{\Lambda}(q, n) = \frac{1}{5} \sum_{j=1}^{30} [1 - \cos(q w_j \cdot n)].$$

The exact eighth moment is

$$M_8(n) = \frac{10}{3} - \frac{8}{15} \mathcal{I}_6(n).$$

Define

$$P_6(q, n) = q^2 - \frac{q^4}{20} + \left(\frac{1}{840} - \frac{\mathcal{I}_6(n)}{12600}\right) q^6,$$
$$P_8(q, n) = P_6(q, n) + \left(-\frac{1}{60480} + \frac{\mathcal{I}_6(n)}{378000}\right) q^8.$$

Uniformly for $0 \leq q \leq 1$,

$$|\widehat{\Lambda} - P_6| \leq \frac{7}{388800} q^8, \qquad 0 \leq \widehat{\Lambda} - P_8 \leq \frac{7}{34992000} q^{10}, \qquad \frac{19}{20} q^2 \leq \widehat{\Lambda} \leq q^2.$$

In the convention

$$\Lambda_a(k, n) = k^2 + C_4 k^4 + [B_0 + B_6 \mathcal{I}_6(n)] k^6 + O(a^6 k^8),$$

the low-momentum coefficients obey

$$C_4 = -\frac{a^2}{20}, \qquad B_0 = \frac{a^4}{840}, \qquad B_6 = -\frac{a^4}{12600}.$$

Proof. The exact edge-orbit moments through order eight reduce to the isotropic line and the unique rank-six invariant. The alternating Taylor bounds for cosine, together with the exact range of $\mathcal{I}_6$, give the two remainder inequalities and the positive quadratic bounds. The moment identity and the remainder arithmetic are independently replayable in exact arithmetic and checked by the Lean theorem chain [20]. □

Proposition 7.17 (Conditional basis-free transverse oscillator)**.** *For the nonzero coordinate dilation of Theorem 7.14 and nonzero $k \in \mathbb{R}^3$, let*

$$T_k = \{v \in \mathbb{R}^3 : k \cdot v = 0\}.$$

Then $\dim_{\mathbb{R}} T_k = 2$, and the basis-free orthogonal projector onto T_k,

$$\Pi_T(k)v = v - \frac{k \cdot v}{k \cdot k} k,$$

is idempotent. The scalar symbol $\Lambda_a(k)$ preserves T_k and acts equally on its two dimensions. If one declares the canonical oscillator generator

$$(A_T, V_T)' = (V_T, -\Lambda_a(k)A_T),$$

then

$$A_T'' + \Lambda_a(k)A_T = 0, \qquad \omega_a(k)^2 = \Lambda_a(k).$$

The spatial symbol vanishes at $k = 0$, hence its nonnegative square-root branch satisfies $\omega_a(0) = 0$. The algebraic first variation of

$$E_k = \frac{1}{2}\left(|V_T|^2 + \Lambda_a(k)|A_T|^2\right)$$

vanishes on that generator.

The proposition chooses the spatial symbol as an oscillator stiffness. It constructs no physical dynamics, covariance, clock, or particle content. The zero mode is an exact property of the declared mathematical oscillator. A physical photon statement requires the missing sector, action, clock, position, frame, quantization, and pole attachments.

There is nevertheless one exact first-order factorization of this modal oscillator that is useful for locating the remaining electromagnetic gap. Complexify T_k, and, for $k \neq 0$, define

$$D_{a,k}v = i\,\frac{\omega_a(k)}{\sqrt{k \cdot k}}\, k \times v.$$

The coefficient is the committed nonnegative square root $\omega_a(k)^2 = \Lambda_a(k)$; it is not a new kinetic symbol.

Proposition 7.18 (Exact modal Maxwell-shaped factorization and boundary)**.** *For every nonzero k, $k \cdot D_{a,k}v = 0$. On the complexified transverse fibre,*

$$D_{a,k}^2 v = \Lambda_a(k)v.$$

Consequently the opposite-sign paired generator

$$\mathcal{G}_{a,k}(E, B) = (D_{a,k}B, -D_{a,k}E)$$

obeys

$$\mathcal{G}_{a,k}^2(E,B) = -\Lambda_a(k)(E,B),$$

and preserves both modal transversality constraints. The same-sign mutation $\mathcal{G}_{a,k}^+(E,B) = (D_{a,k}B, D_{a,k}E)$ *instead squares to* $+\Lambda_a(k)(E,B)$*; whenever the spatial action on the tested amplitude is nonzero, it cannot obey the displayed oscillator equation.*

Proof. The vector triple-product identity gives $k \times (k \times v) = -|k|^2 v$ on T_k, while $k \cdot (k \times v) = 0$. Multiplication by $i\omega_a(k)/|k|$ therefore gives $D_{a,k}^2 = \omega_a(k)^2 = \Lambda_a(k)$ on that fibre. Squaring the two off-diagonal sign choices proves the paired statements. The universal identities, scalar-extension bridge, transversality receipts, and mutation control are checked in Lean; an independent off-axis Gaussian-rational replay checks both sign and normalization mutations [20]. □

This is an exact pointwise Fourier-modal, hence generally pseudodifferential, factorization. The factor $\omega_a(k)/|k|$ is momentum dependent, so the proposition does not prove a local position-space curl operator. It supplies neither a reality pairing between the k and $-k$ modes nor an assembled real field; it identifies neither amplitude as a physical electric or magnetic field. It also supplies no source-produced dynamics, U(1) potential or gauge quotient, Maxwell action, bridge from the rational seam-incidence Gauss receipts to this complex modal divergence, conserved physical current or source coupling, Lorentz covariance, continuum control, or laboratory readout. Propositions 7.20 through 7.22 commit exact finite seam-carrier counterparts of the gauge quotient, a local face-curvature action with conserved-source coupling, and the canonical Gauss solution; the construction supplies no bridge from those receipts to this complex modal divergence or any physical attachment listed above. The result depends on every stated premise and implies no empirical prediction. The construction supplies no physical local-field, gauge-action, source-current, continuum, or readout attachment.

Theorem 7.19 (Continuum Maxwell limit of the complete seam symbol)**.** *Supply one Euclidean domain* $\mathbb{R}^3$ *with Lebesgue measure, the complete equal-weight seam symbol* Λ_a *at every* $a > 0$*, and the reversible curl-pair generator above. Write* $r = |k|$*,* $\omega_a = \sqrt{\Lambda_a}$ *and* $P_T(k)$ *for the orthogonal transverse projection on both field components. Use the convention* $P_T(0) = 0$*. For* $r > 0$ *let* $\mathcal{J}(k) = \mathcal{G}_{0,k}/r$*, where* $\mathcal{G}_{0,k}(E,B) = (ik \times B, -ik \times E)$*; set* $\mathcal{J}(0) = 0$*. On the full six-component space define*

$$U_a(t,k) = I + (\cos(t\omega_a) - 1)P_T + \sin(t\omega_a)\mathcal{J}.$$

At $k = 0$ *this is the identity. These multipliers define strongly continuous unitary groups on* $L^2(\mathbb{R}^3;\mathbb{C}^6)$ *preserving real fields and both divergence-free constraints when those constraints hold initially. For every* k *and* $a > 0$*,*

$$0 \le r^2 - \Lambda_a(k) \le a^2 r^4/20, \tag{2}$$

$$0 \le r - \omega_a(k) \le a^2 r^3/20, \tag{3}$$

$$\|U_a(t,k) - U_0(t,k)\| \le \min\{2, |t|a^2 r^3/20\}. \tag{4}$$

Here U_0 *is the local Maxwell group, with longitudinal components fixed. In particular, for* $F \in H^3(\mathbb{R}^3;\mathbb{C}^6)$*,*

$$\sup_{|t|\le T} \|(U_a(t) - U_0(t))F\|_2 \le \frac{Ta^2}{20}\, \||D|^3 F\|_2, \qquad T \ge 0,$$

where $|D|^3$ *has Fourier multiplier* r^3*. For every* $F \in L^2$ *the left side tends to zero as* $a \to 0$*, without an imposed momentum cutoff.*

For the same initial datum F and the same prescribed forcing $f \in L^1([0,T];H^3)$ on every scale, the mild solutions satisfy

$$\sup_{0\le t\le T} \|F_a(t) - F_0(t)\|_2 \le \frac{a^2}{20}\left(T\,\||D|^3F\|_2 + \int_0^T (T-s)\,\||D|^3 f(s)\|_2\,ds\right).$$

Strong convergence also holds for $F \in L^2$ and $f \in L^1([0,T];L^2)$. For $f = (-j, 0)$ and a prescribed charge obeying $\partial_t\rho + \nabla\cdot j = 0$, the initially satisfied constraints $\nabla\cdot E = \rho$ and $\nabla\cdot B = 0$ propagate, in the distributional sense, at every scale and in the limit.

Proof. The complete source support has $\sum_e (k\cdot w_e)^2 = 10r^2$ and $\sum_e (k\cdot w_e)^4 = 6r^4$. The global inequalities

$$1 - x^2/2 \le \cos x \le 1 - x^2/2 + x^4/24$$

give (2) after summing the full cosine symbol. The upper cosine inequality follows by integrating $\cos x \ge 1 - x^2/2$ twice from zero and using evenness; it requires no small-argument restriction. Positivity of Λ_a is termwise. For $r > 0$, factor $r^2 - \omega_a^2 = (r-\omega_a)(r+\omega_a)$ and use $r + \omega_a \ge r$ to obtain (3); at zero both frequencies vanish. These universal inequalities and their sine/cosine propagator consequences are checked in `Lean/Screen/SeamMaxwellContinuum.lean`.

The Fourier curl is Hermitian, so $\mathcal{J}^* = -\mathcal{J}$, $\mathcal{J}^2 = -P_T$, and $\mathcal{J}$ vanishes on the longitudinal sector. The displayed formula is therefore $\exp(t\omega_a\mathcal{J})$, is unitary, and obeys the group law. Its transverse eigenvalues are $e^{it\omega_a}$ and $e^{-it\omega_a}$. Thus its difference from U_0 has norm $2|\overline{\sin(t(\omega_a - r)/2)}|$, proving (4). The evenness of ω_a and P_T, and $\mathcal{J}(-k) = \overline{\mathcal{J}(k)}$, preserve $\widehat{F}(-k) = \overline{\widehat{F}(k)}$ and hence real fields. The zero mode is fixed, not divided by its frequency. Pointwise time continuity and the bound two give strong continuity by dominated convergence.

Apply Plancherel to (4) for the H^3 bound. For arbitrary L^2 data, its square is dominated by $4|\widehat{F}(k)|^2$ and tends to zero uniformly for $|t| \le T$ at each fixed k. Dominated convergence proves the uniform strong limit. Apply the same bound inside Duhamel's integral for the forcing estimate; the $L^1_tL^2_x$ result follows with the bound $2\|f(s)\|_2$. Finally, $k\cdot(k\times v) = 0$ gives $\partial_t\nabla\cdot E = -\nabla\cdot j$ and $\partial_t\nabla\cdot B = 0$; the stipulated charge continuity gives Gauss propagation. The field assembly and functional-analytic limit are the paper argument just given, rather than additional Lean declarations. Independent matrix-exponential controls test the full propagator, constant forcing, reality, zero mode, high momenta, and sign and normalization mutations. □

The theorem recovers a local continuum field equation from the complete source-shaped operator after specifying a common spatial domain and reversible dynamics. The scale maps act on that same domain; they are not refinements of authenticated source logs. The finite-scale curl is generally pseudodifferential and no exact finite propagation cone is inferred from its frequency bound. The Markov generator also available from the seam action does not select the reversible curl-pair dynamics. The supplied current is not generated by observer repair, and no equality with the finite seam-incidence Gauss/action/readout packet is proved. Physical position, operational time, charge units, detector records, and an empirical realization of this field family require additional maps.

The two seam-carrier packages are premise-free exact finite mathematics on the committed thirty-seam, twelve-port incidence table. Write ∂ for the committed rational seam boundary, d for its exact transpose, $\langle x,y\rangle = \sum_e x_e y_e$ for the equal-weight seam pairing, and $\|F\|^2 = \langle F,F\rangle$ for the seam energy.

Proposition 7.20 (Exact rational Coulomb Green function of the committed seam Laplacian)**.** *The vertex Laplacian $L = \partial d$ of the committed incidence gives every port degree five, and an* *explicit symmetric rational Green matrix G, with integer entries over the common denominator* 180*, satisfies exactly*

$$LG = GL = I - \tfrac{1}{12}\,\mathbf{1}\mathbf{1}^{\mathsf{T}}, \qquad G^{\mathsf{T}} = G, \qquad G\mathbf{1} = 0,$$

and the kernel of L is exactly the constant loads, proved along the committed spanning tree through the energy identity. For every neutral rational port load ρ, the seam field $F_\rho = d(G\rho)$ solves the committed discrete Gauss problem $\partial F_\rho = \rho$; it is the unique Gauss solution orthogonal to every source-free cycle, and it is the unique Gauss solution of minimal seam energy, by the exact Thomson decomposition

$$\|F\|^2 = \|F_\rho\|^2 + \|F - F_\rho\|^2 \qquad \text{for every solution } F \text{ of } \partial F = \rho.$$

This supplies the canonical choice absent from the committed discrete Gauss layer of Theorem 7.13: each nineteen-dimensional solution fibre carries exactly one cycle-orthogonal point, equivalently exactly one energy-minimal point. The exact adjacent-port dipole receipt across seam zero has potential classes indexed by the pair of graph distances to the two poles, taking exactly the values

$$\frac{11}{60}, \qquad \frac{1}{20}, \qquad \frac{1}{60}, \qquad 0$$

and their negatives, with the displayed list strictly decreasing, and the single-source Green values by graph distance are

$$\frac{7}{36}, \qquad \frac{1}{90}, \qquad -\frac{7}{180}, \qquad -\frac{1}{18},$$

strictly decreasing. Finally, the committed uniform seam-repair operator acts on real port loads as $1 - \frac{1}{60}L$ for the real scalar extension of this same Laplacian, and its fixed loads are exactly the constants, that is, exactly the kernel of L, exactly the Gauss-obstruction directions.

Proof. The Laplacian matrix is derived from the committed table alone, and the Green identities are kernel-checked integer matrix identities over the denominator 180, cast exactly to $\mathbb{Q}$ and $\mathbb{R}$. The kernel characterization chains constancy along the eleven committed tree seams through the energy identity. The Thomson clauses compose the committed Gauss-fibre description with the displayed orthogonal energy decomposition. The dipole and single-source values are re-derived from the committed incidence by a two-step hop classification proved to coincide with graph distance, and the repair bridge is an exact operator identity against the committed uniform repair. The chain is machine-checked in `DiscreteCoulombGreen` (`laplacian_mul_green`, `greenMatrix_symm`, `greenMatrix_row_sum`, `laplacian_eq_zero_iff`, `coulombField_gauss`, `thomson_unique_orthogonal`, `thomson_minimum`, `dipole_value_strict_ordering`, `green_distance_strictly_decreasing`, `uniformSeamRepair_eq_id_sub_realLaplacian`, `uniformSeamRepair_fixed_iff_constant`) with no native decision procedure [20]. ☐

The same Green matrix assembles a global real Hodge-projection package on the committed seams. Because the Green matrix is global, this is not a local curvature action.

Proposition 7.21 (Global Hodge-projector action and separate Coulomb solution)**.** *Let real position-space seam fields be real-valued functions A on the thirty committed seams, and let a real port gauge function χ act by $A \mapsto A + d\chi$ through the real cast of the committed incidence. The field-strength projector $P = 1 - dG\partial$ is idempotent and self-adjoint for the equal-weight seam pairing; its kernel is exactly the image of d, and its range is exactly the kernel of ∂, the cycle space, of real dimension nineteen. The projected cycle representative PA is gauge invariant and* *determines A exactly up to gauge, and the quotient of position fields by gauge is linearly equivalent to the cycle space, in exact dimension agreement with the committed finite $U(1)^{19}$ chord-holonomy classification, of which it is the real Lie-algebra/dimension companion rather than an identification with the compact torus. For every source J on the seams, the sourced quadratic action*

$$S_J(A) = \frac{1}{2}\,\|PA\|^2 - \langle J, A\rangle$$

is gauge invariant exactly when $\partial J = 0$, in both directions; it obeys the exact quadratic expansion

$$S_J(A+h) = S_J(A) + \langle PA - J,\, h\rangle + \frac{1}{2}\,\|Ph\|^2$$

with no remainder; its first variation at A vanishes identically exactly when $PA = J$; the stationarity equation is solvable exactly when J is conserved, and then $A = J$ is a solution; any two solutions differ exactly by a gauge; and every solution is a global minimum, with equality exactly on the gauge orbit. Separately, for a neutral rational port load, the real cast of the Coulomb field of Proposition 7.20 is the unique minimal-energy solution of the Gauss problem. It is a coboundary and hence obeys $PE_\rho = 0$: it is not a stationary electrostatic sector of the displayed sourced Hodge action.

Proof. Idempotence, self-adjointness, and the kernel and range identifications are exact consequences of the Green identities and the adjointness of d and ∂; the dimension count follows by rank-nullity through the projector and agrees with the committed chord count and the committed rational cycle rank. The gauge-invariance equivalence and the displayed expansion and variational clauses are exact for the global Hodge action. The separate scalar-extended Gauss/Thomson statement of Proposition 7.20 is exact real linear algebra over the same incidence; their non-join is pinned by $PE_\rho = 0$. The chain is machine-checked in `PositionSpaceMaxwellAction`. Its component theorems are:

```
fieldProjector_idempotent, fieldProjector_selfAdjoint, ker_fieldProjector,
range_fieldProjector, positionQuotientEquivCycles, cycleSpace_finrank_eq_chord_
count, fieldStrength_gauge_invariant, sourcedAction_gauge_invariant_iff,
sourcedAction_expansion, stationary_iff, stationary_solvable_iff, solutions_
differ_by_gauge, sourced_global_minimum, fieldStrength_realCoulombField_zero,
electrostatic_join.
```

The composed theorem `positionSpaceMaxwellCoulomb_receipt` carries all clauses as a single typed conjunction. A producer re-derives the Green, Coulomb, Thomson, and repair values from its two byte-pinned Lean sources, and an independent verifier replays that exact-rational receipt with different linear algebra; neither certifies the separate real Hodge module. A pytest suite with mutation guards binds the scoped replay to `discrete_coulomb_green_receipt.json` [20]. ☐

The committed oriented twenty-face incidence supplies the local operator that the Green-built Hodge projector does not. Let $C : \mathbb{R}^{30} \to \mathbb{R}^{20}$ be its signed face–seam matrix. Each face row has three entries, each seam occurs in two faces with opposite orientations, and the boundary-of-boundary identity is $Cd = 0$.

Proposition 7.22 (Local face-curvature action and separate scalar Coulomb sector)**.** *On the committed oriented face complex, C has rank nineteen and*

$$\ker C = \operatorname{im} d.$$

The local kinetic operator $H = C^{\mathsf{T}}C$ has exactly five nonzero entries in every seam row, 150 nonzero entries in its 30×30 matrix, and diagonal entry two. For a seam current J,

$$S_J^{\text{face}}(A) = \frac{1}{2}\|CA\|^2 - \langle J, A\rangle$$

is gauge invariant under $A \mapsto A + d\chi$ exactly when $\partial J = 0$. Its stationarity equation $HA = J$ is solvable exactly for those conserved currents; stationary points are global minimizers and are unique modulo gradients.

Independently, for a port load ρ, the scalar action

$$S_\rho^{\text{scal}}(\phi) = \frac{1}{2}\|d\phi\|^2 - \langle \rho, \phi\rangle$$

has stationarity equation $L\phi = \rho$. For neutral ρ, the canonical potential $\phi = G\rho$ is stationary and globally minimizing, uniquely modulo constants. The direct-product static action therefore combines a local vector face-curvature sector and the exact scalar Coulomb sector while retaining J and ρ as different typed sources.

Proof. The rank calculation is exhibited by an explicit dual-tree section of the codimension-one face-sum space. Rank–nullity and $Cd = 0$ then identify the kernel with the eleven-dimensional gradient space. The local support counts, adjoint identity, range characterization, gauge equivalence, stationarity, and minimum statements are checked in `LocalFaceMaxwellAction`. An independent exact-rational replay reconstructs both incidences, their ranks and nullspaces, the complete support patterns, the Green solve, and the characteristic factors of CC^{T} and $\partial\partial^{\mathsf{T}}$. Erasing the signed orientation of one face boundary produces a machine-checked nonzero boundary-of-boundary control. ☐

Proposition 7.23 (Conditional temporal identities and continuity on the committed carrier)**.** *Index seam potentials $A_n \in \mathbb{R}^{30}$ and port potentials $\phi_n \in \mathbb{R}^{12}$ by a declared evolution step $n \in \mathbb{N}$, and set*

$$E_n = -(A_{n+1} - A_n) - d\phi_n, \qquad B_n = CA_n,$$

so that E sits on half steps and B on integer steps. The leapfrog Ampère law $E_{n+1} - E_n = C^{\mathsf{T}}B_{n+1} - J_n$ for a seam-current history J is the unit-step instance of the declared discrete action (Proposition 7.24). Then Faraday's law $B_{n+1} - B_n = -CE_n$ holds identically. Along the declared evolution, the Gauss constraint $\partial E_n = \rho_n$ propagates from step n to step $n+1$ exactly when the discrete continuity equation

$$\rho_{n+1} - \rho_n + \partial J_n = 0$$

holds, in both directions; the port load ρ and the seam current J remain distinct types, and this equivalence is their only bridge. The staggered quadratic form $\mathcal{E}_n = \frac{1}{2}\|E_n\|^2 + \frac{1}{2}\langle B_n, B_{n+1}\rangle$ satisfies the exact per-step balance $\mathcal{E}_{n+1} = \mathcal{E}_n - \frac{1}{2}\langle E_n + E_{n+1}, J_n\rangle$, hence is exactly conserved for $J = 0$. No positivity claim is made. The time-dependent gauge transformation $A_n \mapsto A_n + d\chi_n$, $\phi_n \mapsto \phi_n - (\chi_{n+1} - \chi_n)$ leaves E, B, the evolution law, and $\mathcal{E}$ invariant. A constant-in-time solution with $J = 0$ has $B_n = 0$, a forced-neutral load, and E_n exactly the canonical Coulomb field $dG\rho$; conversely every neutral ρ admits such a static solution. Zero-current evolution obeys the exact discrete wave law

$$A_{n+2} - 2A_{n+1} + A_n + C^{\mathsf{T}}CA_{n+1} + d(\phi_{n+1} - \phi_n) = 0.$$

Proof. The Faraday, continuity-equivalence, Gauss-propagation, quadratic-form, gauge, static-join, and wave clauses are checked in `TemporalMaxwellEvolution` as one composed receipt from a single typed antecedent bundle; the neutral-load static existence and an explicit nonstatic inhabitant are separate theorems of the same module, the inhabitant being zero initial data kicked by the boundary cycle of one committed face, with magnetic field zero at step zero and three on that face at step one. Faraday's law and the Gauss propagation use $Cd = 0$ and its exact transpose $\partial C^{\mathsf{T}} = 0$; the balance identity uses the adjointness of C and C^{T} under the committed equal-weight pairing together with the staggered magnetic cross term $\langle B_n, B_{n+1}\rangle$, which makes the balance exact. This conservation law is not a stability theorem at unit step: the committed exact spectrum contains 5 and $3+\sqrt{5}$, both above the unit-step leapfrog threshold 4, so growing modes exist. The timestep-scaled continuation with the sharp threshold is the next proposition. □

Proposition 7.24 (Action origin and sharp stability of the step-scaled evolution)**.** *Fix a real step $h > 0$ and set $E_n = -(A_{n+1} - A_n)/h - d\phi_n$, $B_n = CA_n$. Let S_h be the window sum of $\frac{h}{2}\|E_n\|^2 - h\,\mathcal{S}_{\mathrm{loc}}(J_n, A_{n+1}) + h\langle\rho_n, \phi_n\rangle$, with $\mathcal{S}_{\mathrm{loc}}$ the local sourced face action above. The exact expansion of S_h has a linear part and an explicit quadratic remainder; stationarity under variations of A vanishing at both window endpoints is equivalent to $E_{n+1} - E_n = h\,(C^{\mathsf{T}} B_{n+1} - J_n)$ at every interior step, and stationarity under variations of ϕ is equivalent to $\partial E_n = \rho_n$ at every window step. The form $\mathcal{E}_n = \frac{1}{2}\|E_n\|^2 + \frac{1}{2}\langle B_n, B_{n+1}\rangle$ obeys $\mathcal{E}_{n+1} = \mathcal{E}_n - \frac{h}{2}\langle E_n + E_{n+1}, J_n\rangle$, equals $\frac{1}{2}\|E_n\|^2 + \frac{1}{8}\|C(A_n + A_{n+1})\|^2 - \frac{h^2}{8}\|CE_n\|^2$, and, whenever $\|Cv\|^2 \le \Lambda\|v\|^2$ with $h^2\Lambda < 4$ and $J = 0$, bounds $\|E_n\|^2 \le 8\mathcal{E}_0/(4 - h^2\Lambda)$ and $\|B_n\|^2 \le 16\mathcal{E}_0/(4 - h^2\Lambda)$ at every step. On the committed carrier the sharp constant is $\Lambda = 3 + \sqrt{5}$: the five spectral projectors of CC^{T} are explicit rational matrices whose identities are kernel-checked, and the golden sector obeys $\|(N-3)u\|^2 = 5\|u\|^2$. Conversely, an eigenvector v of $C^{\mathsf{T}}C$ with eigenvalue λ and $h^2\lambda > 4$ gives the zero-current solution $A_n = r^n v$ with $r < -1$ and unbounded $\|E_n\|^2$. At $h^2\lambda = 4$, the generalized mode $A_n = (-1)^n nv$ has electric seam energy proportional to $(2n+1)^2$. On the committed carrier, for every nonzero h, uniform boundedness of electric seam energy for all zero-current data is therefore equivalent to the strict inequality $h^2(3+\sqrt{5}) < 4$; within that window the magnetic face energy is uniformly bounded as above. No bound on the gauge potential A follows, since kernel histories may shear. Explicit eigenvectors with eigenvalues 5 and $3+\sqrt{5}$ are exhibited, and the unit step is unstable. The kinetic term and the step are declared; the step index is not physical time; the construction supplies no source, frame, continuum, or readout attachment.*

Proof. Every clause is a theorem of `ScaledMaxwellStability`: the action expansion and the two stationarity equivalences by summation by parts with a delta variation at each interior node, the balance and the energy identity by the adjointness of C and C^{T} under the committed pairing, the bounds by conservation and the energy identity, the sharp constant by five explicit rational spectral projectors of CC^{T} checked by kernel decision on integer tables together with a Cauchy–Schwarz transport through C^{T}, and the unbounded modes by explicit eigenvectors and the real root $r < -1$ of $r^2 - (2 - h^2\lambda)r + 1 = 0$. The composed receipt `scaledMaxwellStability_receipt` carries the clauses from one typed bundle, with an explicit inhabitant at $h = 1/2$. □

The stability layer carries one committed certified step. The declared step $h = 4/5$ satisfies $(4/5)^2(3+\sqrt{5}) \le 84/25 < 4$, through $\sqrt{5} < 9/4$, with rational margin $16/25$; at that step the staggered form is nonnegative for every history, conserved for zero current, obeys the balance with literal coefficient $2/5$, and every zero-current solution obeys $\|E_n\|^2 \le \frac{25}{2}\,\mathcal{E}_0$ and $\|B_n\|^2 \le 25\,\mathcal{E}_0$ at every step. Any positive step strictly below the threshold yields the same certified package, and a second inhabitant at $h = 1/2$ shows the committed choice is a declared selection, not forced. A typed design-only handoff lists what a preregistered instrument run must bind: the certified instrument, the carrier constant with its Courant proof, a declared readout map, declared control histories, and a nonempty rational kill band. No run, seed, or frozen prediction exists, and the committed step also inhabits the antecedent bundle above on a nonstatic demonstration history (`CertifiedScaledStepInstrument`).

The frozen dispersion prediction of the ladder carries a schema-level arming checklist. Each prose clause of its comparison protocol, the source-derived homogeneous edge position action with equal weights, the continuous-field same-operator sector attachment, cofinal gluing, the declared finite carrier scale, coherent frame and boost transport, the frozen nuisance and coverage readout rules, and the positive exclusivity lower bound, is one named placeholder field and one atomically registered premise. The committed results prove the coefficient identities as functions of an unfilled positive scale and the control-support certificate. The module also constructs stipulated mock records by setting proposition fields to true and choosing arbitrary scale, lower-bound, and natural-number digest labels. Their nonemptiness and packaging equivalence are not physical eligibility, externality, or custody theorems. No source-derived, artifact-bound evidential inhabitant is present, but a future source scale, lower bound, or attachment is not ruled out. No comparison data enters and the frozen bytes are untouched (`DispersionArmingInterface`).

All four propositions are exact finite mathematics on the committed carrier, not a physical Maxwell theory. The equal-weight pairing coincides with the committed equal seam-counting selection, and the temporal proposition consumes the declared update, so an instrument construction built on either reads the same pairing. No physical field, charge, current, potential, or photon is identified with a laboratory object; port loads are not charges, seam values are not fields or fluxes, the Green matrix is not a laboratory potential, and the word Coulomb names the canonical discrete Gauss solution of the committed finite table only. The evolution step index is a declared parameter, not physical time, and Ampère names the declared discrete update while Faraday names the proved backward-difference identity. No Lorentz covariance, continuum control, physical source production, or laboratory readout is constructed. The construction supplies no attachment to measured clock and energy standards or to a source-produced gauge connection, current, and action, and no theorem gives stable calibrated propagation. The canonical receipt records exact rational observables and a replay template for an instrument model; it makes no empirical prediction.

Proposition 7.25 (Exhaustive fixed-design recovery calibration)**.** *Consider the twelve design rows formed by*

$$q \in \left\{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\right\}, \qquad \mathcal{I}_6 \in \left\{1, -\frac{5}{9}, -\frac{5}{16}\right\},$$

and the three-column model for C_4/a^2*,* B_0/a^4*, and* B_6/a^4*. Enumerate all* $2^{12} = 4096$ *independent sign-error vectors with amplitude* $1/200$*. Exact ordinary least squares and exhaustive* $19/20$ *interval calibration give coordinate coverage counts* $3904/4096$*,* $3892/4096$*, and* $3892/4096$*. The corresponding nonzero-signal detection counts are* $4096/4096$*,* $234/4096$*, and* $195/4096$*. The joint higher-order pair has coverage* $3892/4096$ *and detection count* $209/4096$*. The exact remainder of Proposition 7.16 cannot erase the leading-coefficient detection margin in this design.*

This is a finite synthetic calibration with no sampling error. The sign-error law is a declared dimensionless stress model. The result gives no continuous-direction coverage, detector response, nuisance model, tail calibration at discovery significance, or experimental sensitivity claim. It rejects zero for the leading coefficient throughout this design and does not determine the linked higher-order pair.

One conditional metric theorem records the exact relation available if a single physical similarity metric is attached to the normalized port frame, the primitive vertex action, and the seam-current action. Let a_{cell} be one physical cut-area element, let $s^2 > 0$ be the squared physical length represented by one response-metric unit, and suppose a port-dual sector obeys $12a_{\mathrm{cell}} = 4\pi s^2$. If the vertex action has squared step $a_v^2 = s^2$ and the seam action uses the same metric, then

$$a_v^2 = \frac{3}{\pi} a_{\mathrm{cell}}, \qquad a_e^2 = \frac{3}{\pi}\left(2 - \frac{2}{\sqrt{5}}\right) a_{\mathrm{cell}}, \qquad \frac{a_e^2}{a_v^2} = 2 - \frac{2}{\sqrt{5}}.$$

With the additional definition $a_{\mathrm{cell}} = P\ell^2$, where $\ell > 0$,

$$\frac{a_v^2}{\ell^2} = \frac{3}{\pi} P, \qquad \frac{a_e^2}{\ell^2} = \frac{3}{\pi}\left(2 - \frac{2}{\sqrt{5}}\right) P.$$

No source theorem constructs this joint attachment or identifies ℓ with a physical length. Equal normalized port weights and seam weights do not imply it.

The physical comparison routes have different dependency surfaces. A direct arrival-time comparison uses one-particle propagation. Electron and positron actions play no role in that calculation. It requires a physical position map, a physical clock, a field-sector and propagation equation, a controlled frequency remainder, a frame and boost law, cosmological transport, an emission-time model, detector timing, nuisance treatment, and an exclusive readout. A pair-production threshold or opacity comparison additionally requires physical electron and positron dispersion, energy-momentum conservation for the shared action, pair-production kinematics and interactions, a cross section, background-radiation transport, and source composition. The finite seam-current theorems supply none of these physical attachments.

Proposition 7.26 (Conditional photon-decay exclusion and electron-positron threshold)**.** *Let the nonnegative square root of (1) be attached to a physical photon frequency in the Euclidean carrier metric, with nonzero a. For $\mathbf{k} = kn$, $k \geq 0$, and $|n| = 1$, the exact finite symbol satisfies*

$$0 \leq \Omega_\gamma(k,n)^2 = \Lambda_a(k,n) \leq k^2.$$

Suppose energy and momentum compose additively in the declared frame and the electron and positron have positive mass and Lorentz-invariant positive-energy dispersion. A photon cannot decay into an electron-positron pair. For two incoming photons, any declared final-state energy within the seam-current incoming-energy budget does not exceed the Lorentz-invariant incoming-energy budget at the same momenta. The exact upper bound is global for nonzero a; the separately certified Taylor-remainder interval uses $0 \leq |a|k \leq 1$.

For a separate leading ultrarelativistic calculation, use $E_i^2 = p_i^2 + m_i^2 + \delta_{i,2}E_i^4$ through the retained order, with $i \in \{\gamma, +, -\}$, $m_\gamma = 0$, $m_+ = m_- = m_e$, E the hard-photon energy, and ϵ the soft-photon energy. Treat the soft background photon as Lorentz invariant at leading order and allow independent dimension-six coefficients $\delta_{\gamma,2}, \delta_{+,2}, \delta_{-,2}$. Head-on incoming momenta, collinear outgoing momenta, additive conservation, and a fixed positron energy share $0 < x < 1$ give

$$\left[\delta_{\gamma,2} - x^3\delta_{+,2} - (1-x)^3\delta_{-,2}\right] E^4 + 4\epsilon E - \frac{m_e^2}{x(1-x)} = 0.$$

At $x = 1/2$, the visible coefficient is

$$\delta_{\gamma,2} - \frac{\delta_{+,2} + \delta_{-,2}}{8}.$$

This map has rank one and a two-dimensional coefficient fiber. On the Lorentz-invariant charged-lepton branch, $\delta_{+,2} = \delta_{-,2} = 0$, and for $m_e > 0$, $0 < x < 1$,

$$\frac{m_e^2}{x(1-x)} - 4m_e^2 = \frac{m_e^2(2x-1)^2}{x(1-x)}.$$

Hence equal sharing is the unique global maximizer of the leading head-on, collinear threshold residual on this branch.

On the separate universal negative-coefficient branch $\delta_{\gamma,2} = \delta_{+,2} = \delta_{-,2} = -d$, $d > 0$, put $u = x(1-x)$. The same residual vanishes exactly when

$$\epsilon = \frac{m_e^2}{4Eu} + \frac{3dE^3}{4}u, \qquad 0 < u \leq \frac{1}{4}.$$

Every such value is bounded below by the reciprocal-linear envelope. Equal sharing minimizes in the endpoint regime $3dE^4 \leq 16m_e^2$; above the transition $E_t = 2\sqrt{m_e}(3d)^{-1/4}$, the interior balance $u_* = m_e/(\sqrt{3d}\,E^2)$ gives $\epsilon_{\mathrm{env}} = (\sqrt{3}/2)m_e E\sqrt{d}$. A soft energy below this envelope cannot solve the declared leading residual at any open share. At and above the transition the balanced equality witness is itself realized by an open share, so the envelope is the constrained physical-share minimum there.

Proof. The elementary inequality $1 - \cos y \leq y^2/2$, together with the exact second moment of the thirty normalized seam directions, gives $\Lambda_a(k,n) \leq k^2$ in the same Euclidean metric used by the carrier completion. A positive-mass lepton pair with Lorentz-invariant positive-energy dispersion has energy strictly greater than the magnitude of its total momentum. The triangle inequality gives the decay exclusion and the incoming-energy-domain inclusion. Expanding the three dispersion relations through first order in the dimension-six coefficients and enforcing the declared head-on momentum balance gives the displayed threshold equation. Its equal-share row is $(1, -1/8, -1/8)$, which has rank one and a two-dimensional kernel. The displayed square identity proves the Lorentz-invariant-lepton optimizer. The common-coefficient cubic identity reduces the coefficient to $-3dx(1-x)$. The bound $0 < u \leq 1/4$, an exact square-factor AM–GM identity, its balanced witness, the endpoint comparison, transition, and no-solution statement prove the universal-branch envelope; a separate existence theorem closes physical-share attainment above the transition. The use of E_i^4 in this threshold equation is the retained first-order, ultrarelativistic rewrite of the carrier's p_i^4 correction, not an exact identity of the finite dispersion. The complete identities and metric typing are checked in Lean, with deterministic receipt calculations for the coefficient algebra and representative roots [20]. □

The conditional kinematic theorem supplies no interaction, rate, or detector content for physical photons or charged leptons. The fixed-share leading expansion retains independent charged-lepton coefficients. The Lorentz-invariant-lepton specialization has an exact global share optimizer, and the common negative-coefficient specialization has an exact share envelope. The calculation supplies neither the general independent-coefficient share optimization nor the full anisotropic minimization over outgoing three-momentum. It does not identify the three dispersion coefficients separately and supplies no vertex, rate, opacity, source, shower, detector response, or empirical exclusion.

Theorem 7.27 (Icosahedral face-corner bundle)**.** *On the icosahedral screen-sieve branch, the twenty outward-oriented triangular faces form the homogeneous orbit*

$$\mathsf{F}_{20} \simeq A_5/C_3.$$

The stabilizer of a face acts regularly on its three corners. Hence the face orbit carries an A_5-associated bundle of local three-dimensional permutation fibers with cyclic shift $R^3 = 1$. A fiberwise Hermitian operator C that is transported A_5-equivariantly across the face orbit and commutes fiberwise with this shift has the circulant form

$$C = a1 + bR + \bar{b}R^2,$$

and its unordered real spectrum is independent of the chosen face representative. Reversing the face orientation exchanges R and R^2 and preserves that unordered spectrum.

Proof. The icosahedron has $(V, E, F) = (12, 30, 20)$. The orientation-preserving icosahedral group A_5 is transitive on the outward-oriented faces. By orbit–stabilizer the face stabilizer has order $60/20 = 3$, hence is C_3, and its nontrivial rotations cyclically permute the three corners. The commutant of the regular cyclic shift on one fiber is the circulant algebra; Hermiticity gives the displayed coefficients. The stated A_5-equivariance transports that operator to every other face by conjugation and therefore preserves the spectrum. Without this equivariance, independent coefficients could be chosen on different faces. Orientation reversal complex-conjugates the transported presentation and again preserves the unordered real eigenvalues. □

Remark 7.28 (Face-carrier boundary). The theorem supplies local geometric C_3 fibers, not one canonical global three-dimensional matter-family space. The sixty face-corner flags form a free transitive A_5-set, whose linearization is the regular sixty-dimensional A_5 representation. A physical family claim therefore requires a quotient-visible section, connection, or intertwiner from this bundle to the relevant matter multiplicity space. Cyclic covariance alone also leaves a, $|b|$, and $\arg b$ free. In particular this theorem does not emit a charged-family phase, determinant normalization, or mass coordinate.

Remark 7.29 (Engineered charged face-quotient model at fixed cutoff). An auxiliary finite construction shows that a stipulated charged face-quotient packet is algebraically nonempty. Inside a bounded patch it realizes eight connected matrix-register graphs, 6,467 matrix units, eight declared paths, rank-one local events, and a central accepted/rejected readback. For one event projection P, write the central two-point record algebra as

$$\mathcal{D}_2 := \mathbb{C}|a\rangle\langle a| \oplus \mathbb{C}|r\rangle\langle r|.$$

The binary event-to-record channel is

$$\mathcal{B}(H_r) \longrightarrow \mathcal{B}(H_r) \otimes \mathcal{D}_2, \qquad \rho \longmapsto P\rho P \otimes |a\rangle\langle a| + (I-P)\rho(I-P) \otimes |r\rangle\langle r|.$$

Sixty proper-icosahedral charts verify the declared graph intertwiners, and the evidence artifact verifies normalized-trace invariance under inert ancillary stabilization. The construction proves schema existence plus a fixed-cutoff bridge from noncentral event to central public record. The register dimensions, path automaton, coupling character, grading, unit clock, and scalar response are authored inputs. The evidence checker does not enforce every source-law and provenance field, and its negative-control wrapper can count an unrelated nonzero exit as a successful rejection. Path exhaustion is internal to that automaton; inert stabilization supplies no cofinal physical screen refinement. The construction has no precommitted source packet excluding target dependence. A global mutually exclusive response law, physical charged-source selection, recovery dynamics, family attachment, and a pole map are outside this model. A separate conditional nature/pole interface shows how a supplied chiral three-family carrier and exact renormalized kernel would transport the face operator to a charged singularity readout. One premise identifies the physical Yukawa response with the face response, and another identifies the Dyson readout. It therefore adds a precise downstream certificate format without deriving either attachment from this screen model.

Remark 7.30 (Hardware-to-QFT export boundary). A screen or hardware evidence bundle exports only the bounded observer-like object it actually instantiates: local state, ports and boundaries, readback, records, repair or feedback moves, and public evidence. Even a complete echosahedral selector receipt does not thereby export a source-selected G_6 action, chiral measure or constrained Hamiltonian, BV/ST restoration transcript, dressed-current amplitude, nonperturbative observable tower, or resonance sheet. The quantization-step implications for those typed inputs are proved in the compact and particle papers; construction of the inputs is a separate producer problem.

On the certified echosahedral lineage, the declared integer counting and normalized readback-cost realization gives the unit lines. Oriented incidence gives the inverse pairs, proper A_5 action, and rank-three frame. On other declared cost/selector branches, the variational screen-sieve theorem provides a sufficient alternative. These results give the reference patch enough boundary structure to support nontrivial overlap comparison, symmetry tests, verifier-shadow behavior, and cross-patch routing while remaining small enough for explicit evidence records. The theorem selects the exposed interface architecture. The dimension of the observer's internal algebra belongs to a separate carrier theorem over ports, records, repair maps, and refinement.

Remark 7.31. The word "echosahedral" is intentionally implementation-facing. In a mathematical theorem, one uses the tuple above. In a public hardware evidence bundle, one may instantiate the tuple using a specific twelve-port body, wiring map, controller, firmware hash, readout protocol, and evidence manifest.

8 Toroidal Recurrence Is Local

Toroidal hardware supplies a local recurrence and mode-competition surface. A toroidal subchannel is a recurrent internal path inside a patch or between a bounded set of ports. It can recycle boundary information, support settling dynamics, and expose winding-like or phase-locking observables.

Global scalar order parameters are often misleading in recurrent local systems. A toroidal or ring-like patch may fail to show global alignment while selecting a stable winding class, a local coherence island, or a persistent twisted state. The validation metrics for toroidal subchannels should include:

1. local coupling matrices, with total brightness treated as a secondary summary;
2. recurrence and ring-diversity statistics, with global response treated as a secondary summary;
3. winding-sensitive or phase-lock-sensitive summaries when phase data are available;
4. matched controls that distinguish chamber-mediated dynamics from host-side filtering;
5. exact-verifier receipts for task-level claims.

Proposition 8.1 (Conditional phase-lock synchronization bridge)**.** *For every routed interface $e = ((i,a),(j,b))$, suppose a source-produced phase process provides an independently calibrated time parameter, frequency entrainment, a stable declared phase offset, and held-out controls. Suppose further that the locked phase selects a commensurability map for the exposed port packets and that every resulting write enters the consensus ledger through semantic-dependency-complete read sets, atomic revalidation, coherent union-collar payloads, and the quotient local-diamond and repair-completeness checks. Then the phase process supplies a physical synchronization parent for the federated consensus theorem.*

Proof. The phase certificate fixes when the two port readouts are commensurable. The other hypotheses are exactly the premises under which the accepted transaction relation terminates and has the declared schedule-independent quotient normal form. The phase process supplies the interface timing and comparison map; the consensus theorem supplies public agreement. □

Phase locking without the transaction, record, and clock packet is compatible with two oscillators that entrain while reporting different semantic records. It is therefore neither an observer test nor a consensus theorem by itself. No physical phase-lock receipt is asserted in this paper.

The OPH interpretation is simple: toroidal recurrence supplies local memory and mode competition inside the patch federation. Echosahedral symmetry supplies a reference and consensus geometry. The universe, at this level of description, is a federated repair system with many local carriers.

9 Patch Federation and Overlap Synchronization

Given a federation of echosahedral patches, overlaps are formed by routing ports into declared interface pairs or interface hyperedges. For an edge $e = \{(i,a),(j,b)\}$, the exposed packets are

$$x_{i,a} = \rho_{i,a}(s_i), \qquad x_{j,b} = \rho_{j,b}(s_j),$$

and the edge mismatch is a nonnegative function

$$\Phi_e(x_{i,a}, x_{j,b}) \geq 0.$$

The total visible mismatch is

$$\Phi(s) = \sum_{e \in E} \Phi_e\big(\rho_{i,a}(s_i), \rho_{j,b}(s_j)\big).$$

Accepted repairs do not increase the declared touched-overlap mismatch, and exact fixed-cutoff branches lower the relevant mismatch unless the local visible datum is repaired. The implementation is a bounded patch operation:

1. read the exposed overlap packets;
2. compare them through a declared commensurability map;
3. choose an allowed local update or rerank move;
4. write a record of the move;
5. expose the resulting port packet;
6. let the exact verifier decide any high-level task claim.

Proposition 9.1 (Federated synchronization contract). *Suppose a finite patch federation has a declared mismatch functional Φ with values in a finite ordered set and a finite repair menu. Treat a repair of a locally repaired datum as the identity, not as a step. Assume every accepted nonidentity repair strictly lowers Φ, and every state outside the declared visible local normal forms admits such a repair. Then every maximal accepted-repair sequence is finite and ends at a visible local normal form. If the union-collar gluing and repair-completeness hypotheses of the OPH consensus theorem hold together with protected-support-complete transactions, protected-conflict-complete dependencies, and an independently checked quotient local diamond, the terminal physical observable state is schedule-independent on that carrier. The carrier is thereby a finite constraint-code implementation surface; no QECC distance, min-cut formula, spectral mixing rate, or wall-clock liveness bound follows unless the corresponding extra certificate is supplied.*

Proof. The first conclusion is the finite Lyapunov argument: the declared score strictly decreases on every allowed step, so an infinite accepted-repair sequence is impossible; completeness makes a terminal state a declared local normal form. The second conclusion consumes the quotient-local confluence package of the OPH consensus theorem. Atomic commits alone do not supply its protected-support, coherent-payload, or local-diamond receipts. The result applies to the federated carrier and visible interface defined above. □

The schedule-independent conclusion above is a same-source statement. Equality of endpoints reached from different interiors exposing the same protected boundary requires injectivity of the boundary map on the consistent quotient, as characterized in Ref. [13]; weak normalization and all-schedule liveness are separate hypotheses. The same reference shows that a total exact collar-preserving local repair exists only when every admissible collar value has a locally consistent extension. Neither condition follows from finite descent alone.

10 Records, Observers, and Checkpoints

An observer is not added as a metaphysical extra. In this paper an observer is an operational pattern in a patch or patch subfederation with persistent access to:

1. an observer-facing local algebra;
2. a record algebra;
3. a stable readout/update interface;
4. enough checkpoint data to define future observer-accessible probabilities.

Definition 10.1 (Federated observer checkpoint)**.** *For an observer-supporting patch subfederation O, a checkpoint at semantic cut D is*

$$\mathrm{Chk}_O(D) = \big(\mathcal{R}_O(D), \rho_O^{\mathrm{acc}}(D), \mathfrak{I}_O^{\mathrm{ext}}(D), \mathcal{L}_O^{\mathrm{sem}}(D), \mathfrak{B}_O(D)\big),$$

where $\mathcal{R}_O(D)$ is the observer-accessible record algebra, $\rho_O^{\mathrm{acc}}(D)$ is the state restricted to observer-accessible records and visible interfaces, $\mathfrak{I}_O^{\mathrm{ext}}(D)$ is the external port-interface tuple, $\mathcal{L}_O^{\mathrm{sem}}(D)$ is the semantic future-law class for the same physical instruments and event algebra, and $\mathfrak{B}_O(D)$ is the public or internal provenance bundle needed to replay the checkpoint at the declared accuracy. Worker IDs, queue positions, retry counters, repair-cycle indices, timestamps, and packet latencies live in $\mathfrak{B}_O(D)$ unless a branch declares them physical inputs; they are not observer identity or observer time.

Theorem 10.2 (Checkpoint continuation at fixed cutoff)**.** *If two federated observer checkpoints agree exactly on $\mathcal{R}_O(D)$, $\rho_O^{\mathrm{acc}}(D)$, $\mathfrak{I}_O^{\mathrm{ext}}(D)$, and $\mathcal{L}_O^{\mathrm{sem}}(D)$, then they induce the same future probability law on the observer-accessible event algebra. If their accessible states differ by trace distance at most ε, then the induced future history laws differ in total variation by at most ε.*

Proof. All future observer-accessible probabilities are computed by applying the same completely positive update maps and the same event-readout maps from the same semantic future-law class to the same accessible algebra and external interface data. Exact equality gives equality of all future event probabilities. In the approximate case, contractivity of trace distance under completely positive trace-preserving maps gives the stated total-variation bound. □

Definition 10.3 (Public checkpoint capacity interface)**.** *For a finite federated checkpoint family, the capacity-facing export is the packet*

$$\mathsf{PubChk}_{r,D} = \big(\{X_O\}, \{X_e\}, \{r_{Oe}\}, \mathsf{Reach}, \mathfrak{P}, \mathfrak{K}, \mathcal{H}_{\mathrm{cap},r,D}, \{P_x\}, \mathsf{Ext}, \mathsf{Ref}\big).$$

Here $X_O = \mathrm{At}\,\mathcal{R}_O$ is the local central-record atom set, X_e is the interface-record atom set, and $r_{Oe} : X_O \to X_e$ is the completed-instrument atom readout. The semantic event DAG Reach *witnesses* *endogenous public-record reachability;* $\mathfrak{P}$ *freezes the authorized collective, universal-local, or quorum publicness policy; and* $\mathfrak{K}$ *contains the source-derived joint checkpoint kernels with the required local marginals. The declared capacity carrier has* $\dim \mathcal{H}_{\mathrm{cap},r,D} = D$*, and* P_x *is the nonzero record projection for reachable public class* x*. The final two entries are capacity-extension and fixed-capacity refinement packets, kept as different types.*

The packet fields are an interface contract, not automatic outputs of a local checkpoint. In particular, the generic visible algebra maps $\pi_{i,e} : \mathcal{A}_i \to \mathcal{I}_e$ do not automatically supply the record-atom maps r_{Oe}, and observer-by-observer future-law marginals do not determine the joint kernels $\mathfrak{K}$. The source must therefore provide record-atom restriction maps, endogenous public-record reachability, the frozen publicness policy, a globally coupled checkpoint family, and a faithful representation of public records on the capacity carrier. The fixed-cutoff checkpoint and central-record theorems are their immediate parents. The correctable-code capacity and finite-size closure theorems are carried by Ref. [17].

On that synthesis surface the packet defines the multiplicative readback $M_0(q) = \alpha(G_q)$, where G_q is the compound confusability graph of reachable public records. When the complete terminal fiber scalarizes, $M_0(\mathfrak{U}_N) = \widehat{F}_{r,0}(e^N)$, and the universe-level equation is

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

Thus M_0 counts records and N is the logarithmic capacity; the checkpoint packet supplies the finite data needed to evaluate the count.

11 Central Records and Measurement

The measurement package is a finite central-record package. Records are exposed by observer-facing patch subfederations.

Let $\mathcal{Z}_{\mathrm{rec}}(t)$ be the commutative algebra generated by the completed, observer-accessible record projectors at cycle t. An event E is a projector in $\mathcal{Z}_{\mathrm{rec}}(t)$. Declare a finite-dimensional $*$-algebra $\mathcal{A}_t$, a normalized state ρ, and an embedding of $\mathcal{Z}_{\mathrm{rec}}(t)$ into its center. These quantum representation data are inputs rather than outputs of the repair process. The declared operational measurement rule is:

$$\Pr(E) = \mathrm{Tr}(\rho E), \qquad \rho \mapsto \frac{E\rho E}{\mathrm{Tr}(\rho E)} \quad \text{when } \Pr(E) > 0.$$

Theorem 11.1 (Conditional federated central-record measurement)**.** *On a fixed-cutoff federated patch carrier, suppose a completed write/verify slice exposes a finite commutative central record algebra* $\mathcal{Z}_{\mathrm{rec}}(t)$ *with the declared algebra-state representation above. The Born probability and declared Lüders conditioning rule above define the operational measurement package on the observer-accessible event surface. Re-reading the same completed record event has probability one, conditional on that record event.*

Proof. The theorem is the standard finite-dimensional algebra-state argument. Because all declared record events commute and belong to the observer-accessible center for the completed slice, they define an ordinary finite classical event algebra. Probabilities are the supplied Born traces on that event algebra, and conditioning on an event uses the declared Lüders update. The effect projectors alone do not select this update; same-effect non-Lüders instruments exist. After conditioning on E, the same central projector E is true with probability one. Each finite-matrix step of this argument (commutativity and centrality of the record span, Born traces, the Lüders update with its fixed-point law, and the collapse of record conditioning to the normalized projector) is machine-checked in the companion Lean 4 event-algebra development [14]. ∎

11.1 Record-conditioned modular cap responses

For a quotient-visible record token i seen by observer O, let $P_{i,O}$ be its central, or declared approximately central, record projector with $\omega_O(P_{i,O}) > 0$. The conditioned record state is

$$\omega_{i,O}(A) := \frac{\omega_O(P_{i,O} A P_{i,O})}{\omega_O(P_{i,O})}.$$

At fixed cutoff this is the usual Lüders state

$$\rho_{i,O} := \frac{P_{i,O}\rho_O P_{i,O}}{\mathrm{Tr}(\rho_O P_{i,O})}.$$

For a cap C, let $M_{C,0,O}$ be the declared cap-response probe observable and set

$$M_{C,t,O} := \sigma_t^{C,O}(M_{C,0,O}), \qquad R_i(C,t,O) := \omega_{i,O}(M_{C,t,O}).$$

On the support-visible geometric branch, the dimensionless modular-parameter convention is the same 2π-normalized cap flow used by the spacetime and Einstein paper:

$$\sigma_t^{C,O} = \alpha_{\lambda_C(2\pi t)}.$$

The point-source cap-response model is a branch hypothesis or finite receipt, not an automatic consequence of a record projector. A passing source-localization branch must exhibit

$$R_i(C,t,O) = b(C,t,O) + a(i,C,t,O)\,\psi_{C,t,O}\left(\frac{\eta(X_i(t), n(C,t,O))}{R_H}\right) + \xi(i,C,t,O),$$

where gains, baselines, kernel shape, cap normal, modular normalization, and the total error budget are declared. The approximate centrality defect of $P_{i,O}$, calibration error, modular transport error, kernel mismatch, finite-record noise, and point-model error all contribute to the single localization error norm σ_{record} used by the compact localization theorem.

Two carrier presentations that induce the same quotient-visible record token, conditioned record state, cap probe, cap geometry, modular-parameter normalization, and response calibration must produce the same calibrated response vector and the same H^3 localization ball. A fitted point that does not clear held-out cap residuals, mixture controls, and the declared error bound is a failed or approximate point-source branch, not a populated-H^3 theorem receipt.

12 Edge Sectors and Casimir Continuation

The edge heat-kernel/Casimir package is a fixed-cutoff theorem package tied to an overlap collar with a finite exposed sector algebra.

At fixed cutoff, let α label a finite set of exposed edge sectors on one declared overlap collar. Suppose the local thermalized edge dynamics has stationary weights

$$\pi_\beta(\alpha) = \frac{d_\alpha e^{-\beta C_2(\alpha)}}{Z(\beta)}.$$

For finite groups, $C_2(\alpha)$ is the declared sector penalty or finite-group Casimir surrogate. For compact groups, the Peter–Weyl lift belongs to the companion compact-gauge construction and cannot be supplied by hardware evidence.

Theorem 12.1 (Fixed-cutoff edge-sector law)**.** *If a declared finite overlap collar has sector labels* α*, degeneracies* d_α*, and a local repair/thermalization generator whose detailed-balance stationary law is* $\pi_\beta(\alpha) \propto d_\alpha e^{-\beta C_2(\alpha)}$*, then the collar gives the fixed-cutoff Casimir edge law used by the compact-gauge construction. The compact-group heat-kernel lift requires the separate companion theorem.*

Proof. This is a finite Markov or finite instrument stationary-measure statement on the declared sector algebra. The theorem gives only the finite overlap law. The compact-group lift uses the Peter–Weyl construction and normalization conventions supplied by the companion compact-gauge construction. □

13 Bell/CHSH Event Surfaces

The Bell package is fixed-cutoff and event-surface-local. A federated carrier may contain two observer-facing wings L and R with commuting setting and outcome records on one compare slice. If the source-specified joint law is the usual two-wing quantum law, the CHSH and Tsirelson statements are carried on that declared event algebra. Neither the two-wing representation nor the joint quantum law follows from the commuting record indicators alone.

The binary-icosahedral spinor adapter gives one exact finite candidate. Its unique invariant singlet and an incidence-defined (120)-row setting family give

$$|S_{\mathrm{CHSH}}| = 1 + \frac{3}{\sqrt{5}} > 2.$$

All (720) Pauli covariance identities and the joint law are checked by an independent exact verifier. The complete twelve-port census has (960) maximizing quadruples, so the (120)-row family is not selected uniquely. The source packet contains no completed two-wing record instrument or source-selected setting mechanism. The result is a finite projective candidate, without a physical Bell prediction.

Laboratory echosahedral hardware carries its own public evidence rule for Bell-style claims. The theorem package here is a finite event-algebra statement inside the OPH microphysics surface.

14 Relation to Yang–Mills

The compact-gauge theorem gives the Yang–Mills repair-gap argument. Echosahedral geometry supplies:

1. the fixed-cutoff patch, overlap, record, and repair interface;
2. the finite edge-sector Casimir law on declared collars;
3. the local repair semantics that the compact-gauge branch reads in controlled quotient form.

The finite cylinder system and its projective weak-* / GNS extraction are proved under the compact-gauge premises. Identification with the four-dimensional Yang–Mills state, reflection-positive Osterwalder–Schrader reconstruction, noncollapse, and equality between the Yang–Mills gap and the repair gap require separate receipts. Hardware evidence can test implementation discipline. It does not supply the Clay-admissible construction. A public evidence bundle that is used for any continuum or Lorentzian claim must therefore include the multiresolution regulator certificate: factor manifest, presentation-circuit hash, refinement/expectation identity defects, transported-state errors, renormalized-observable tails, and positive-transfer or reflected-Gram receipts. Conventional free-field or lattice-gauge vacuum baselines used for calibration are reference ensembles only. An OPH-native vacuum identification requires the quotient-ensemble selector, source Euclidean slab data, and transfer/reflection-positive reconstruction under the compact-gauge premises.

15 Relation to Hadrons and Strong-Binding Execution

First-principles hadron masses require a working OPH strong-binding construction with:

1. Ward-projected hadronic spectral data;
2. provenance tying every spectrum to body, controller, firmware, geometry, and scorebook;
3. finite-volume, continuum, chiral, and production-systematics fields where applicable;
4. exact replay or verification receipts for the calculation being claimed;
5. an explicit scope boundary for surrogate or calibration runs.

Echosahedral or related hardware can be described as a candidate construction family only through public evidence bundles. Without those bundles, it is an architectural target and a design motivation. A first-principles hadron theorem requires the evidence listed above.

16 Validation criteria

Validation has two independent parts.

16.1 Mathematical and Digital Calibration

The octahedral $\mathbb{Z}_2/S_3$ finite-group model is a digital calibration model. It gives exact finite groups, explicit patch covers, controlled defects, frustrated cycles, record writes, repair schedules, and negative controls.

> The calibration model checks the finite interface logic. The primary physical picture is the federated patch-carrier architecture.

16.2 Public Hardware Evidence

A public hardware run may support only the support level its evidence bundle can justify. The minimal claim form is:

> Module set M produced candidate enrichment or reproducible readout signature on benchmark T, under controls C, and exact verifier V accepted the reported hits.

The rejected form is:

> The optical chamber solved the hard problem or proved OPH.

Required checks include dark baseline, low-power sweep, coupling matrix, discharge-timing or MDD trace where applicable, ring-diversity or recurrence-sensitive metrics for toroidal bodies, duplicate-body checks, symmetric-reference or echosahedral shadow checks, exact-verifier receipts, and negative controls against hidden duplicate amplification or host-side filtering.

Material, plasma, and topological-phase claims need the same discipline. The evidence bundle must name the physical quotient, source action or repair ledger, readouts, controls, and physical-identification conditions. A carrier that passes self-reading tests certifies the carrier branch only; it does not by itself certify a material selector, nuclear yield, delivered power, or condensed-matter order.

17 Finite Packet Source Bridge Schema

Cosmology-facing evidence bundles may not infer stress data from scalar source rows. A packet-source bridge exported from the microphysics layer must carry the primitive data needed by the finite covariant parent contract:

$$(\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \mathsf{Read}_r).$$

In practical evidence bundles this means:

1. finite cell/face incidence, four-volumes, normals, areas, causal adjacency, and parallel transport maps;
2. metric and tetrad data with the declared $(-+++)$ convention;
3. packet sectors, local momenta, masses, positive invariant weights, occupations, and mass-shell residuals;
4. signed face fluxes and reaction-channel stoichiometry with transported four-momentum residuals;
5. quotient maps, local-frame covariance checks, and restriction/refinement maps;
6. readout fields for stress moments, variational stress, exchange currents, and gauge-invariant or rest-frame perturbation variables.

The corresponding finite evidence receipt is residual-based. A row labelled "recipient", a raw Newtonian/synchronous gauge comparison, a producer-declared detailed-balance boolean, or a frozen likelihood hash is not a substitute for primitive packet, stress, exchange, causal-response, and refinement evidence.

18 Finite Quotient Ensemble Compatibility

Finite quotient ensemble theorem. Fix a finite regulator r. The physical presentation space is Σ_r, the presentation redundancy groupoid is Γ_r, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior ν_r on Q_r and set $\mu_r = (n_r)_\# \nu_r$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any retraction $N : Q \to Q_{\mathrm{nf}}$ onto a subset $Q_{\mathrm{nf}} \subseteq Q$, that is, any map whose restriction to Q_{nf} is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#}\mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X. If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q, positive m, and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' : c_{sr}(q') = q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q. Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#}\mu_{k+1} - \mu_k\|_{\mathrm{TV}}\,,$$

then

$$\|(c_{nr})_{\#}\mu_n - \mu_r\|_{\mathrm{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0}\left[\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r\right]$$

to equal $\kappa_{sr}(\theta)+R_{sr}(\theta)\cdot F_r(q_r)$. If the residual is uniformly bounded by ε, the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_{\#}\mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\widetilde{P}(x, y)$ descends to Q_r only when

$$P_Q(q, q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C (I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C) f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound γ_*, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt.

Finite evidence accuracy. For bounded coarse observables O, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \leq \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to ϵ_{ref}, then

$$\left| \mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O] \right| \leq 2\|O\|_\infty (\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

Physical-vacuum requirement. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \geq 0$, local potential V_r, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q')(f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac}. A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

Primordial and cosmological prediction firewall. A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance.

Interpretation classes and required receipts. Every ensemble-facing run records its ensemble id, interpretation class, regulator, representative schema, gauge action, canonicalizer, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are

seed, proposal, or repair noise,
conventional reference ensemble,
OPH-native quotient ensemble,
OPH physical vacuum,
OPH primordial field,
observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

19 Conclusion

OPH microphysics is a federation of finite observer patches. The sphere is a regulator and symmetry chart for observer-facing cuts. A_5-icosahedral and E_8-type structure belong to the geometry data and representation-closure data. The echosahedral patch is the reference local interface: a bounded multi-port patch with symmetry, records, readout, repair, and checkpoint data. Toroidal subchannels supply local recurrence and winding-sensitive dynamics. The mathematical outputs are fixed-cutoff patch-net embedding, an edge-sector Casimir law, central-record measurement on a declared algebra-state representation, declared two-wing Bell/CHSH event surfaces, and checkpoint/restoration.

The carrier architecture is physical data up to its declared quotient. Hidden labels and material presentations may vary, while the visible incidence, response, repair, record, clock, and refinement signature may not be discarded. The local twelve-port theorem constrains the A_5-current branch. A separate carrier-to-support bridge constrains the spherical Lorentz/Einstein branch. The consensus normal form is their common finite hinge. Operational observer-like behavior requires readback, durable records, feedback or repair, prediction/control, and checkpoint continuation. The source-derived incidence nerve passes those observer controls together with twenty nonvacuous face cocycles and a refinement-natural oriented support limit. The finite source inhabitation result adds an exact informational event complex, observer-atlas transitions, a seam-sign layer, local section spaces, and finite operator identities. Its absent continuum identification prevents it from supplying the physical source-causal continuum, BW/KMS, physical-scale, or laboratory bridges; those bridges are premises wherever a stronger reading consumes them. Complete reversible port response and endogenous overlap transport force the local Standard Model gauge Lie algebra. The declared matrix current and matter action are conditional. Their common $\mathbb{Z}_6$ kernel is exact for the declared tensor table. The twelve-port orbit carries an exact multipole fixed point: ranks one through five annihilate, and the unique rank-six invariant survives with a certified sixty-two-point Morse census and a universal least-rank residue on the invariant carrier class. On the declared equal-weight cosine branch, the full kernel has exactly the same sixty-two stationary directions for $0 < |ak| \leq 1$. Selecting the physical global form, identifying laboratory currents, attaching the finite response band to physical matter poles, and realizing the construction in a continuum quantum field theory all require input this paper does not supply.

Hardware can guide the architecture and supply public evidence through hash-stable evidence bundles. The mathematical conclusions stand on finite algebras and their declared branch assumptions.

A Evidence Bundle Sketch

A minimal hardware evidence bundle should use a structure like:

```
evidence/hardware/<bundle-id>/
  manifest.json
  README.md
  body/
    mesh_hashes.txt
    photos/
    measurements.csv
  controller/
    firmware_sha256.txt
    wiring_map.csv
  calibration/
    dark_scan.csv
    low_power_sweep.csv
    coupling_matrix.csv
    mdd_trace.csv
    ring_diversity.csv
  task/
    task.json
    scorebook.json
    candidates.jsonl
    verifier_receipts.jsonl
  controls/
    shuffle_replay.jsonl
    abba_controls.csv
    negative_controls.md
  claim.md
```

The manifest should state the strongest supported result and its exclusions. The evidence bundle supports only that stated result.

B Digital Calibration Compatibility

The octahedral $\mathbb{Z}_2/S_3$ build has the following reading rule:

1. it validates finite patch/overlap/record/repair bookkeeping;
2. it tests frustrated-cycle and defect behavior in an exact digital setting;
3. it calibrates the edge-sector law on a finite declared interface;
4. it supplies calibration data for the interface layer;
5. it leaves the physical carrier role with the federated echosahedral patch architecture.

References

[1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59**, 521–524 (1987). https://doi.org/10.1103/PhysRevLett.59.521.

[2] S. Surya, *The causal set approach to quantum gravity*, Living Rev. Relativ. **22**, 5 (2019). https://doi.org/10.1007/s41114-019-0023-1.

[3] L. Bombelli and D. A. Meyer, *The origin of Lorentzian geometry*, Phys. Lett. A **141**, 226–228 (1989). https://doi.org/10.1016/0375-9601(89)90474-X.

[4] G. Brightwell and R. Gregory, *Structure of random discrete spacetime*, Phys. Rev. Lett. **66**, 260–263 (1991). https://doi.org/10.1103/PhysRevLett.66.260.

[5] S. Major, D. Rideout, and S. Surya, *On recovering continuum topology from a causal set*, J. Math. Phys. **48**, 032501 (2007). https://doi.org/10.1063/1.2435599.

[6] S. Major, D. Rideout, and S. Surya, *Stable homology as an indicator of manifoldlikeness in causal set theory*, Class. Quantum Grav. **26**, 175008 (2009). https://doi.org/10.1088/0264-9381/26/17/175008.

[7] L. Bombelli, J. Henson, and R. D. Sorkin, *Discreteness without symmetry breaking: A theorem*, Mod. Phys. Lett. A **24**, 2579–2587 (2009). https://doi.org/10.1142/S0217732309031958.

[8] S. W. Hawking, A. R. King, and P. J. McCarthy, *A new topology for curved space–time which incorporates the causal, differential, and conformal structures*, J. Math. Phys. **17**, 174–181 (1976). https://doi.org/10.1063/1.522874.

[9] D. B. Malament, *The class of continuous timelike curves determines the topology of spacetime*, J. Math. Phys. **18**, 1399–1404 (1977). https://doi.org/10.1063/1.523436.

[10] L. Glaser and S. Surya, *Towards a definition of locality in a manifoldlike causal set*, Phys. Rev. D **88**, 124026 (2013). https://doi.org/10.1103/PhysRevD.88.124026.

[11] D. D. Reid, *Manifold dimension of a causal set: Tests in conformally flat spacetimes*, Phys. Rev. D **67**, 024034 (2003). https://doi.org/10.1103/PhysRevD.67.024034.

[12] M. Roy, D. Sinha, and S. Surya, *Discrete geometry of a small causal diamond*, Phys. Rev. D **87**, 044046 (2013). https://doi.org/10.1103/PhysRevD.87.044046.

[13] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observable_normal_forms.pdf.

[14] B. Müller, *Verified Projection-Event Calculus in Lean 4: Bundled Arbitrary-Partition Pinching, Lüders Retractions, and CHSH Interoperability*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/machine_checked_finite_event_algebras.pdf.

[15] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/reality_as_consensus_protocol.pdf.

[16] B. Müller, A. Osika, M. Poneder, K. Xue, P. Nguyen, and D. Matscheko, *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf.

[17] B. Müller, A. Osika, M. Poneder, K. Xue, B. Cassie, P. Nguyen, J. Kim, D. Matscheko, J. Hill, W. T. Glynn, M. A. Visser, K. A. Anirudha, and B. de La Fournière, *From Observer Consensus to Standard Physics*. Available at https://wkaxfdgxoqmghwgshymt.supabase.co/storage/v1/object/public/papers/from_observer_consensus_to_standard_physics.pdf.

[18] B. Müller, A. Osika, M. Poneder, and K. Xue, *Deriving the Particle Zoo from Observer Consistency*. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[19] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Source available at https://github.com/muellerberndt/oph-physics-sim. Finite provenance and operator producers, independent verifiers, and receipts: https://github.com/muellerberndt/oph-physics-sim/tree/6080f3a4045b7cbd821476a0a8fa0a6b57eeeeab. Large-run source revision: https://github.com/muellerberndt/oph-physics-sim/tree/ce17921eb7504106fef1ba445e1349b5367aa676. Exact-federation archive: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/exact_federation_L6_canonical_20260909. Closure-loop archive: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/closure_loop. Causal-poset archive: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence/source_net_causal_poset. Evidence index: https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence.

[20] FloatingPragma (2026). Observer Patch Holography: Lean library, exact code receipts, finite certificates, and claim ledgers. Available at https://github.com/FloatingPragma/observer-patch-holography.

[21] B. Müller and J. Hill, *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics: A Support-Visible OPH Route to the Clay Problem*. Available at https://github.c

om/FloatingPragma/observer-patch-holography/blob/main/extra/yang_mills_gap_clay_problem.pdf.

[22] B. Müller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.tex.

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-5.6-Sol, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.